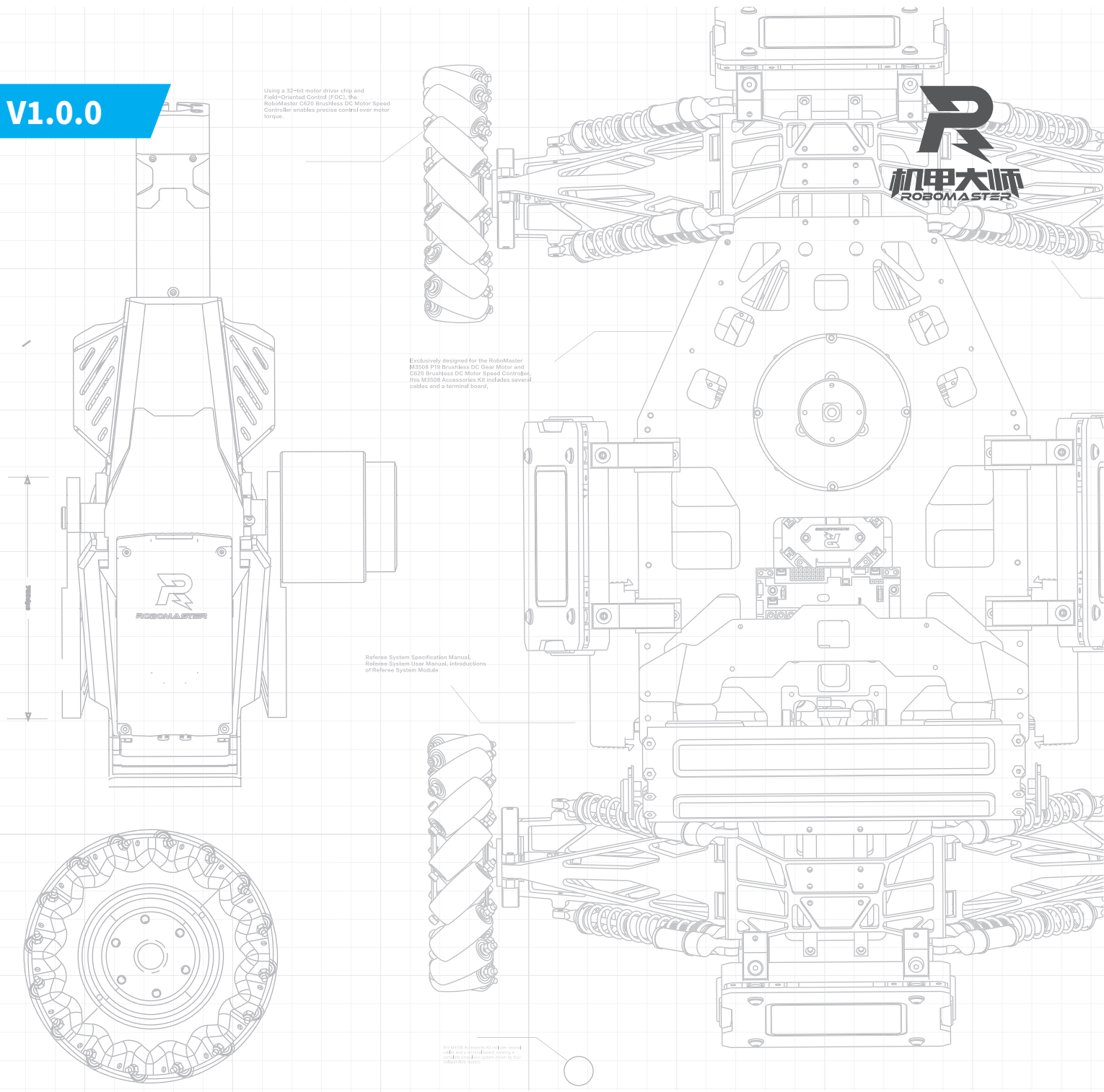
V1.0.0

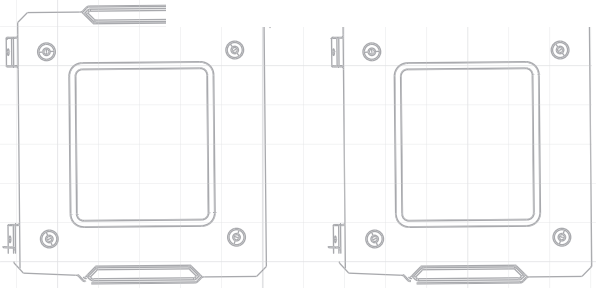


26th National University Student Robot Competition

ROBOMASTER2027

RoboMaster University Championship

Competition Rules Manual



Compiled by the RoboMaster Organizing Committee

Published October 2026

Intellectual Property Statement

The RoboMaster Organizing Committee (RMOC) encourages and advocates technical innovation and open-source technology, and respects the intellectual property of participating teams. All intellectual property developed by a team during the competition belongs to that team. The RMOC does not handle intellectual property disputes among members within a team. Teams must properly handle ownership of intellectual property among their school members, corporate members, and members of other affiliations.

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The owners of intellectual property will pursue legal liability for any infringement of the intellectual property of competition education products provided by the Organizing Committee and host organizations.

For recommendations on open-source materials, see: https://bbs.robomaster.com/article/4820

Reading Notes

The RoboMaster competition is a robot competition whose purpose is to cultivate young engineering talent. Related documents may contain sensitive wording, names, operations, technologies and other content. This content does not reflect any technology or action that exists in reality, nor is it intended to refer to any specific country, organization or individual. This manual explains the robot competition rules. It must not be used for any purpose other than the RoboMaster competition and should not be considered associated with any real event or entity.

Version Number Description

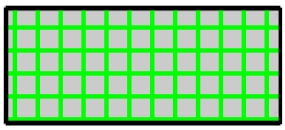
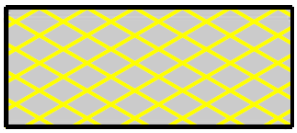
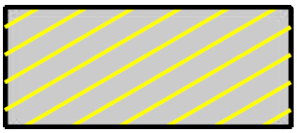
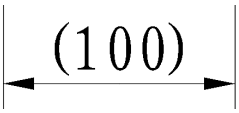
The version numbers of competition rules documents use the semantic versioning format X.Y.Z, with the following meanings:

- X: Updates for different competition stages (fixed at 1 for Regional Competitions and 2 for the Final Tournament; the specific year is shown in the file name and on the cover).
- Y: Updates to important content such as mechanisms and the field, for example, changes to competition mechanisms or robot attributes.
- Z: Corrections that do not change the existing mechanisms or field, such as correcting a typo, revising an ambiguous sentence, or replacing an incorrect drawing.

Examples:

- The first version of a season's manual (for Regional Competitions) is numbered V1.0.0.
- If some competition mechanisms are adjusted on the basis of the first version (for Regional Competitions), the version number is V1.1.0.
- On the basis of the second version (for Regional Competitions), three typos were corrected; the version number is V1.1.1.
- If some competition mechanisms are adjusted on the basis of the third version (for the Final Tournament), the version number is V2.0.0.

Field Drawing Legend

		
Buff Point for both sides	Buff Point for one side	Shared Restricted Zone
		
One-side Restricted Zone	The horizontal plane on which the Battlefield floor lies; it is the lowest plane of the Battlefield.	Dimensions are for reference only.

Revision Log

Date	Version	Revision Record
2026.10.15	V1.0.0	First release

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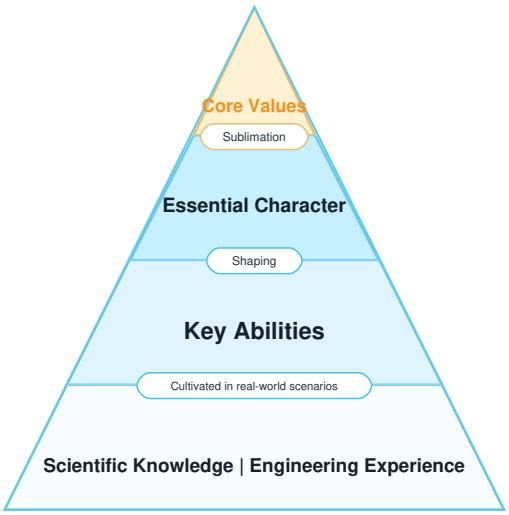
1 Preface

1.1 About the Competition

The National University Student Robot Competition RoboMaster University Championship (RMUC) is initiated by DJI. Its core format is tactical combat between robots. Participating teams must independently design, develop and build multiple robots that meet the specifications, then field them as a robot team. During a seven-minute match, the two sides engage in tactical combat by operating their robots; destroying the opposing Base wins the match.

RMUC aims to cultivate outstanding young engineers and encourages participants to combine what they have learned with practical applications and continue to grow through participation.



- **Core Values**
  Investigating things to extend knowledge, striving for excellence, reflecting on oneself
  - Develop the habit of deep thinking and the ability to locate the root causes of problems
  - Pursue excellence driven by high goals and maintain a positive mindset
  - Face oneself honestly and be aware of one's own limits
- **Essential Character**
  Proactive, positive, seeking truth from facts, upholding fairness
- **Key Abilities**
  Thinking, communication and expression, problem solving, self-directed learning, forward design

Figure 1-1 Talent Cultivation Philosophy

1.2 Major Changes in the New Season

Compared with RMUC 2026, RMUC 2027 makes the following main adjustments:

Robots and Operators

- The former Hero Robot and Engineer Robot are merged into the Heavy Robot.
- The former Aerial Robot is changed to the Aerial System (comprising the Aerial Robot and Hangar).
- The former Custom Controller and Custom Client are merged into the Custom Terminal.
- A Commander and Safety Officer are added.
- The Gimbal Operator and Drone Pilot roles are removed.

Battlefield

- The former Launch Ramp is changed to a Tunnel.
- The overall layout of the Operator Room is adjusted.
- A High Point, Hangar platform and Workbench are added.
- The Flight Zone, Wireless Charging Device Placement Zone and Landing Pad are removed.

Competition Mechanisms

- The HP deduction and exceeding-the-limit penalty mechanisms are adjusted.
- The Economy System is adjusted.
- The Dart System special mechanisms are adjusted.
- The Radar System special mechanisms are adjusted.
- The Adaptive Armor mechanism is added.
- Special mechanisms for the Heavy Robot are added.
- The special mechanisms for the Hero Robot, Engineer Robot and Aerial Robot are removed.

Competition Procedure

The number of backup robots is adjusted.

Violations and Penalties

Some penalty provisions are adjusted to fit the rules.

1.3 About the Competition Rules Documents

The competition rules documents include the Rules Manual, Participant Manual, Robot Building Specifications Manual, Communication Protocol and supplements to these documents. They apply to all participating teams, referees, event staff and other partners.

- During preparation, the Organizing Committee will update the competition rules documents from time to time according to the circumstances.
- During the competition, to allow participating teams to fully demonstrate their technical achievements, the Organizing Committee may adjust the rules at the following points, but these adjustments will not involve direct structural changes to robots.
  - During Regional Competitions, after the matches at an individual site have concluded.
  - During the Final Tournament, after a stage of the schedule (for example, the group stage or the Round of 16 to 8) has been fully completed.

The Organizing Committee has final interpretive authority over the competition rules documents. During the competition, only the Chief Referee may explain the documents on behalf of the Organizing Committee. Questions about the competition rules documents may be directed only to the Chief Referee.

For more RMU reference materials, see the “Rules/Materials Center,” built and maintained by the RoboMaster Organizing Committee. It provides official materials needed for participation in one place to help teams find information efficiently. Its coverage includes:

- Competition rules: Rules Manual, Robot Building Specifications Manual, Participant Manual and Communication Protocol.
- Competition products: Referee System (including onboard modules and the competition engine), robot kits and accessories, RM Assistant product details, document downloads, and purchase and after-sales policies.
- Technical Assessment: Technical Assessment specifications and stage instructions.
- Brand promotion: Brand promotion materials package.
- Sponsorship guidance: Sponsorship Manual and Sponsorship Guide.
- Training system: Participation guides, courses and salons, and open-source materials.

1.4 About Q&A

When participating teams or other relevant personnel have questions about the competition rules documents, they may submit them through official channels for the Organizing Committee to answer.

The process is as follows:

1. The person asking the question completes the questionnaire and submits it:

   https://qingflow.com/f/fpckalpss002?exWsld=1SdM5_PB-NwUtnyPn2b8NA

2. The Organizing Committee posts its responses and updates at:

   https://qingflow.com/appView/fpckalpss002/shareView/fpckam8gs001

Response times:

- Questions submitted during the competition period receive a response within 1 calendar day.
- Questions submitted during the preparation period receive a response within 5 business days.

- Competition period: From the event check-in date through the last day of competition.
- Preparation period: Any time outside the competition period.

Note: Competition and preparation periods are calculated separately for each event and event stage.

Q&A responses have the same effect as the competition rules documents. If a Q&A response conflicts with a rules document, the latest published Q&A response or rules document prevails. Q&A responses for each season apply only to that season.

2 Important Concepts

This chapter briefly explains common important concepts related to the competition rules. For details, consult the relevant chapters using the keyword index.

Table 2-1 Overview of Important Concepts

Concept	Definition
Robot	
Robot	A collective term for the Ground Robots, Combined Systems and Auxiliary Equipment that a participating team brings to compete.
Ground Robot	A robot that operates independently within the competition field, including the Heavy Robot, Infantry Robot and Sentry Robot.
Combined System	A robot made up of two or more system units that competes as a whole, including the Dart System, Radar System and Aerial System. The permitted units and their connection, separation or launch methods are subject to the specific rules for each system.
System Unit	A basic unit that makes up a Combined System and has an independent physical form and function. Unless specific rules provide otherwise, a System Unit must not be further divided into multiple independently moving parts.
Auxiliary Equipment	Equipment that assists Ground Robots and Combined Systems but does not directly participate in competition, including the Custom Terminal.
Robot main states	Surviving: A state in which the robot's Referee System Main Controller Module is connected to the Referee System server and its HP is above 0. Out of Combat: A state in which a surviving robot has not fired a projectile or lost HP for 6 consecutive seconds. Robots are considered out of combat at match start. Defeated: A state in which a robot's HP is 0 (including because an Armor Module was attacked or struck, a Referee System module disconnected, the robot was penalized off the field, or HP was deducted due to a dart). Penalized Off: A state in which a robot receives a red-card warning. Temporarily Activated: A state in which the chassis and gimbal of a defeated robot are temporarily powered on. Its launching mechanism remains powered off. Abnormally Disconnected: A state during a match in which, due to power loss or another reason, the robot's Referee System Main Controller Module cannot connect to the Referee System server.

Concept	Definition
	Note: After a robot is defeated, the Referee System cuts off power output to the robot (except for the Mini PC).
Referee System	The Referee System is an electronic judging system for robot competitions that integrates computing, communication and control. It includes the onboard modules and the competition engine installed on PC hosts (server, referee client, player client), and has functions such as monitoring robot power, projectile launches and damage, and automatically determining the winner according to the competition rules.
Multi-robot communication	The mode of interaction in which robots communicate with each other through the Referee System serial port.
Chassis	A mechanism that carries a Ground Robot's power system and its auxiliary components and supports the robot body.
Chassis Power	The power consumed by a mechanism that uses rotational motion and ground friction to move or rotate the robot horizontally on a flat surface. mechanism's power consumption. See the definition of Chassis Power in the "Referee System Installation Specifications" chapter of the "RoboMaster 2027 University Series Robot Building Specifications Manual."
Launching mechanism	A mechanism that allows projectiles to leave a robot along a fixed path at a certain initial speed.
Launching Mechanism Locked	A locked launching mechanism is in a powered-off state.
Launching Mechanism Unlocked	An unlocked launching mechanism is powered on or off according to the permitted projectile allowance.
Launch	The act of a robot firing a projectile or dart. In the "Competition Mechanisms" chapter below, "launch" means an action detected by the Referee System.
Muzzle Velocity	The speed value detected by the relevant Referee System module after a projectile or dart has finished accelerating.
Shooter Heat	Shooter Heat increases as a robot fires and is used to limit continuous firing.
Projectile Allowance	The number of projectiles the robot is currently allowed to fire.
Initial HP	The HP value assigned to the robot by the Referee System at the start of the match. Note: The Referee System rounds HP to the nearest integer when settling HP.
Current HP	The robot's real-time HP value.
Max HP	The maximum value to which a robot's HP can be restored.
Experience	A value that a robot accumulates to level up; it can be gained in multiple ways.
Invincible	A state in which a robot takes no damage from projectiles, darts or collisions.

Concept	Definition
Attack	An act of a robot firing a projectile or dart, or an Aerial Robot making an impact.
Defeat	When one side's robot attacks the Armor Module of an opposing robot, Base or Outpost, reducing its HP to 0.
Activate	When a surviving robot reaches a Buff Point and its Field Interaction Module detects a Field Interaction Module card in the area, or it meets another activation condition.
Entanglement	A state in which, during a match, a mechanism connects two robots so that moving one robot in any direction keeps the mechanism connected to the other robot.
Collision	An intentional collision by one side's robot during a match.

Battlefield

Buff Point	An area that produces a special effect when activated by a robot during a match.
Restricted Zone	An area that robots are prohibited from entering.
Field Prop	An element of the Battlefield, including but not limited to the Base, Outpost and Tech Core.

Personnel

Arbitration Committee	The Chief Referee and other relevant Organizing Committee officials, responsible for matters such as handling appeals.
Referee	A person who maintains order at the venue and enforces the competition rules.
Chief Referee	The person with final interpretive authority over the competition rules documents during the competition.
Head Referee	The lead referee responsible for maintaining order at the venue and enforcing the competition rules.
Head Inspector	The referee who leads and assigns inspection work and has final interpretive authority over inspection standards.
Participating Team	A team registered for the current season and entered into the registration system.
Participant	A person registered for the current season and entered into the registration system.
On-Site Personnel	Official team members, Team Mentors and Advisors registered for the current season and entered into the registration system who may enter the waiting and competition areas.

Concept	Definition
Operator	On-site personnel responsible for controlling robots during a match, including Ground Robot operators, the Commander and the Safety Officer.
Violating Party	A participating team that commits a violation.
Violating Person	A participant who commits a violation.

Competition Procedure

Match Time Description	Throughout this document, “X minutes Y seconds” means a countdown of (6 − X):(59 − Y); “from X minutes Y seconds to M minutes N seconds” means a countdown from (6 − X):(59 − Y) to (7 − M):(60 − N). Examples: Five minutes after the match starts means a countdown of 1:59. From 2 to 5 minutes after the match starts means a countdown of 4:59–2:00.
Round	A complete match consisting of the Setup Period, Referee System Self-Check Period, five-second countdown and Match Period.
Match	Depending on the format, one match may include several rounds.
Official Technical Pause	A technical pause initiated by the Head Referee during the Setup Period or Referee System Self-Check Period.
Team Technical Pause	A technical pause requested by a participating team during the Setup Period.
Normal Battle Damage	A failure of a Referee System onboard-end module during the three-minute inter-round Setup Period or seven-minute Match Period.

Match Result Determination

Attack Damage	HP deducted when a side’s robot, Base or Outpost is hit by projectiles or darts, or impacted by an Aerial Robot. Note: HP deducted from a side’s robot, Base or Outpost as a penalty counts as attack damage dealt by the opposing side.
Non-Attack Damage	HP deducted from a side’s robot due to an impact on an Armor Module, Referee System module disconnection or another cause.
Total Remaining HP	The total remaining HP of all surviving robots on a side at the end of each round.
Control Level	A measure of a side’s overall control of the Battlefield through occupying positions at the end of each round.

3 Robots and Operators

RoboMaster emphasizes teams of robots competing together and requires balanced cooperation among the robots. For robot building specifications, see
the RoboMaster 2027 University Series Robot Building Specifications Manual.

For each match, teams must bring out the required numbers of each robot type according to the lineup table below.

Table 3-1 Robot lineup

Type	Name	No.	Main lineup quantity	Maximum backup quantity	Maximum Custom Terminals Custom Terminal quantity	Maximum reserve Custom Terminals Custom Terminal quantity	Competition stage
Ground Robot	Heavy Robot	1	1	1	1	1	Throughout RMUC 2027
	Infantry Robot	3/4	2	2	1 per robot	1 per robot	
	Sentry Robot	7	1		0	0	
Combined System	Aerial System	6	Hangar: 1 Aerial Robot: 4	Hangar: 0 Aerial Robot: BO2/3: 4 BO5: 8	1	1	
	Dart System	8	Dart Launching Station: 1 Dart bodies: 4	Dart Launching Station: 0 Dart bodies: BO2/3: 4 BO5: 8			
	Radar System	9	Computing Platform End: 1 Sensor End: 1	Computing Platform End: 0 Sensor End: 0	0	0	

The total number of robots each team may register for an independent competition stage is 9 (backup dart bodies and backup Aerial Robots do not count toward this total).

Minimum lineup for the first round of each match: at least 3 Ground Robots.

The Operator lineup is as follows:

Table 3-2 Operator lineup

Type	Robot operated	Number at full lineup
Ground Robot Operator	Heavy Robot	1
	Infantry Robot	2
	Sentry Robot	0
Commander	Aerial System (optional), Dart System (optional)	1
Safety officer	Aerial Robot (restricted)	1

- Operators must be on-site personnel from the participating team for the current season, excluding Advisors.
- After each round, Operators may be replaced by other on-site personnel from the same team, excluding Advisors.
- Whether or not the Aerial System or Dart System enters the field, the Commander may operate the corresponding Referee System Operator-Side Client, configure the Custom Terminal and wear the corresponding headset. After the three-minute Setup Period ends, the Commander must not change position.

3.1 Robots

3.1.1 Heavy Robot

The Heavy Robot can obtain diverse Team Buffs by acquiring and assembling Energy Units; after meeting certain conditions, it can also fire 42mm projectiles.

Table 3-3 Key characteristics of the Heavy Robot

Key characteristic	Description
Initial zone	Starting Zone
Operating mode	Unrestricted; at most 1 Remote Controller and 1 Custom Terminal may be configured
Multi-robot communication	Allowed
Launching mechanism	1 installable modular 42mm launching mechanism
Initial Launch Speed limit (m/s)	16.5 after the modular 42mm launching mechanism is successfully installed
HP deduction and exceeding-the-limit penalty mechanisms	- Initial Launch Speed exceeding the limit: applies • Barrel Heat exceeding the limit and cooling: applies • Chassis Power exceeding the limit: applies • Attack Damage or collision: applies • Referee System module going offline: applies • Penalty Damage: applies See "5.1 HP deduction and exceeding-the-limit penalty mechanisms"
HP Recovery and Revival mechanisms	- HP Recovery: through permitted Buff Points or remote exchange of HP. • Revival: progress-bar Revival or exchange for immediate Revival. See "5.3 HP Recovery and Revival mechanisms"
Projectile loading and resupply	- During the three-minute Setup Period, the robot may be preloaded with 42mm and 17mm projectiles; the initial Projectile Allowance is 0. • During the seven-minute Round, Projectile Allowance can be exchanged through permitted Buff Points or by remote exchange.

Key Features	Description
Experience and Performance System	Not applicable Note: The attributes of the Heavy Robot change as the match progresses.
Activatable Buff Points	Base Buff Point Central Highland Buff Point Terrain-Crossing Buff Points (Road, Highland and Tunnel) High Point Buff Point Outpost Buff Point Assembly Zone Buff Point Supply Zone Buff Point Fortress Buff Point See “5.6.4 Battlefield Buff Mechanics”.

Table 3-4 Key characteristics of the modular 42mm launching mechanism

Key Features	Description
Starting Zone	Workbench
Projectile loading and supply	During the three-minute Setup Period, 42mm and 17mm projectiles may be preloaded, but the initial Projectile Allowance is 0. During the seven-minute Match Period, Projectile Allowance may be exchanged at permitted Buff Points or remotely.

3.1.2 Infantry Robot

The Infantry Robot can fire 17mm projectiles.

Table 3-5 Key characteristics of the Infantry Robot

Key Features	Description
Starting Zone	Starting Zone
Operating Mode	Unrestricted; at most 1 Remote Controller and 1 Custom Terminal may be configured.
Multi-robot communication	Allowed

Key characteristic	Description
Launching mechanism	1 17mm launching mechanism
Initial Launch Speed limit (m/s)	25
HP deduction and exceeding-the-limit penalty mechanisms	- Initial Launch Speed exceeding the limit: applies • Barrel Heat exceeding the limit and cooling: applies • Chassis Power exceeding the limit: applies • Attack Damage or collision: applies • Referee System module going offline: applies • Penalty Damage: applies See "5.1 HP deduction and exceeding-the-limit penalty mechanisms"
HP Recovery and Revival mechanisms	- HP Recovery: through permitted Buff Points or remote exchange of HP. • Revival: progress-bar Revival or exchange for immediate Revival. See "5.3 HP Recovery and Revival mechanisms"
Projectile loading and resupply	- During the three-minute Setup Period, the robot may be preloaded with 42mm and 17mm projectiles; the initial Projectile Allowance is 0. • During the seven-minute Round, Projectile Allowance can be exchanged through permitted Buff Points or by remote exchange. See "5.4 Economy System"
Experience and Performance System	Applicable Note: Parameters such as Chassis Power limit, initial HP, Max HP, Barrel Heat Limit and Barrel Cooling Rate depend on Level and chassis type; see "5.5 Experience and Performance System".
Activatable Buff Points	- Base Buff Point • Central Highland Buff Point • Terrain-crossing Buff Points (Road, Highland, Tunnel) • High Point Buff Point • Outpost Buff Point • Supply Zone Buff Point

Key characteristic	Description
	Fortress Buff Point See “5.6.4 Battlefield Buff Mechanics”.

3.1.3 Sentry Robot

The Sentry Robot may operate fully or semi-automatically and can fire 17mm projectiles.

Table 3-6 Key characteristics of the Sentry Robot

Key characteristic	Description
Initial zone	Starting Zone
Operating mode	At most 1 Remote Controller, used for debugging
Multi-robot communication	Can send data to all friendly robots, but can only receive data from the Radar
Launching mechanism	1 17mm launching mechanism
Initial Launch Speed limit (m/s)	25
HP deduction and exceeding-the-limit penalty mechanisms	- Initial Launch Speed exceeding the limit: applies • Barrel Heat exceeding the limit and cooling: applies • Chassis Power exceeding the limit: applies • Attack Damage or collision: applies • Referee System module going offline: applies • Penalty Damage: applies See "5.1 HP deduction and exceeding-the-limit penalty mechanisms"
HP Recovery and Revival mechanisms	- HP Recovery: through permitted Buff Points or remote exchange of HP. • Revival: progress-bar Revival or exchange for immediate Revival. See "5.3 HP Recovery and Revival mechanisms"
Projectile loading and resupply	- During the three-minute Setup Period, the robot may be preloaded with 17mm and 42mm projectiles; the initial Projectile Allowance is 300. • During the seven-minute Round, Projectile Allowance can be exchanged through permitted Buff Points, obtained directly, or remote exchange of Projectile

Key characteristic	Description
	Projectile Allowance. See “5.4 Economy System”.
Experience and Performance System	The Experience and Performance System does not apply; performance depends on how the robot is operated, as follows: Fully automatic Sentry Robot • Initial HP / Max HP is 400 • Barrel Heat Limit is 260 • Barrel Cooling Rate is 30/s • Chassis Power limit is 100W Semi-automatic Sentry Robot • Initial HP / Max HP is 200 • Barrel Heat Limit is 100 • Barrel Cooling Rate is 10/s • Chassis Power limit is 60W See "5.7.2 Special Sentry Robot mechanisms"
Activatable Buff Points	- Base Buff Point • Central Highland Buff Point • Terrain-crossing Buff Points (Road, Highland, Tunnel) • High Point Buff Point • Outpost Buff Point • Supply Zone Buff Point • Fortress Buff Point See "5.6.4 Field Buff mechanisms"

3.2 Combined Systems

3.2.1 Aerial System

The Aerial System can perform diverse attack and reconnaissance missions on the Battlefield.

Table 3-7 Key characteristics of the Aerial Robot

Key characteristic	Description
Initial zone	Hangar
Operating mode	Fully automatic, but must be equipped with one Remote Controller for emergency stop
Multi-robot communication	Only indirect communication through the Hangar is allowed

Table 3-8 Key characteristics of the Hangar

Key characteristic	Description
Initial zone	Hangar platform
Operating mode	Unrestricted; at most 1 Remote Controller may be configured
Multi-robot communication	Allowed
Launching mechanism	Not applicable
Projectile loading and resupply	Not allowed

3.2.2 Dart System

The Dart System can attack the opposing Outpost and Base by launching darts.

Table 3-9 Key characteristics of the Dart System

Key characteristic	Description
Initial zone	Dart Launching Station
Operating mode	Unrestricted; at most 1 Remote Controller may be configured
Multi-robot communication	Allowed

3.2.3 Radar System

The Radar can autonomously obtain Battlefield information and transmit it to friendly robots or the Referee System Operator-Side Client through multi-robot communication; it can also intercept the opposing side's information.

Table 3-10 Key characteristics of the Radar System

Key characteristic	Description
Initial zone	Radar Base
Operating mode	Runs automatically; at most 1 Remote Controller, used for debugging
Multi-robot communication	Can send data to all friendly robots, but can only receive data from the Sentry Robot

3.3 Auxiliary Equipment

3.3.1 Custom Terminal

The Custom Terminal (formerly the Custom Controller and Custom Client) is a multipurpose device built by participating teams. It monitors robot movement and status, controls robot actions, receives match information and robot video feeds sent by the Referee System competition engine server, and sends match-specific interaction commands to the server, expanding the possibilities for human-robot interaction.

A typical Custom Terminal may consist of a display device, a computing platform (including memory, controller and processor) and a set of input devices. It may also be integrated, for example, as a handheld tablet with a network port.

Example: A team builds a Custom Terminal that receives robot video and displays it on a custom monitor, overlays a self-made UI shown over an unofficial link, and controls the robot through a matching controller. The team can therefore compete without using the official Referee System Operator-Side Client or Remote Controller.

All Custom Terminals in the Operator Room may be connected to one another by wire and communicate. A Custom Terminal may also connect by wire to the competition engine server through the RJ45 port in the Operator Room to send and receive relevant content.

A Custom Terminal may receive match information displayed by the Referee System Operator-Side Client and available to the team, such as the HP of both sides, the team's Economy, remaining match time and key match events.

A Custom Terminal may send match commands supported by the Referee System Operator-Side Client, such as selecting the Performance System before a match, exchanging Projectile Allowance during a match and intervening in Sentry actions.

4 Battlefield

4.1 Battlefield overview

- Unless otherwise indicated, all dimensions described in this document have tolerances. For structures smaller than 100 mm, dimensional error is within ±5 mm; for dimensions of 100 mm or more, error is within ±5%. Angular error is within ±3° for Battlefield structures and ±1° for Field Props. Dimensions in field drawings are in mm. To ensure field stability and safety, details or construction processes may be adjusted during actual production. Battlefield elements or dimensions not specified in detail in the drawings follow the actual field.

- The Battlefield has a centrally symmetric layout. Where this document describes or illustrates a field module using the red side as an example, the blue side is the same.
  The field environment is complex and may have asymmetric environmental factors such as uneven magnetic fields and venue air-conditioning airflow.
- Renderings are for illustration only; specific field dimensions follow the corresponding drawings. Matters not specified follow the actual conditions during Adaptive Training.
- The Battlefield contains multiple Buff Points and Restricted Zones. To keep this document concise and clear, this chapter does not describe them individually; their definitions are given in “5.6.4.1 Battlefield Buff Point Overview” and “7.2.3.2 Robot Interaction with Field Props”.

The core RMUC competition field is called the “Battlefield.” It is an area 28 m long and 15 m wide, with a wooden internal structure covered by 2 mm PVC flooring or PVC flooring with embedded wiring; some structures have a metal protective layer. It mainly contains the Base Zone, Highland Zone, Road Zone and other areas. A black steel barrier surrounds the Battlefield, with its upper edge 2.4 m above the Battlefield floor.

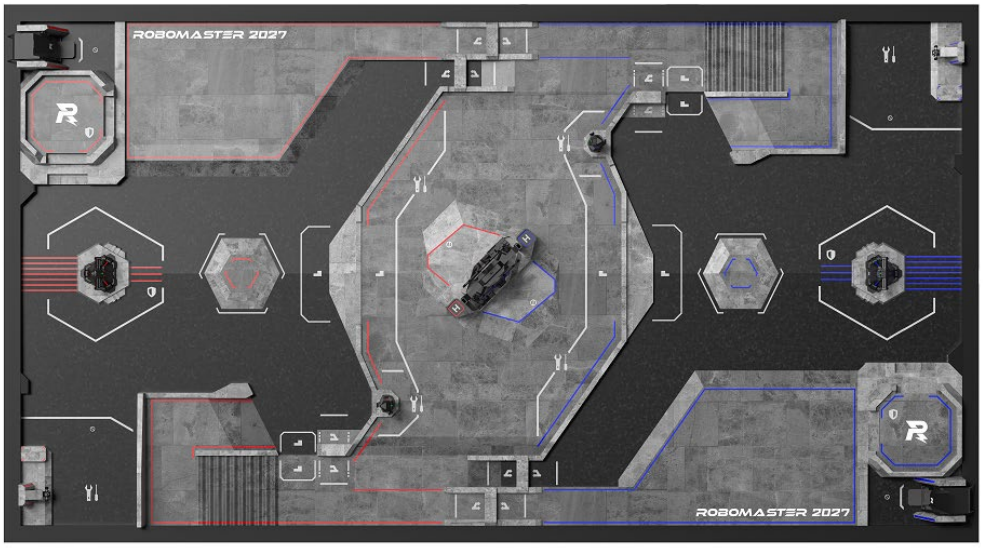
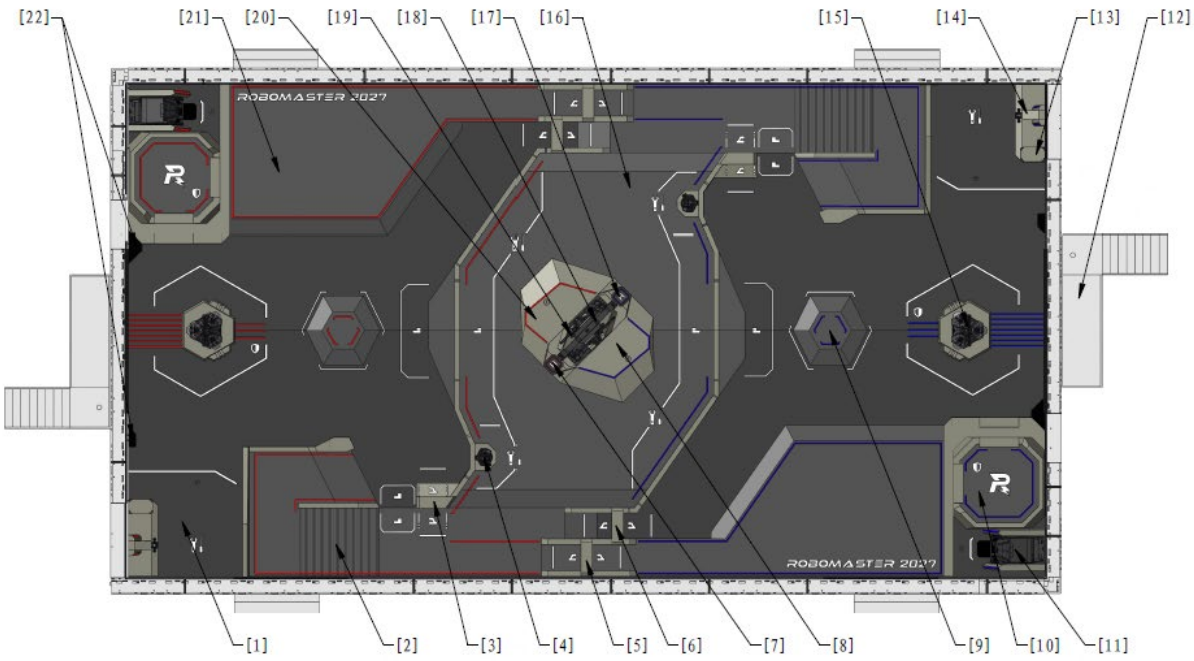


Figure 4-1 Top-view rendering of the Battlefield

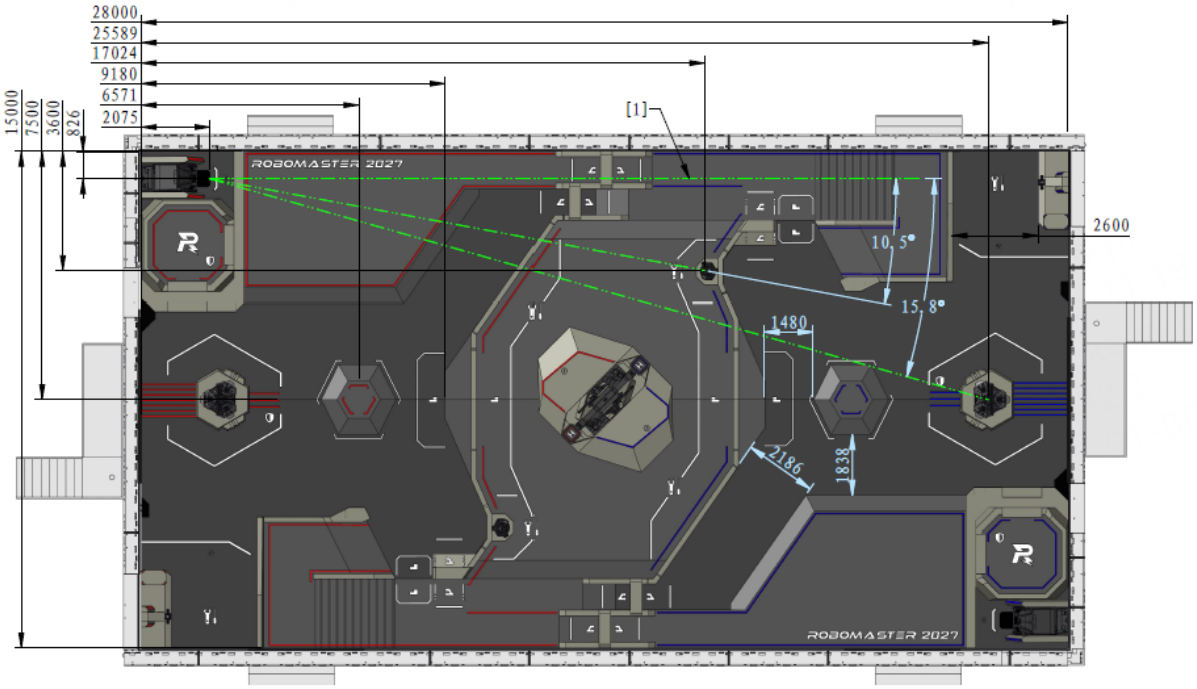
Figure 4-2 Oblique-view rendering of the Battlefield

Figure 4-3 Axonometric rendering of the Battlefield



[1] Supply Zone	[2] Road Zone	[3] Road Tunnel	[4] Outpost
[5] Far Highland Tunnel	[6] Near Highland Tunnel	[7] Red Hangar platform	[8] Blue Assembly Zone
[9] Fortress	[10] High Point	[11] Dart Launching Station	[12] Radar Base
[13] Workbench	[14] Resource Station	[15] Base	[16] Central Highland
[17] Blue Hangar Platform	[18] Energy Mechanism	[19] Tech Core	[20] Red Assembly Zone
[21] Trapezoidal Highland	[22] Fan		

Figure 4-4 Battlefield module diagram

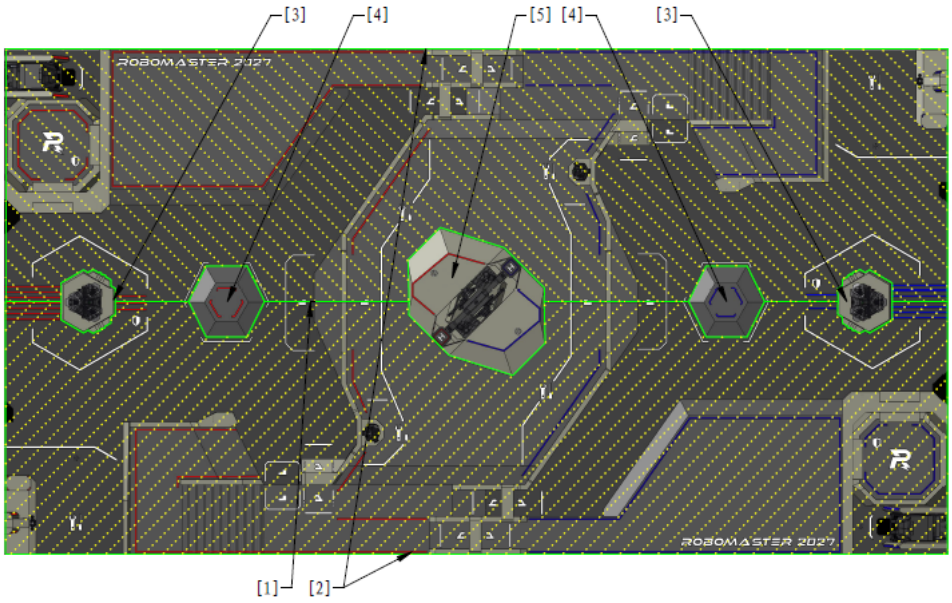


[1] Parallel to the long side of the Battlefield and facing the Dart Launching Station

Figure 4-5 Battlefield module positioning dimension drawing

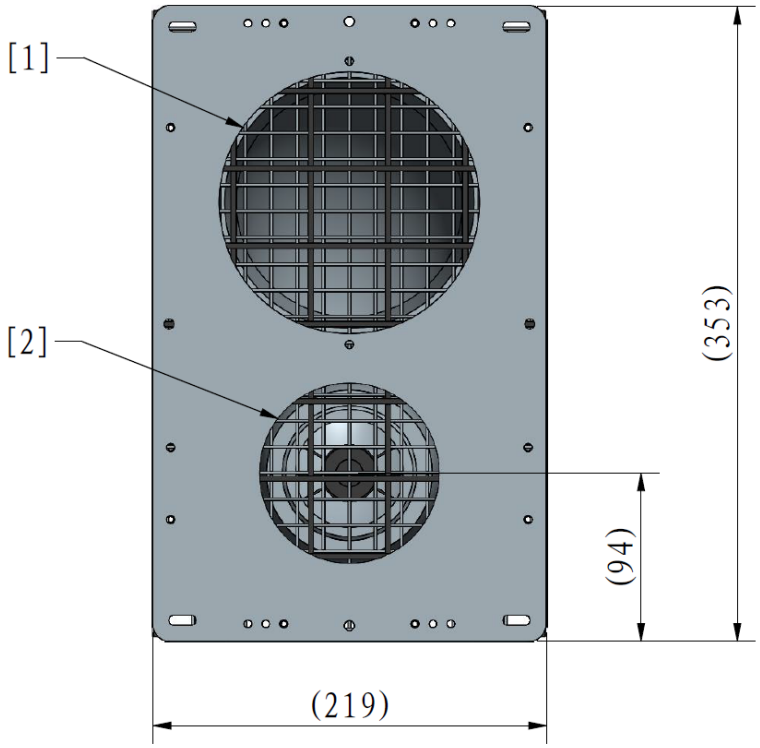
All field dimensions in this document are based on the left and right sloping surfaces. The actual Battlefield floor is not level: it slopes 1°–2° from the central axis toward both long sides, as shown below. The slope causes projectiles to roll toward the two long sides.

To allow projectiles to be recovered, there may be slots approximately 50 mm high where some wooden Battlefield structures meet the floor. After a match, the duct fans inside the wooden structures may be turned on briefly.



[1] Higher side [2] Lower side [3] Base pedestal [4] Fortress pedestal [5] Assembly Zone pedestal

Figure 4-6 Battlefield slope diagram



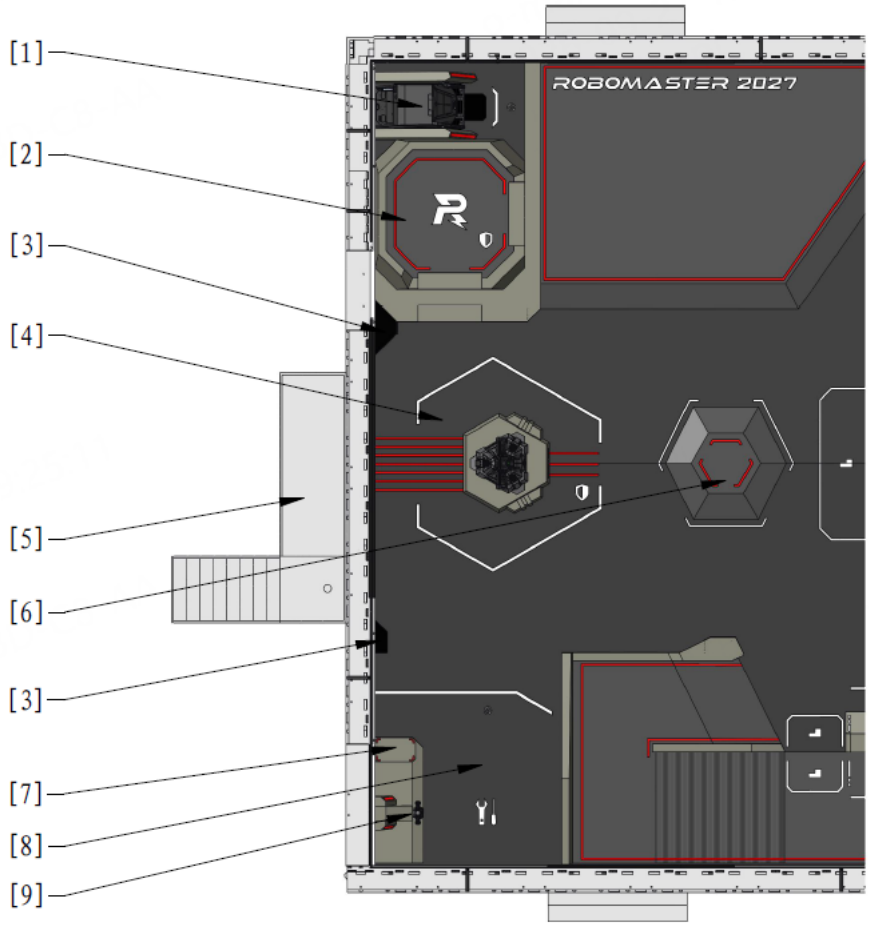
[1] Air inlet [2] Air outlet

Figure 4-7 Fan diagram

The fans are embedded in the wooden structures according to the shape of the wooden structure at the venue.

4.2 Base Zone

The Base Zone includes the Starting Zone, Base, Dart Launching Station, High Point, Radar Base, Supply Zone and Fortress.

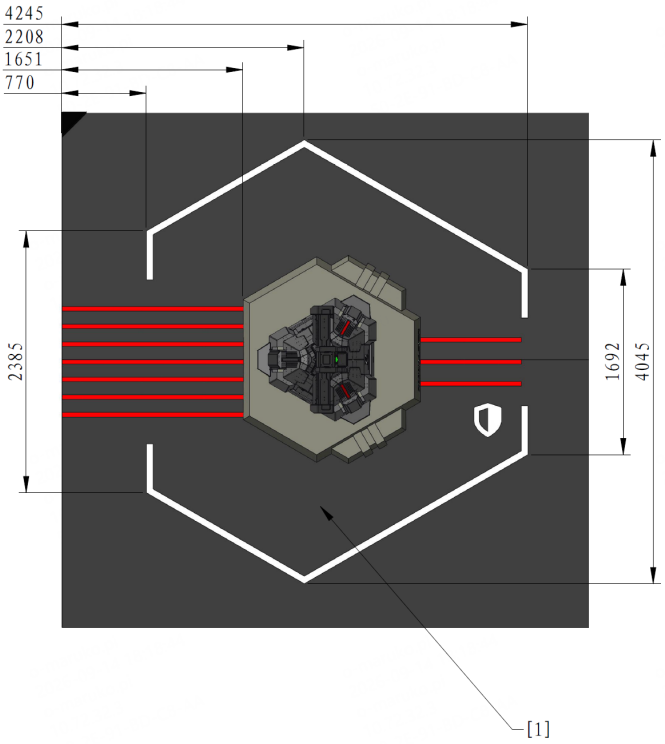


[1] Dart Launching Station	[2] High Point	[3] Fan	[4] Starting Zone	[5] Radar Base
[6] Fortress	[7] Workbench	[8] Supply Zone	[9] Resource Station	

Figure 4-8 Base Zone diagram

4.2.1 Starting Zone

The Starting Zone is a hexagonal area around the Base where Ground Robots are placed before the match starts.



[1] Starting Zone

Figure 4-9 Schematic of the robot Starting Zone

4.2.2 Base

The Base is the core target of both sides' offense and defense. It is placed on the Base pedestal at the center of each side's Starting Zone.

The Base consists of the Base body, six Base Armor Modules, a Dart Detection Module and Base Armor. Base Armor has two states: closed and deployed.

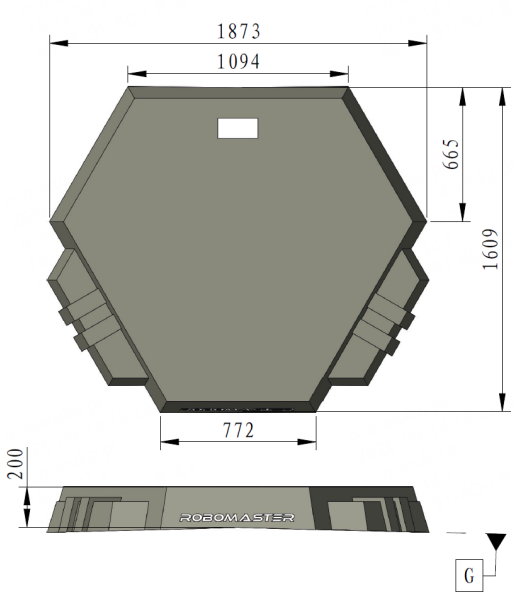
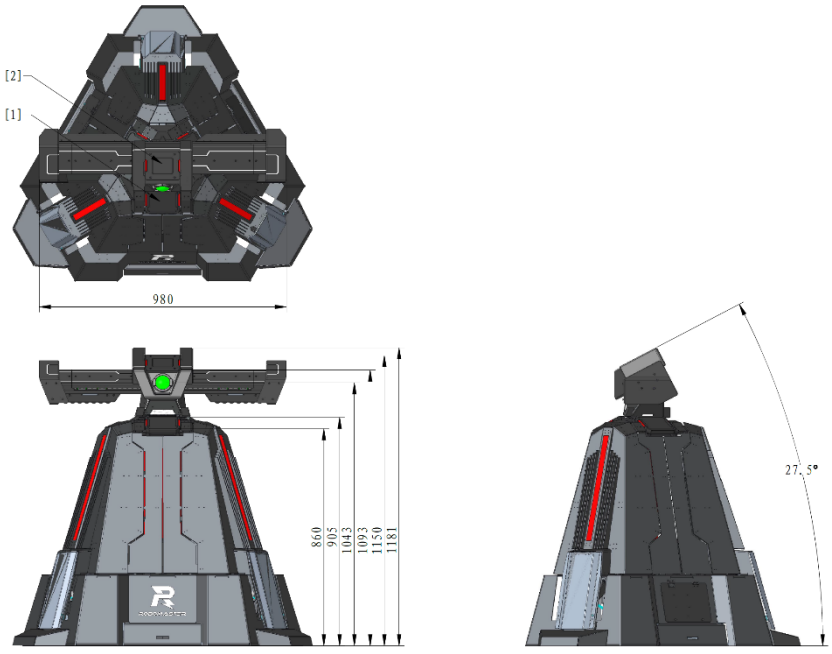


Figure 4-10 Schematic of the Base mount



[1] Base Armor Module [2] Dart Detection Module

Figure 4-11 Dimensions of the Base with armor closed

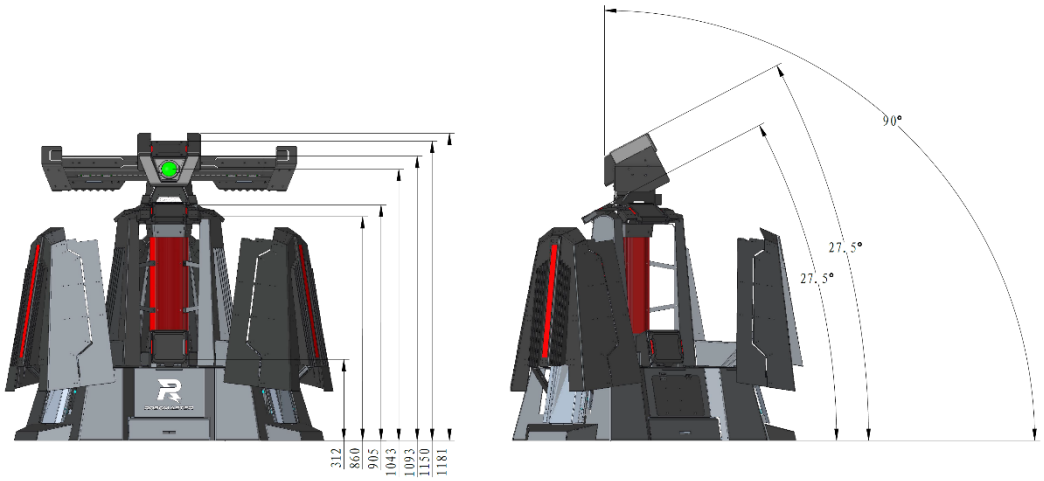
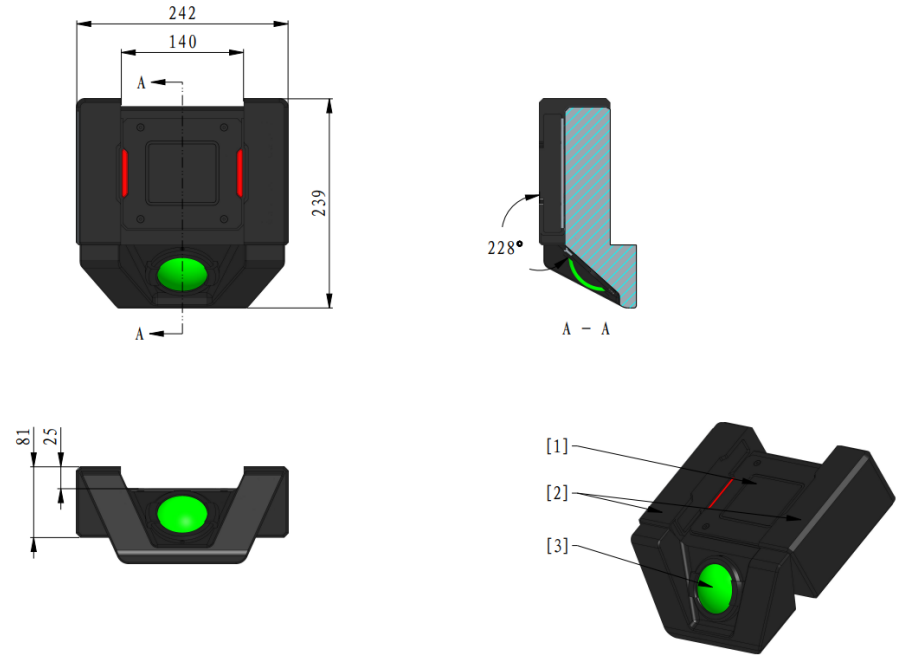


Figure 4-12 Dimensions of the Base with armor open

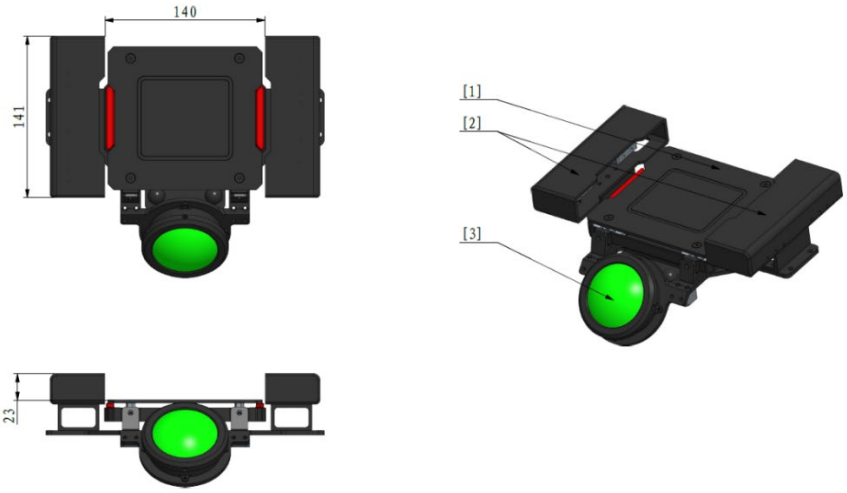
The Dart Detection Module is located at the top of the Base and Outpost. It consists of a Dart Armor Module, Dart Detection Sensor and Dart Guidance Light.

The Dart Armor Module detects dart impacts. The Dart Detection Sensor detects the infrared beam emitted by the Dart Trigger Device. The system considers the module hit by a dart when the projection of the Dart Trigger Device onto the Dart Detection Module falls completely within the Dart Armor Module and the module detects both the infrared beam and an impact. If only an impact is detected, the system considers the module hit by a projectile. The Dart Guidance Light emits green visible light at a wavelength of 520 nm. When approximated as a point source, its luminous intensity is about 10 cd and its illuminated part is about 55 mm in diameter; it guides darts toward the target.

Even when the projection of the Dart Trigger Device onto the Dart Detection Module does not fall completely within the Dart Armor Module, there is still some probability that the Dart Detection Module will identify a hit successfully.



Base Dart Detection Module



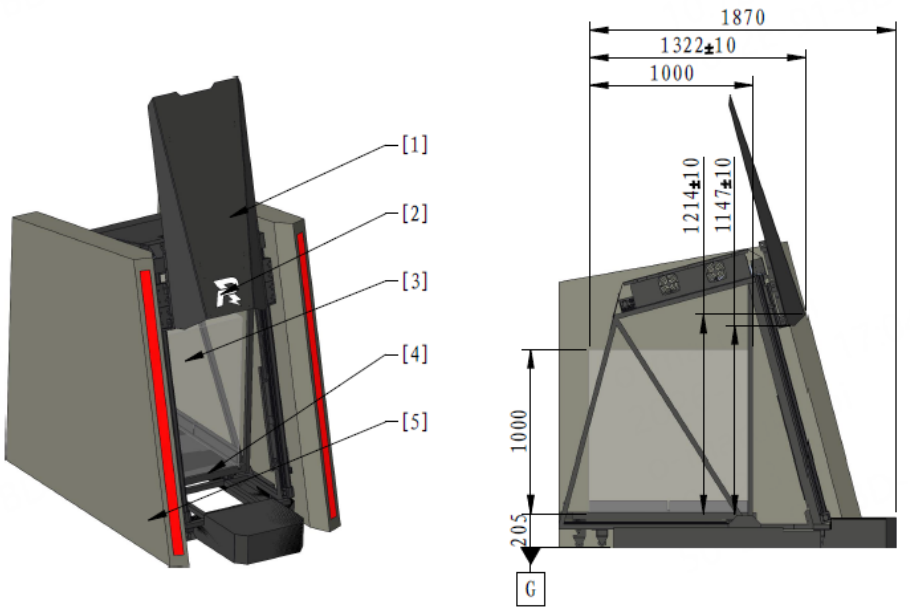
Outpost Dart Detection Module

[1] Dart Armor Module [2] Dart Detection Sensor [3] Dart Guidance Light

Figure 4-13 Schematic of the Dart detection module

4.2.3 Dart Launching Station

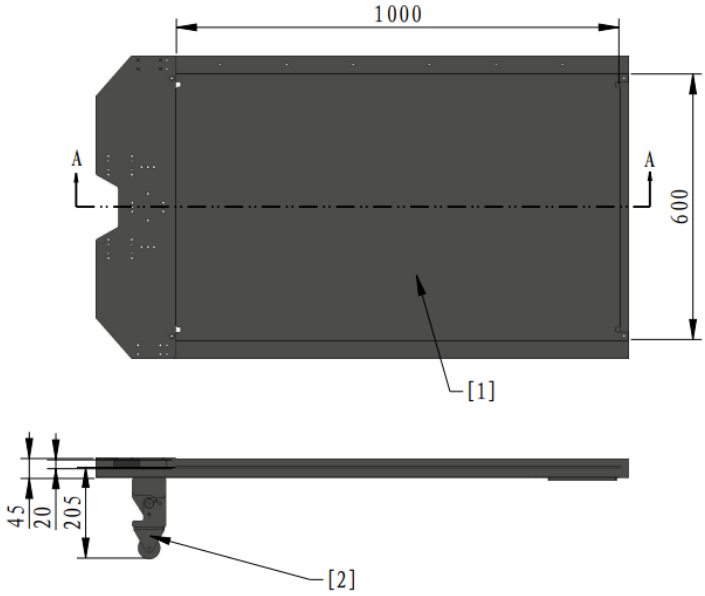
The Dart Launching Station is the only placement zone for the Dart System. It consists of the station body, slide and gate. Lighting inside the station cannot be turned off. During the Referee System Self-Check Period, the gate opens briefly (about 5 seconds); while it is open, the Dart Detection Module remains stationary at the position corresponding to the Base fixed target. During the three-minute Setup Period and 15-second Referee System Self-Check Period, the Dart Guidance Lights on the Outpost and Base are lit.



[1] Gate [2] Status Indicator [3] Dart System placement space [4] Slide [5] Launching Station body

Figure 4-14 Schematic of the Dart Launching Station

Figure 4-15 Schematic of the sliding platform extended



[1] Dart System placement surface

[2] Support wheel

Figure 4-16 Dimensions of the sliding platform

4.2.4 High Point

The High Point is in the Base Zone. Three steps connect the Trapezoidal Highland to the High Point, and robots can climb the steps to enter the High Point.

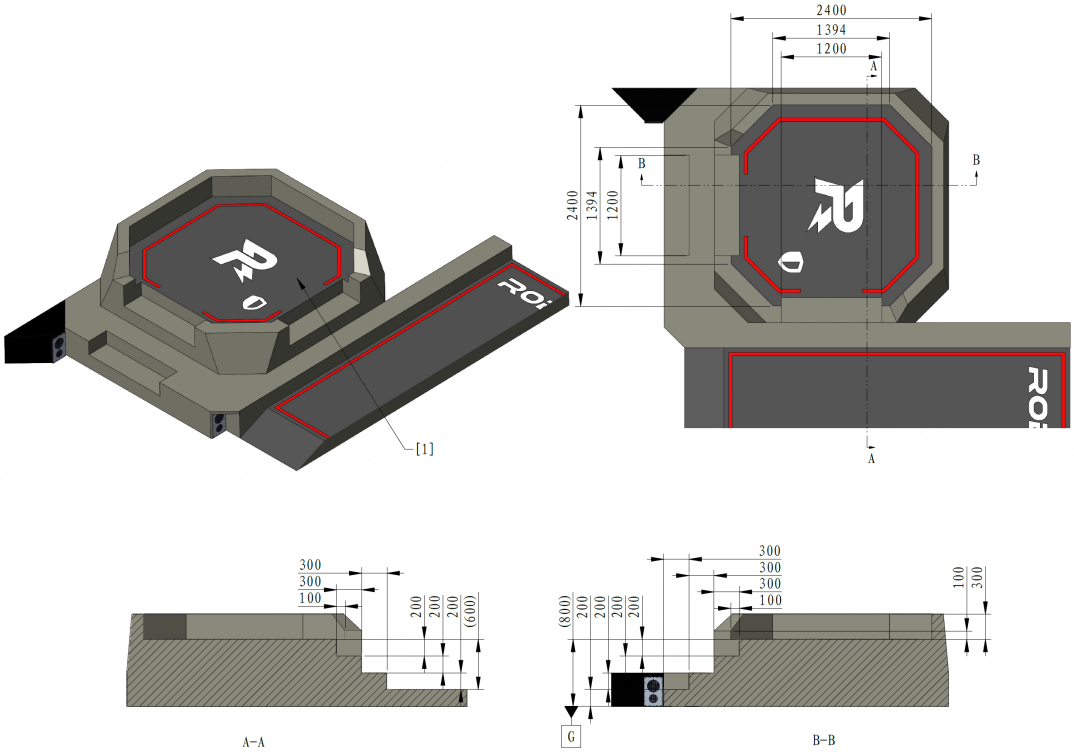


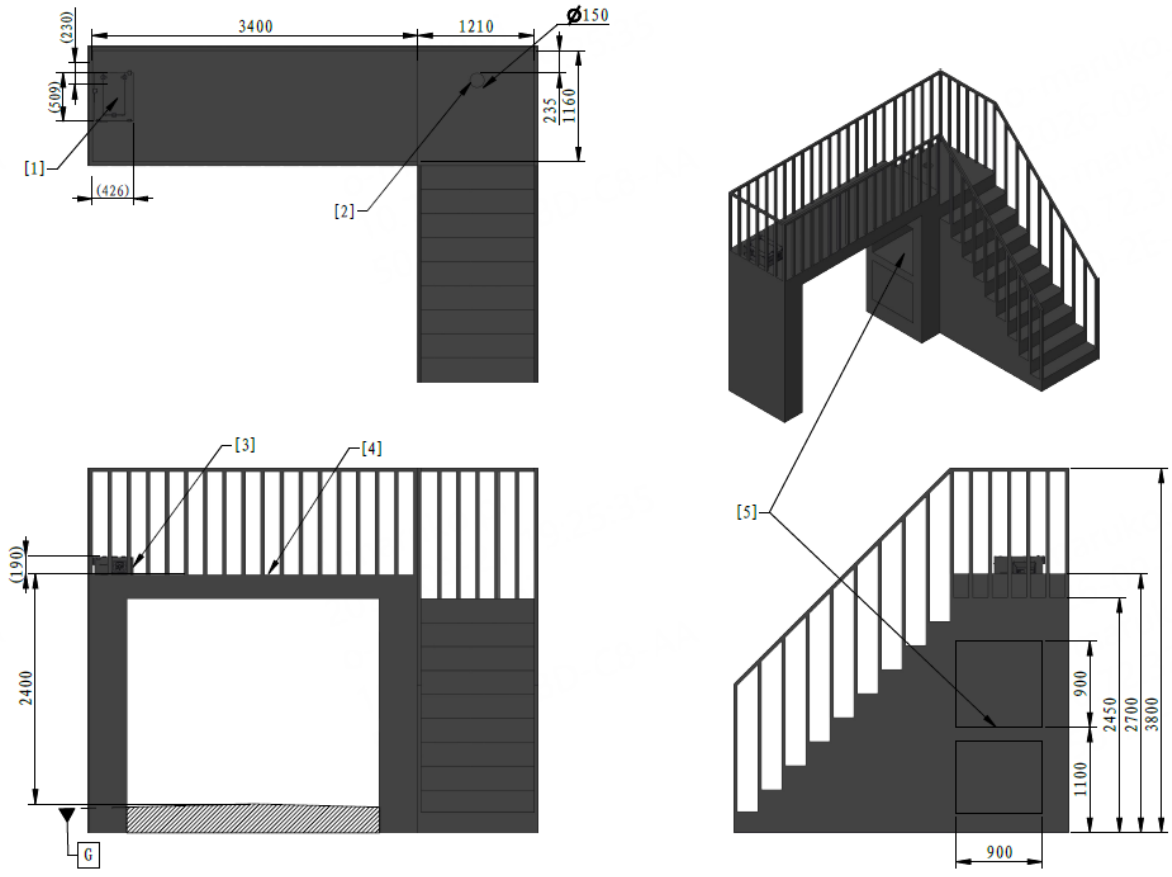
Figure 4-17 Schematic of the High Point

4.2.5 Radar Base

The Radar Base holds the Radar System. Its top is a 3.4 m × 1.16 m platform. The center of the platform is aligned with the Battlefield centerline and is about 2.4 m above the Battlefield floor; a 1.1 m-high railing surrounds it. A cable slot for the sensor data cable is located beside the platform and may be used during a match according to the actual field conditions.

The Radar computing platform end is powered by 220 V mains. The platform is equipped with:

- One official display that supports HDMI input only, with a resolution of 1920 × 1080. Teams may use it to check Radar System operation.
- One HDMI male-to-male signal cable for connecting the Radar System to the official display.
- One fixed five-hole outlet compliant with the new national standard. Its single-phase two-pin and three-pin sockets may be used at the same time to power the Radar System.



[1] Radar Information Wave

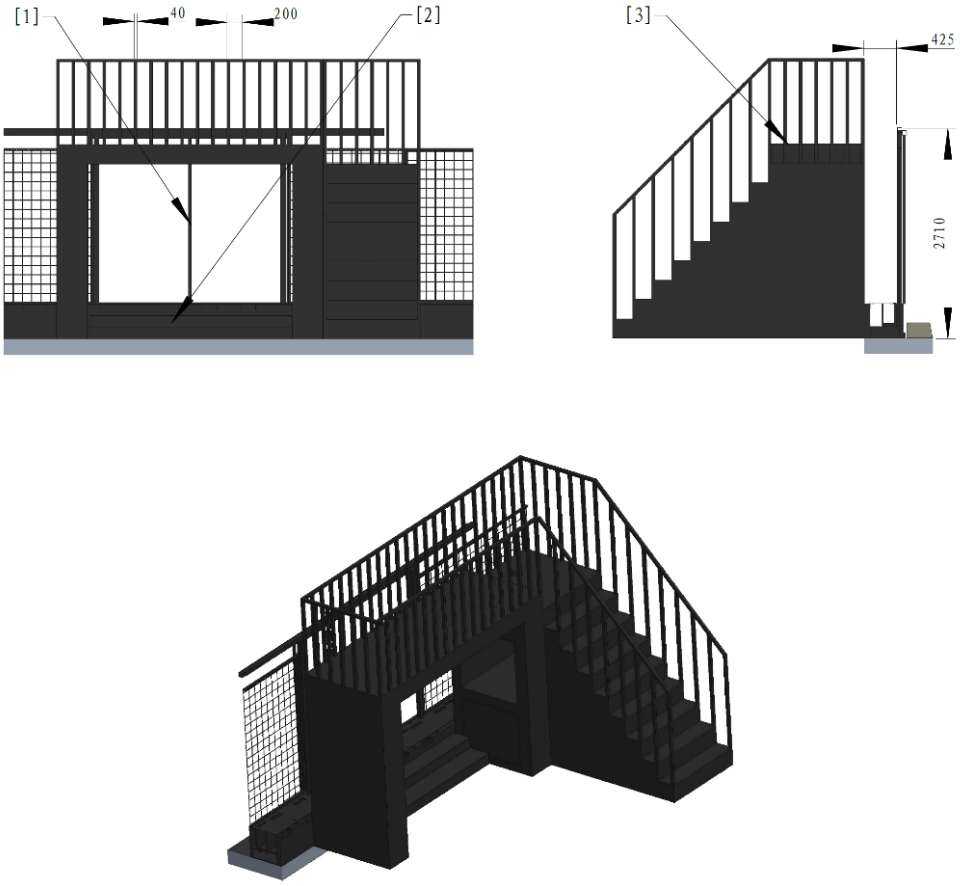
[2] Sensor data cable slot

[3] Radar Information Wave transmitting surface

[4] Radar Sensor placement platform

[5] Radar computing platform placement platform

Figure 4-18 Schematic of the Radar Base

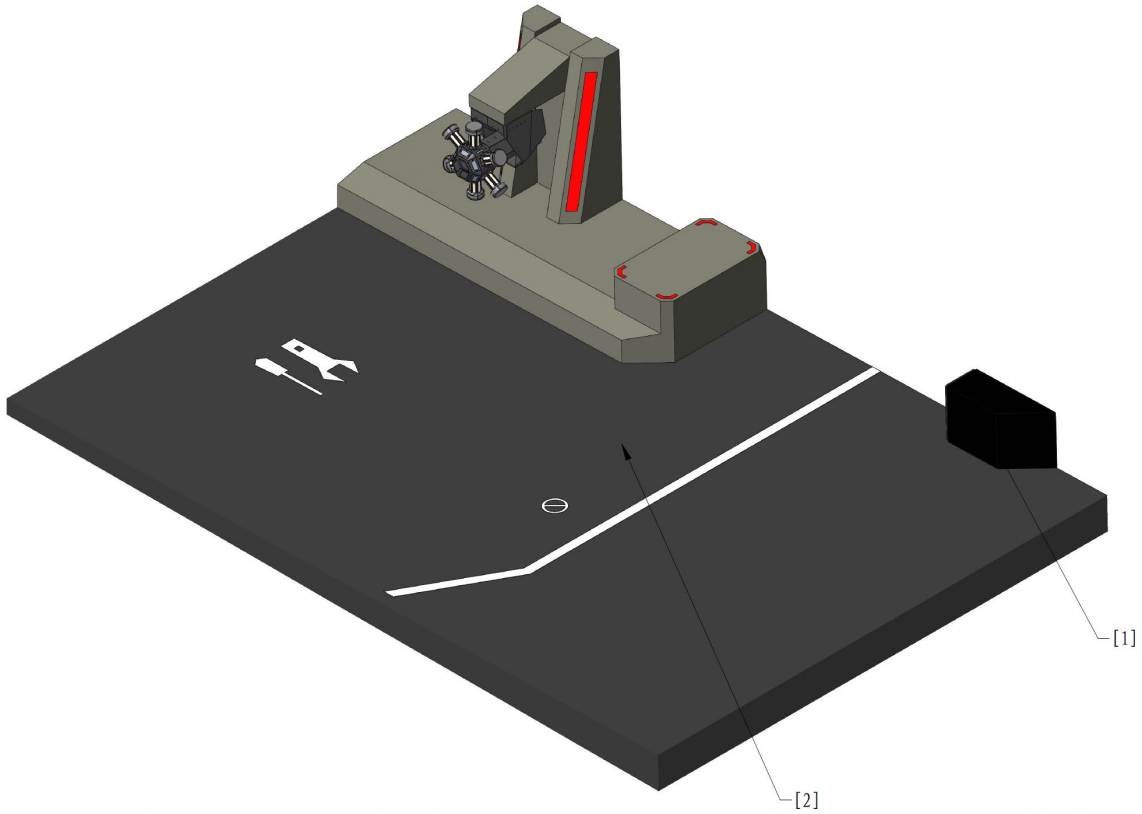


[1] Battlefield entrance	[2] Entrance steps	[3] Radar Base

Figure 4-19 Schematic of the relative position of the Radar Base

4.2.6 Supply Zone

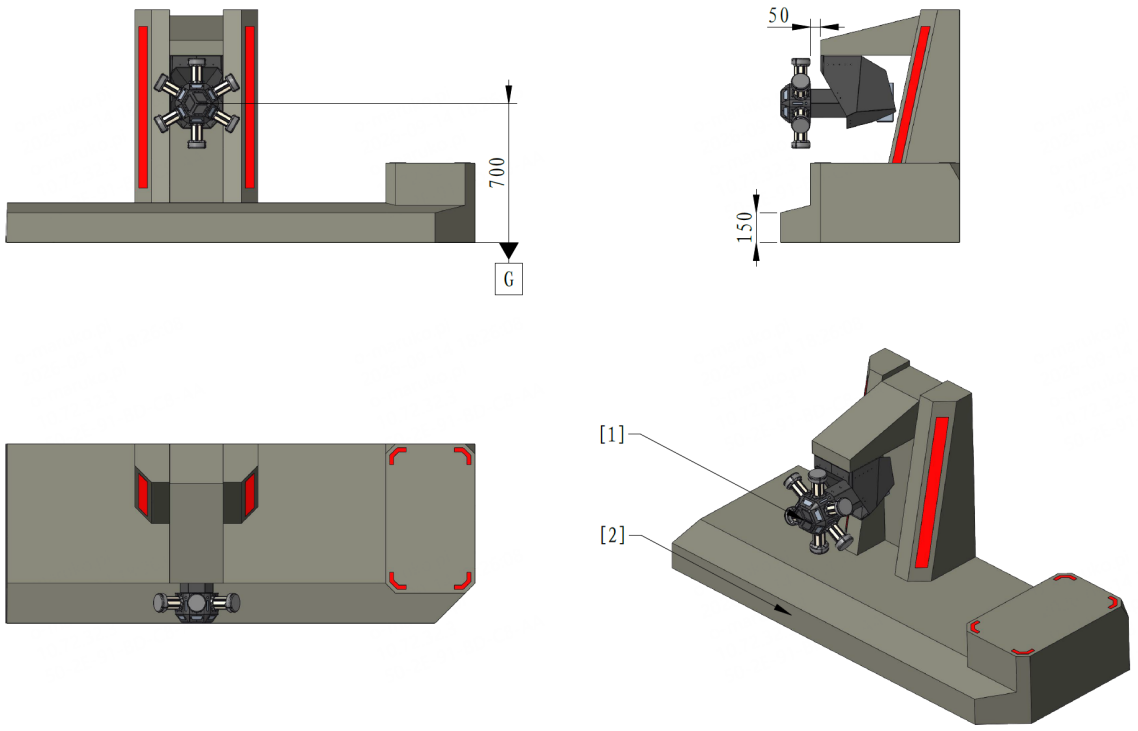
The Supply Zone includes the Resource Station, Supply Zone Buff Point and Workbench. It is an important area where robots recover HP, replenish Projectile Allowance, obtain Energy Units and install the modular 42mm launching mechanism.



[1] Fan [2] Supply Zone

Figure 4-20 Schematic of the Supply Zone

The Resource Station is where the Heavy Robot obtains Energy Units. It contains six Energy Units, each held in its placement zone by magnetic attraction. The Heavy Robot must remove an Energy Unit along the direction of its rotational symmetry axis. In addition to overcoming gravity, a force of up to 10 N must be applied to overcome the magnetic attraction.



[1] Energy Unit placement zone

[2] Baffle

Figure 4-21 Schematic of the resource station

The actual installation angle of an Energy Unit around its axis of symmetry may differ from that shown. The illustration is for reference only; follow the actual placement.

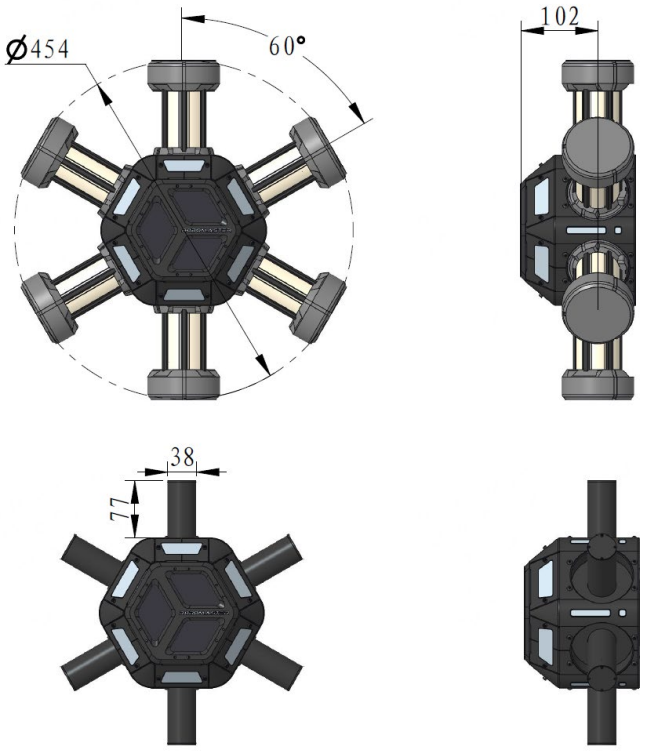


Figure 4-22 Schematic of the Energy Unit placement area

The Workbench is in the Supply Zone. Robots can pick up and install the modular 42mm launching mechanism.

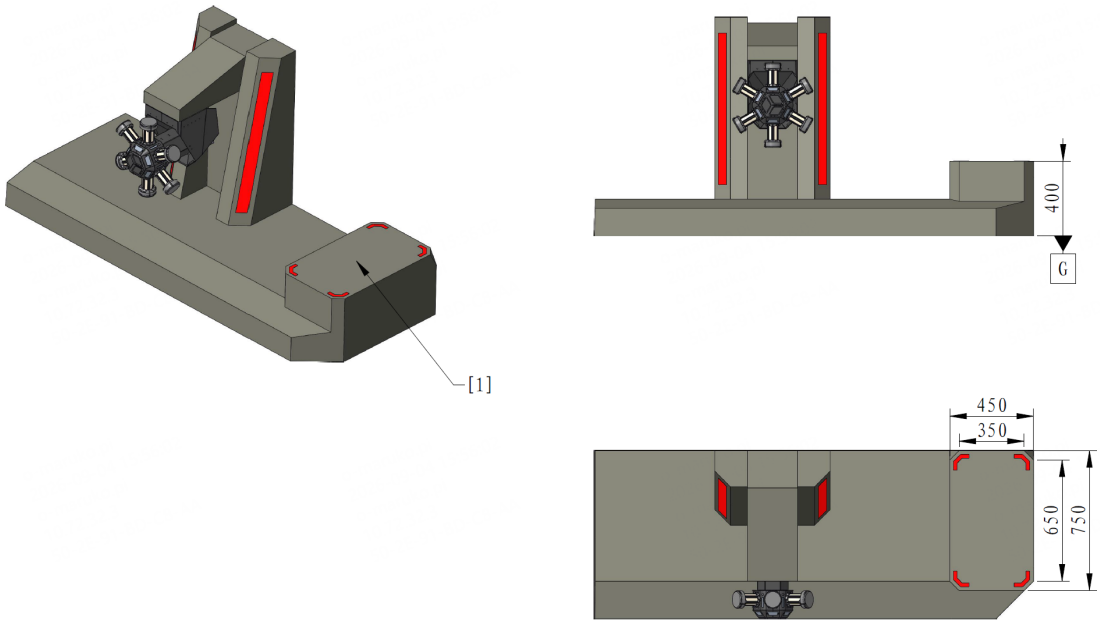
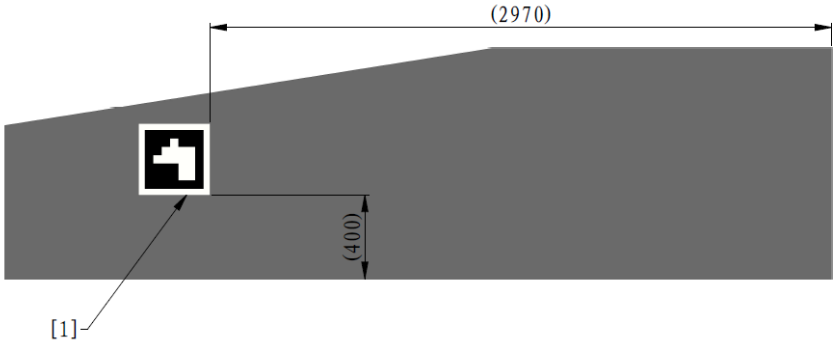
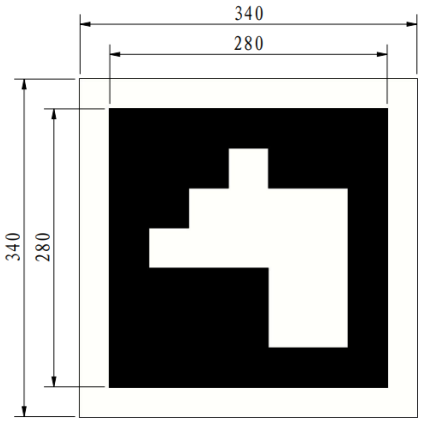


Figure 4-23 Schematic of the Workbench

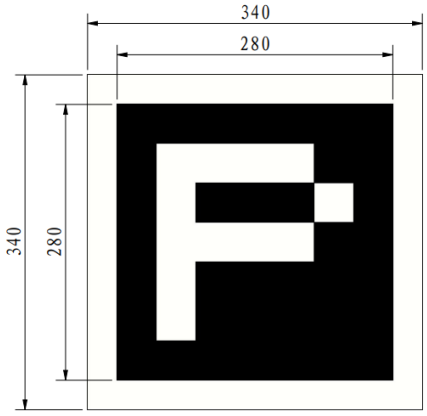
The wall of the Road Zone facing the Supply Zone has a metal visual marker, shown below, for the Camera Video Transmitter Module to determine whether a robot is in the Supply Zone.



[1] Visual marker



Red visual marker

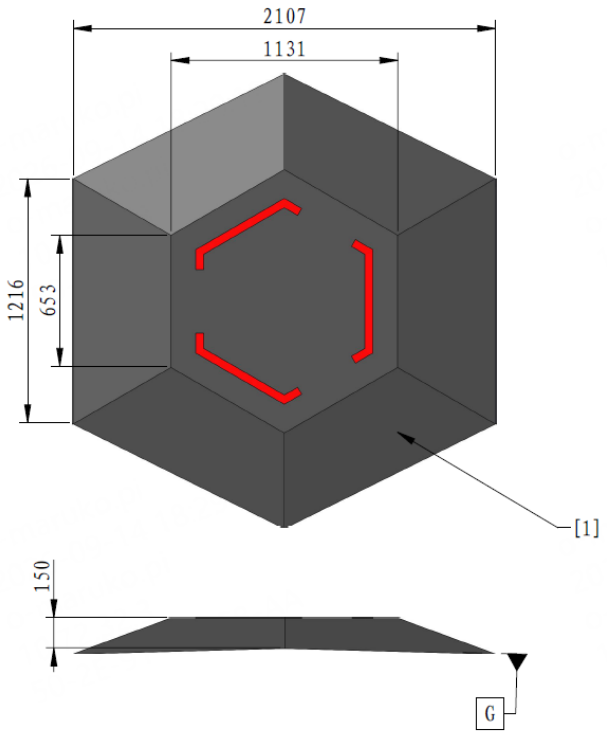


Blue visual marker

Figure 4-24 Schematic of the vision tags

4.2.7 Fortress

The Fortress is a key defensive area in the Base Zone.



[1] 20° slope

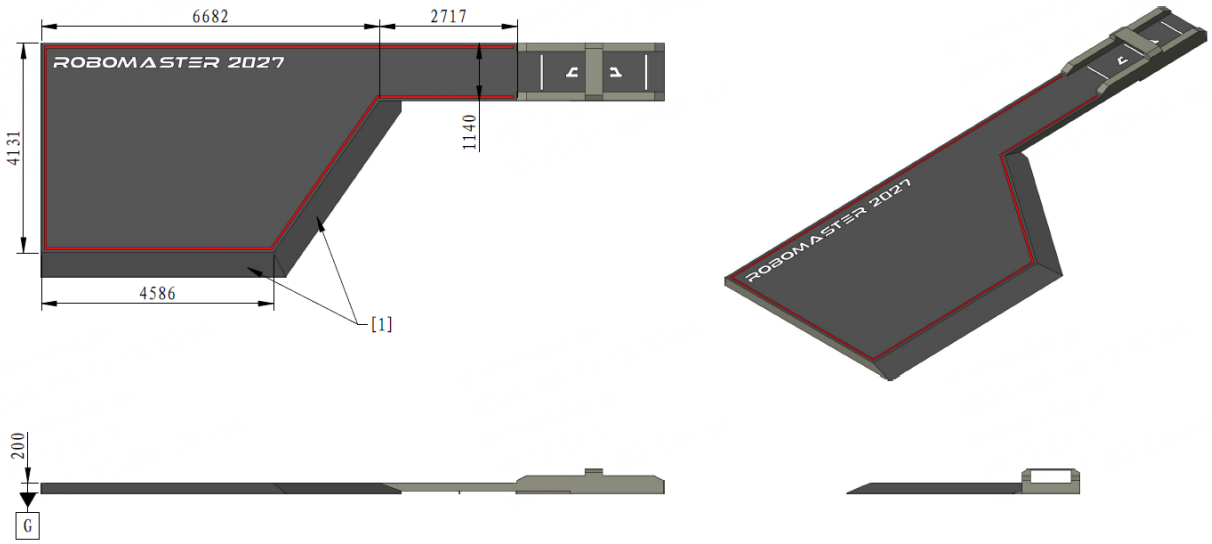
Figure 4-25 Schematic of the Fortress

4.3 Highland Zone

The Highland is an area of the Battlefield above the Battlefield floor. One Trapezoidal Highland lies in each side's half of the field, and the Central Highland lies in the middle. Together they divide the Battlefield into several areas and create a three-dimensional battlefield.

4.3.1 Trapezoidal Highland

The Trapezoidal Highland is near the High Point and is 200 mm above the Battlefield floor.

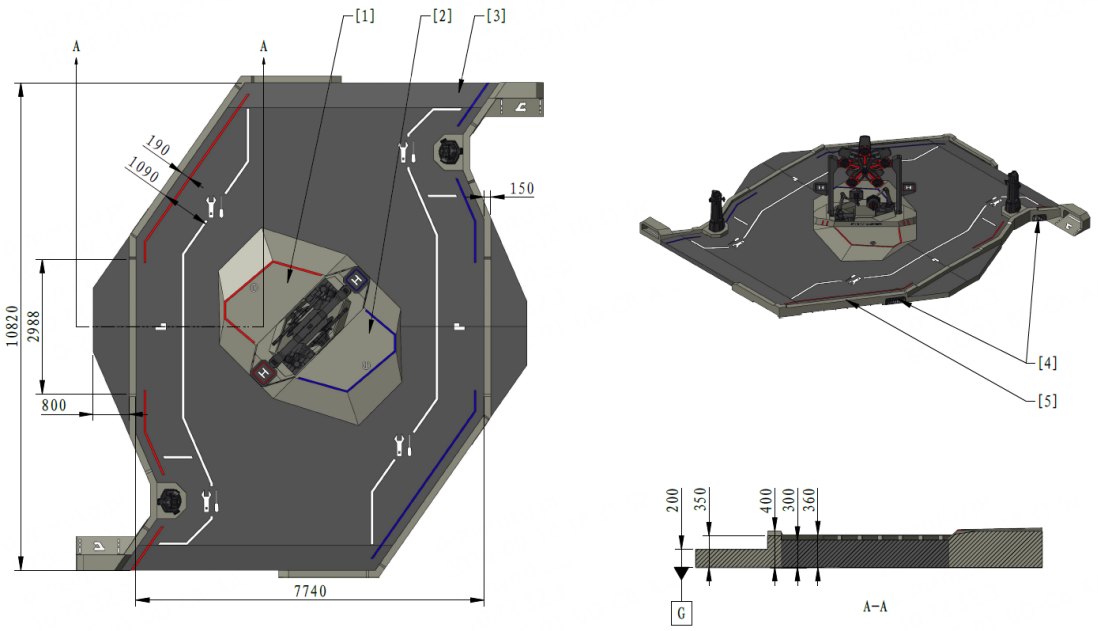


[1] 23° slope

Figure 4-26 Schematic of the Trapezoidal Highland

4.3.2 Central Highland

The Central Highland is in the middle of the Battlefield, with ramps at both ends connecting it to the Road Zone.

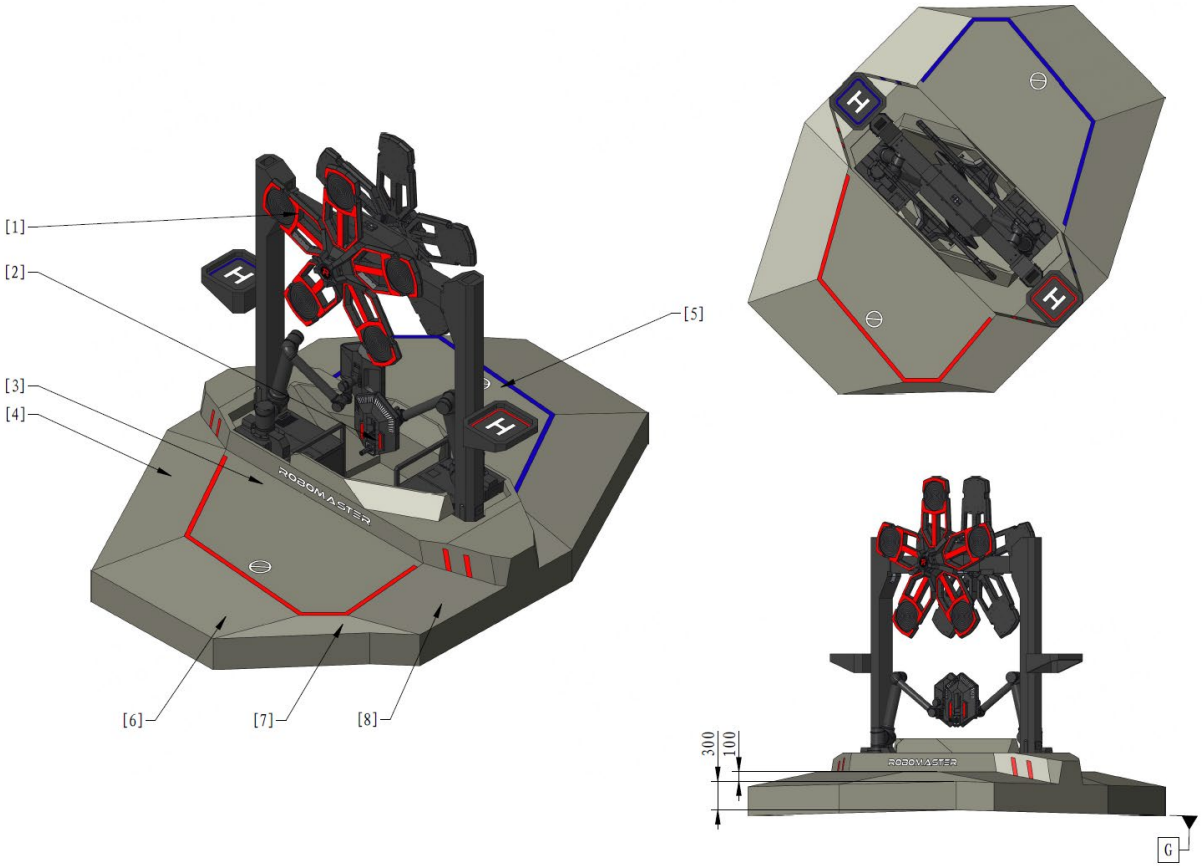


[1] Red Assembly Zone	[2] Blue Assembly Zone	[3] 12° slope
[4] Fan	[5] Metal plate	

Figure 4-27 Schematic of the Central Highland

4.3.2.1 Assembly Zone

The Assembly Zone is in the center of the Battlefield, directly below the Energy Mechanism, and is divided into red and blue sides. A Tech Core is below the Energy Mechanism. The Heavy Robot can assemble Energy Units into the Tech Core, as shown below.



[1] Energy Mechanism	[2] Tech Core	[3] Red Assembly Zone	[4] 14° slope
[5] Blue Assembly Zone	[6] 12° slope	[7] 46° slope	[8] 12° slope

Figure 4-28 Isometric view of the Assembly Zone

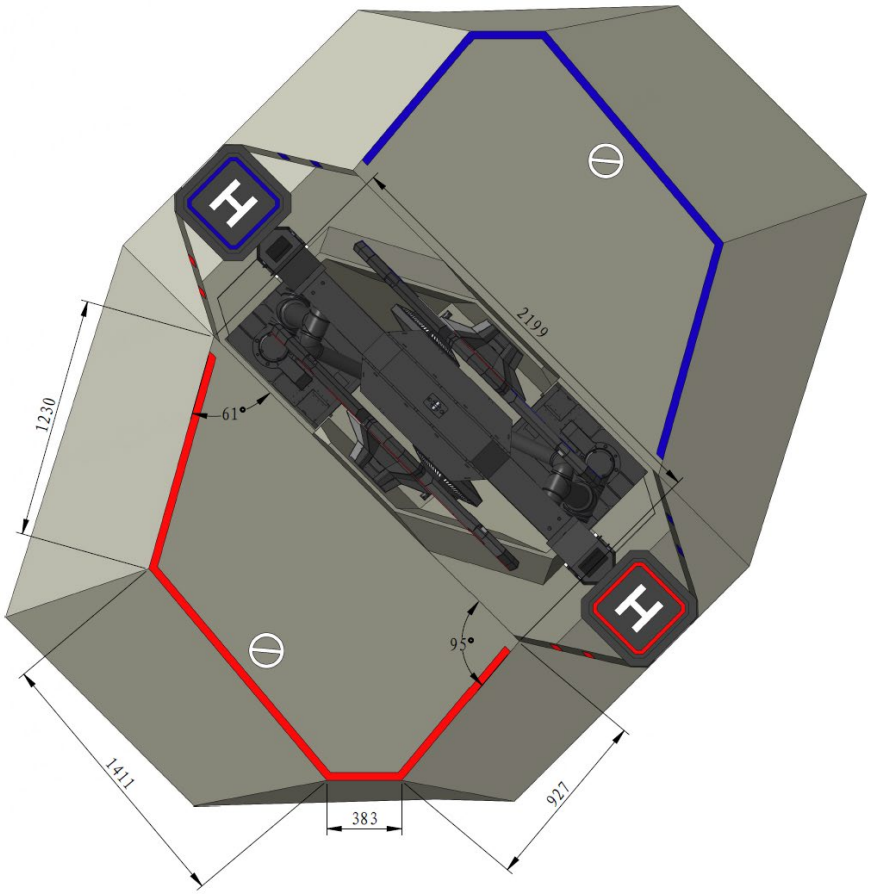
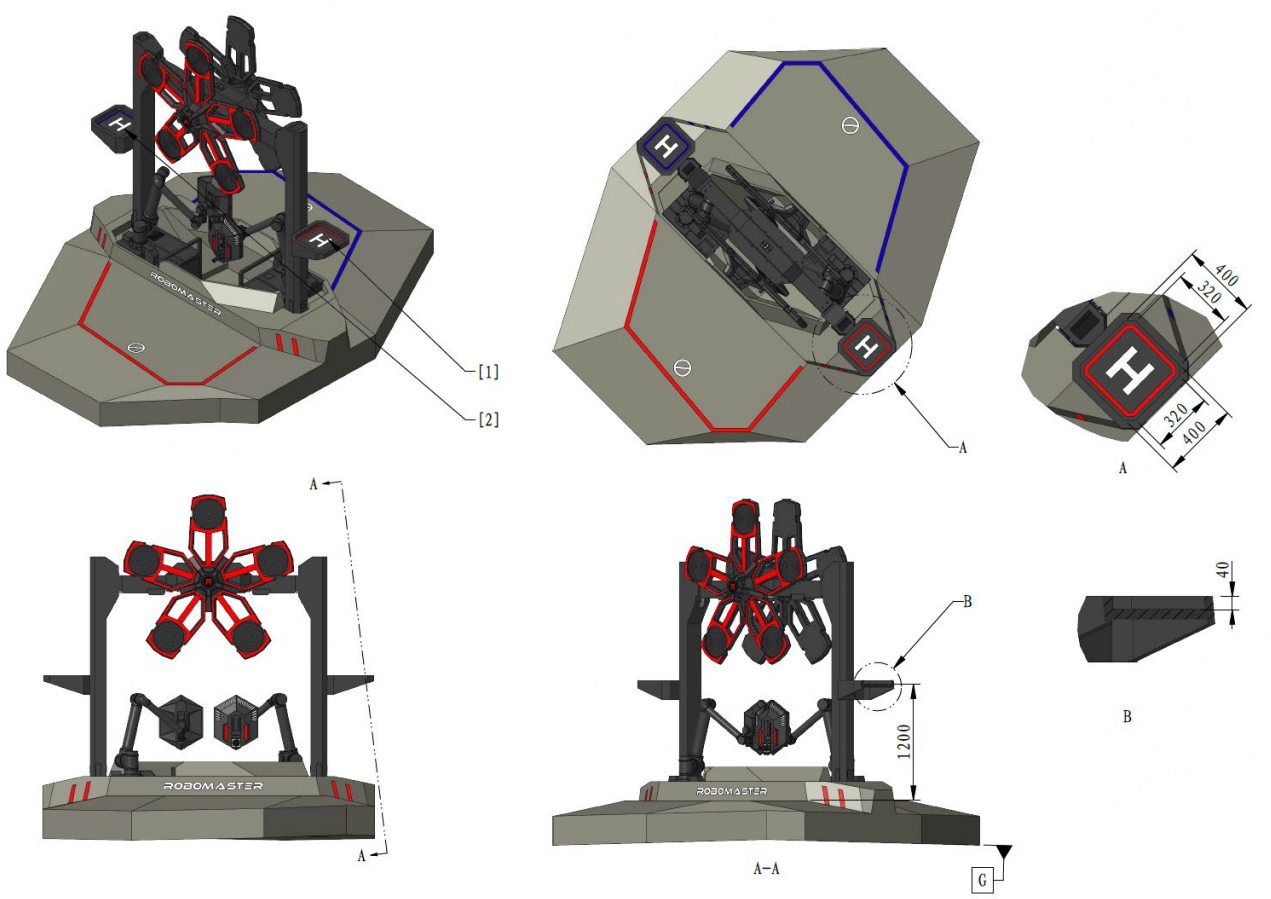


Figure 4-29 Dimensions of the Assembly Zone

4.3.2.2 Hangar Platform

The Hangar platforms are in the center of the Battlefield and are divided into red and blue sides. Each side may install its Aerial System only on its own Hangar platform.



[1] Red Hangar platform [2] Blue Hangar platform

Figure 4-30 Schematic of the Hangar platform

4.3.2.3 Power Rune

The Energy Mechanism is directly above the Assembly Zone. It is motor-driven and rotates synchronously according to a set pattern. The Energy Mechanism has red and blue sides, one for each team.

- Because the center of the Energy Mechanism is heavy, it may sag slightly by 0–50 mm.
- Due to viewing angle and transmission clearance, part of the opposing side's Energy Mechanism may be visible when looking at one side's mechanism.

The Energy Mechanism has five evenly spaced light arms, in the positions and dimensions shown below:

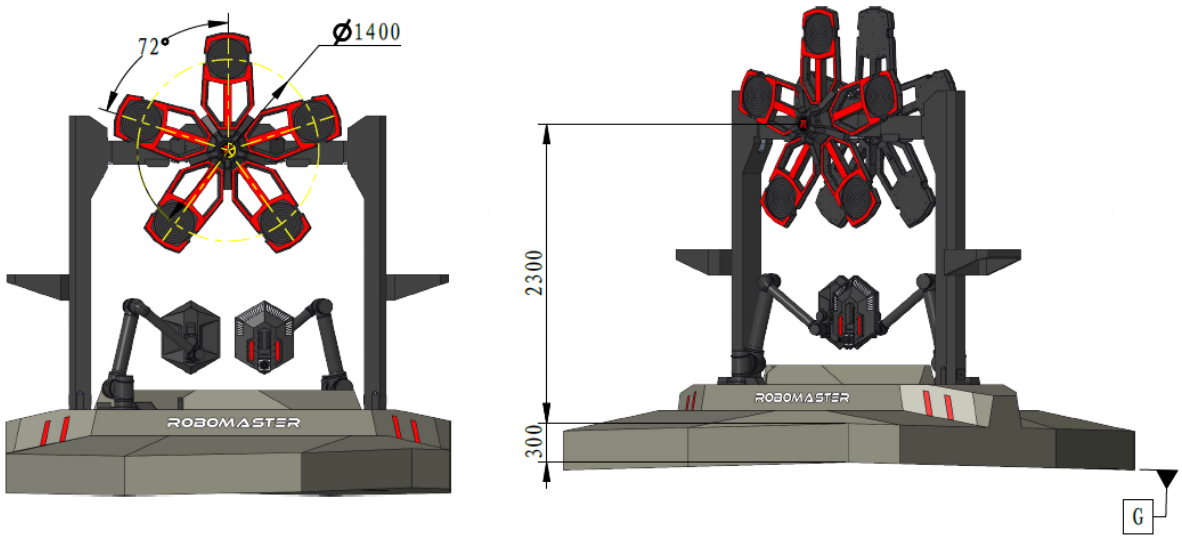


Figure 4-31 Power Rune diagram

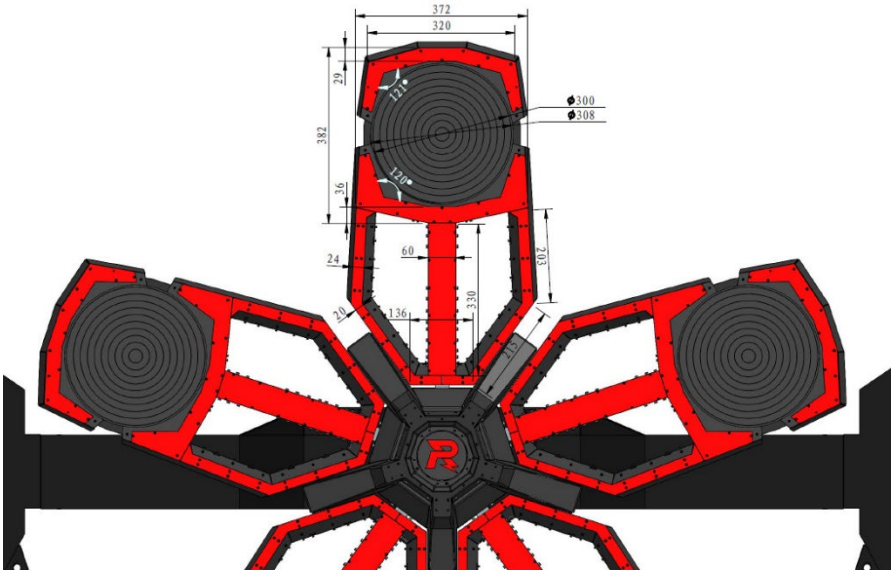


Figure 4-32 Power Rune light arm dimensions diagram

The actual maximum diameter of a light arm's circular target is 308 mm; its actual effective detection diameter is 300 mm.

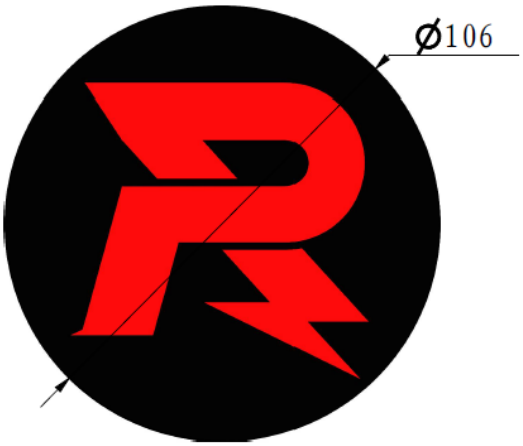
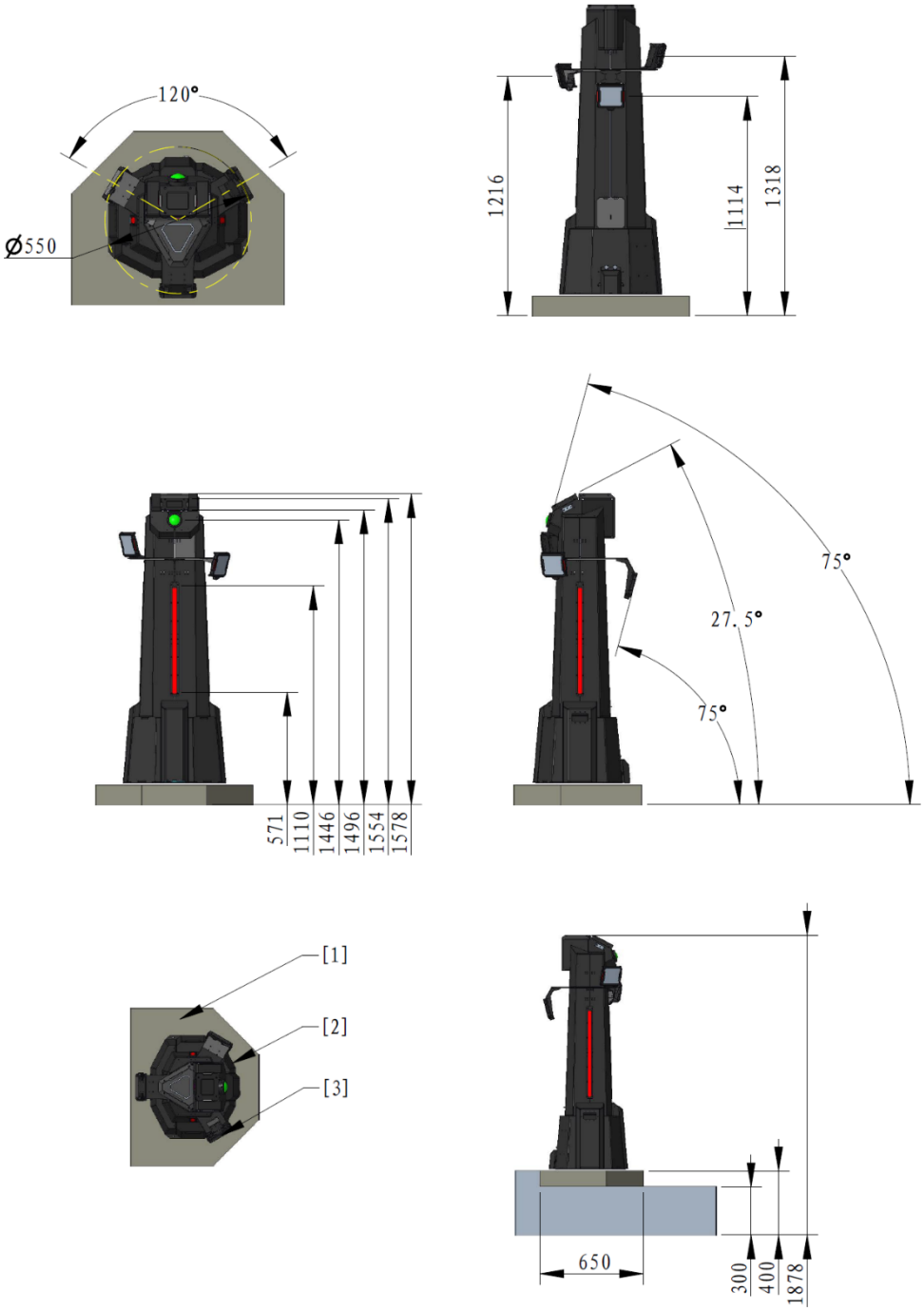


Figure 4-33 Power Rune center marking dimensions diagram

4.3.2.4 Outpost

The Outpost is mounted on the Outpost pedestal and consists of the Outpost body, three Outpost Armor Modules and a Dart Detection Module. For an illustration of the Dart Detection Module, see Figure 4-13. Outpost Buff Points are around the Outpost, as shown below.

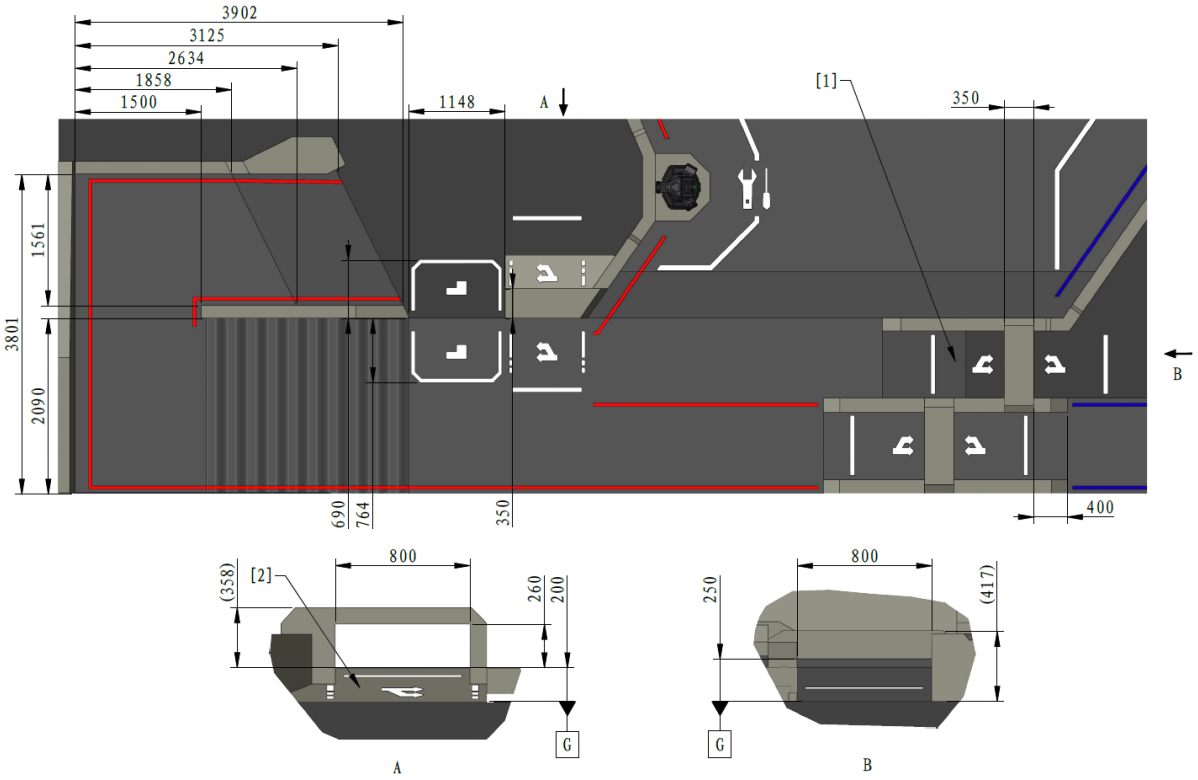


[1] Outpost pedestal [2] Outpost [3] Center Armor Module

Figure 4-34 Outpost diagram

4.4 Road Zone

The Road Zone includes the road, Tunnel and undulating road section.

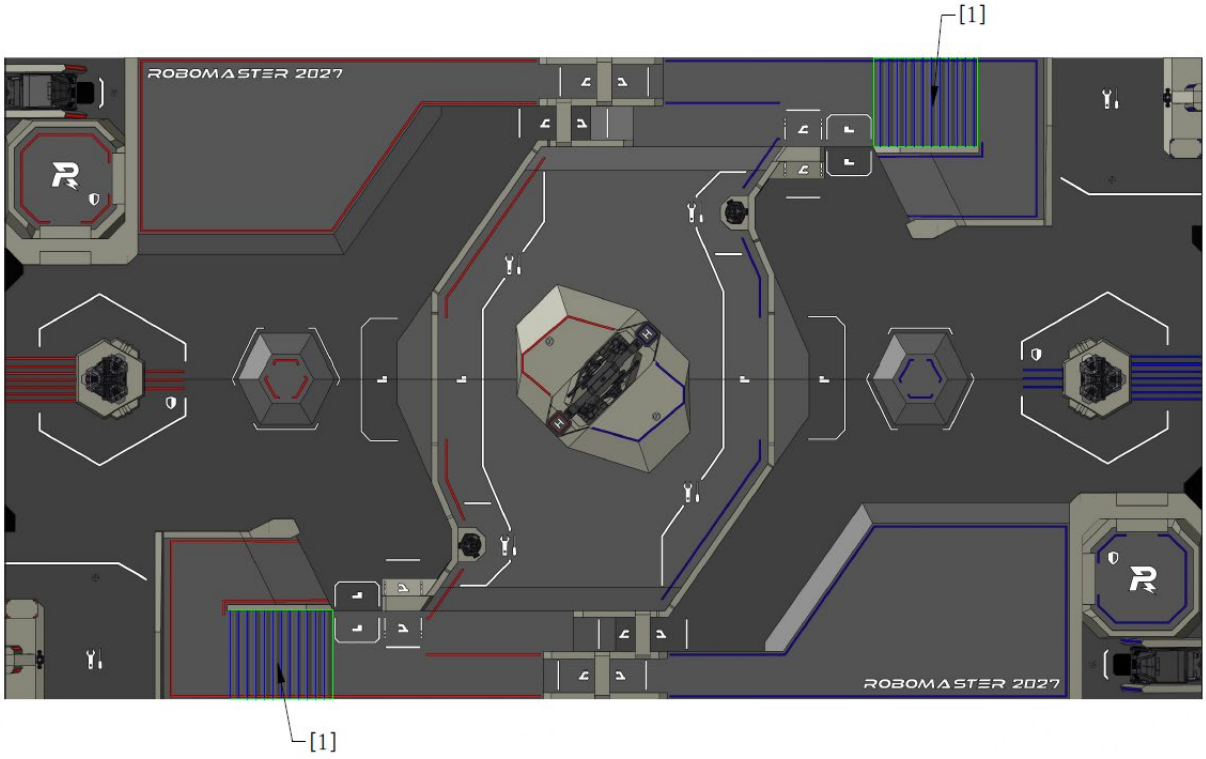


[1] 11° slope [2] 15° slope

Figure 4-35 Road Zone diagram

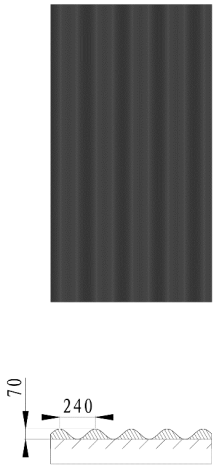
4.4.1 Bumpy Road

The undulating road section is in the Road Zone. Raised features are spaced across its surface in the direction shown below.



[1] Undulating road section

Figure 4-36 Bumpy Road diagram

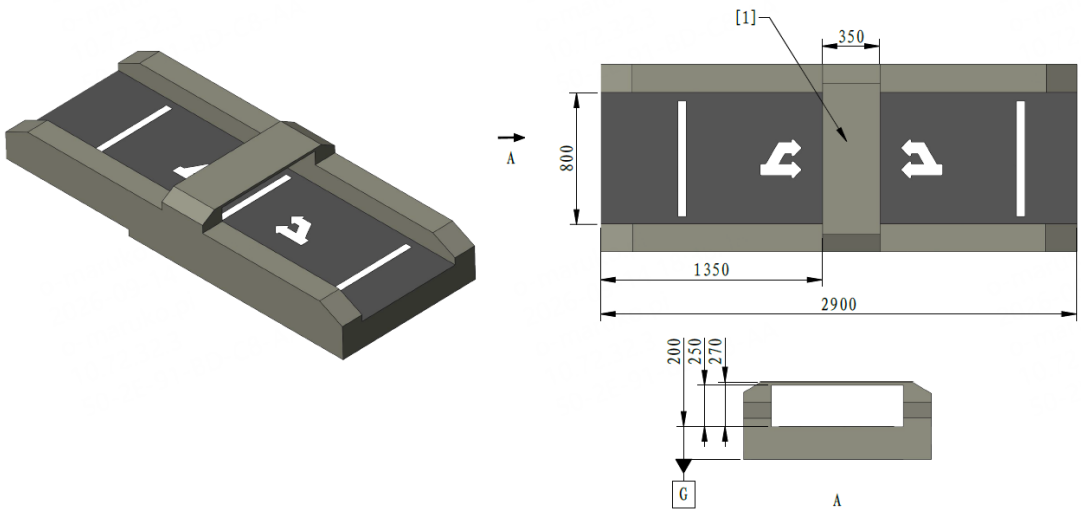


[1] Raised feature

Figure 4-37 Bump diagram

4.4.2 Tunnel

The Tunnel is in the Road Zone. Robots can pass through or leap over it to reach the opposing half of the field quickly.



[1] Metal plate

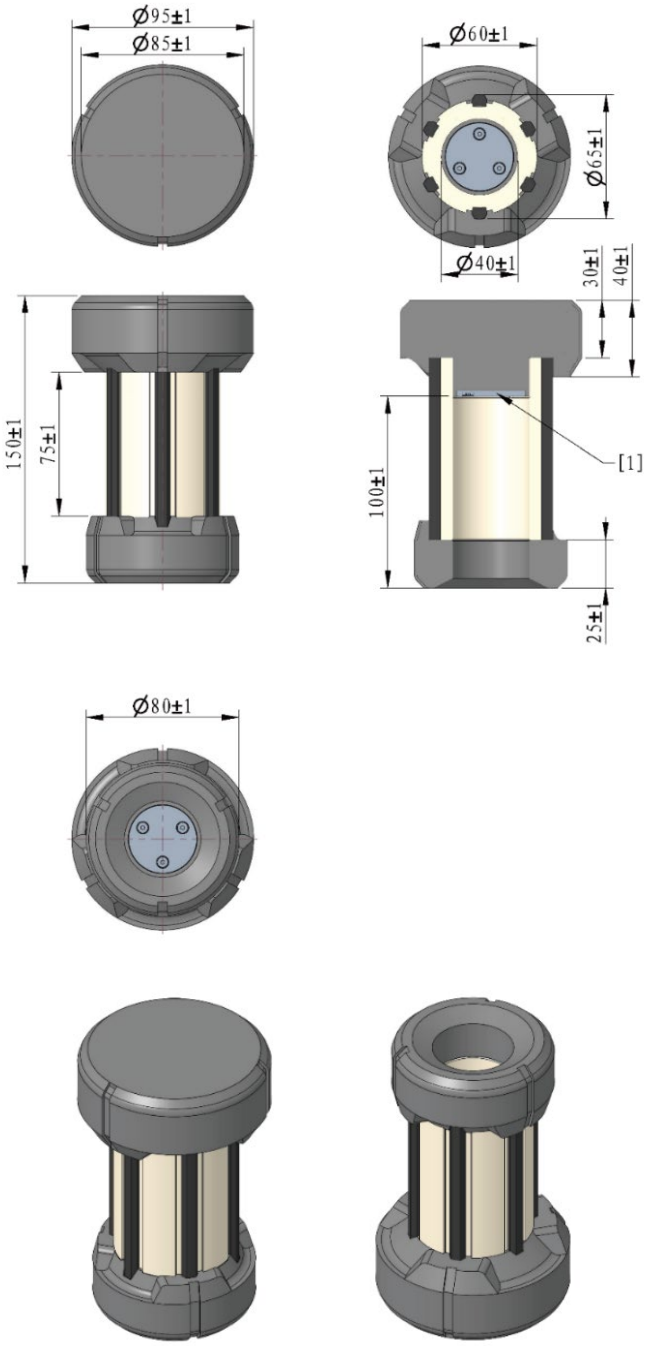
Figure 4-38 Far Highland Tunnel diagram

4.5 Other

4.5.1 Movable Props

An Energy Unit is a movable prop that a robot can pick up and carry. It is a hollow cylinder weighing about 400 ± 50 g, with asymmetrical ends.

As a match progresses, movable props may become slightly damaged or accumulate dust. Teams must adapt to this, but heavily soiled movable props will not be used in official matches.



[1] Metal plate

Figure 4-39 Energy Unit diagram

4.5.2 Projectiles

Robots can attack Armor Modules by firing projectiles and thereby deal damage. The projectile parameters and allocation for the competition are shown below:

Table 4-1 Projectile parameters and usage arrangements

Type	Appearance	Color	Size	Mass	Shore hardness	Material	Usage arrangement	Rated quantity
42mm luminous projectile	Similar in shape to a golf ball	Translucent	42.5mm ±0.5mm	44.5g ±0.5g	90±5A	Plastic TPE	Whole season	100 per round
17mm fluorescent projectile	Spherical	Yellow-green	16.8mm ±0.2mm	3.2g± 0.1g	90±5A	Plastic TPU	Whole season	Limited, but not less than 3000 per round

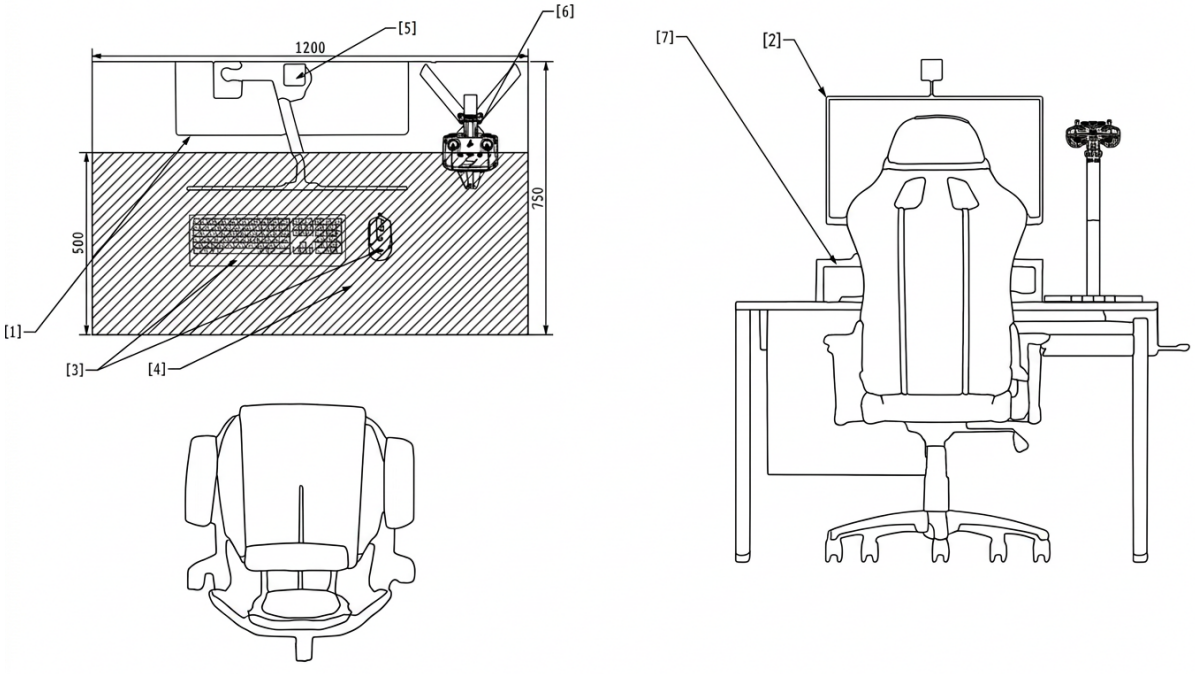
4.5.3 Operator Room

The Operator Room is near the perimeter of the Battlefield and is the activity area for Operators during a match. Each Operator desk is 120 cm long and 75 cm wide, and is equipped with the corresponding number of computers. Each computer has official equipment including a corresponding monitor, mouse, keyboard, Camera Video Transmitter Module (receiver) and its bracket, one 1GbE (RJ45) network port, and a wired headset.

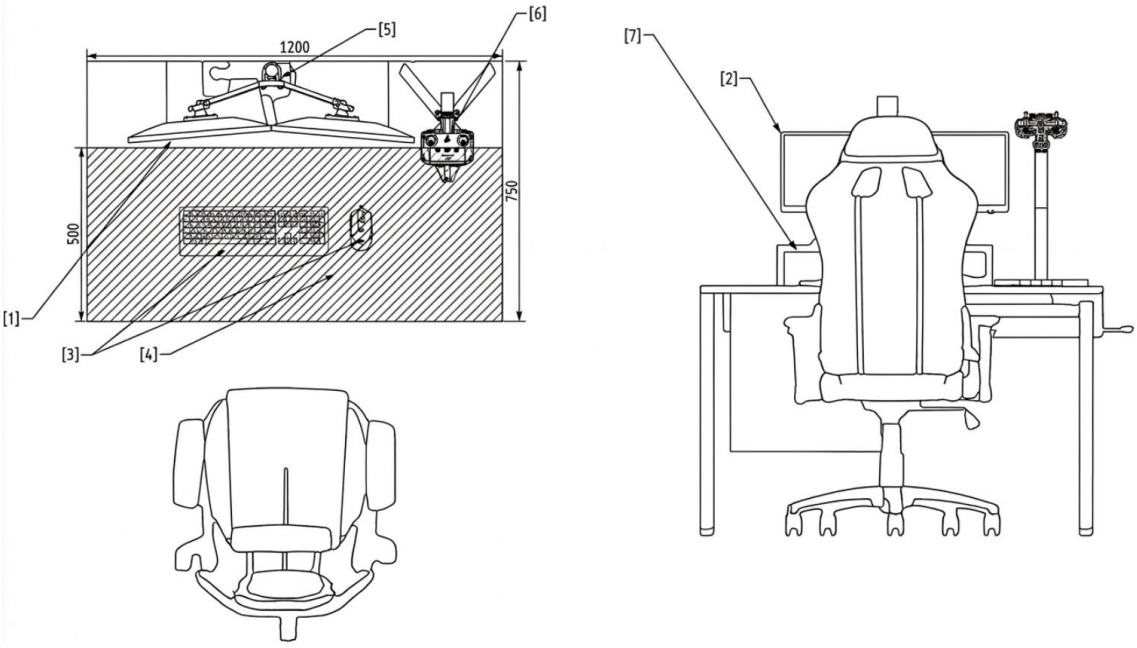
Operators of the Heavy Robot and the Commander are additionally provided with one official display that supports HDMI input only, with a resolution of 1920 × 1080, and one HDMI male-to-male signal cable for connecting a Custom Terminal to the official display. Teams may use it to display the Custom Terminal screen.

The Operator Room is physically divided to separate Ground Robot Operators from the Commander.

The desktop layout of the Operator Room is shown below. The desk has a monitor, broadcast camera, Camera Video Transmitter Module (receiver) and bracket, keyboard, mouse and equipment placement area. The monitor, broadcast camera and ports are fixed and must not be moved. The bracket for the Camera Video Transmitter Module (receiver) is fixed to the desk; Operators may operate the module within the bracket's range of movement. The keyboard and mouse may be moved freely, and Operators may place a Custom Terminal in the equipment area. The specific layout is shown below:



Infantry Robot Operator layout diagram



Heavy Robot Operator and Commander layout diagram

[1] Port location [2] Monitor [3] Keyboard and mouse [4] Placement area

[5] Broadcast camera [6] Camera Video Transmitter Module (receiver) and bracket [7] Storage rack

Figure 4-40 Operator Room layout diagram

- Do not remove the Camera Video Transmitter Module (receiver) from its bracket. The participating team bears all consequences of doing so.

- Robots entering the Battlefield automatically connect to the Camera Video Transmitter Module (receiver) in the Operator Room; Camera Video Transmitter Modules (receivers) brought by teams cannot be used. The default joystick position of the Operator Room's Camera Video Transmitter Module (receiver) is centered, and its default switch position is C.

5 Match Mechanics

5.1 HP Deduction and Limit-Exceeding Penalty Mechanisms

5.1.1 Attack Damage or Collision

When the Armor Modules of a Ground Robot, Base or Outpost are attacked or struck, HP may be deducted. The specific mechanisms are as follows:

The Dart Detection Module detects dart attacks through the combination of an Armor Module and a phototube.

Armor Modules detect projectile attacks and Aerial Robot impacts. The minimum detection interval is 50 ms for 17mm projectiles and 200 ms for 42mm projectiles.

A projectile or Aerial Robot must meet certain requirements when contacting the attack surface of an Armor Module to be detected. The detection requirements for different types of attacks are shown below:

Table 5-1 Effective detection speed of Armor Modules for different types of projectiles

Armor Module	17mm projectile	42mm projectile	Aerial Robot impact
Small Armor Module Base Armor Module Outpost Armor Module	Greater than 12m/s	Greater than 8m/s	Contact duration ∈ [TBD] Cumulative impulse on the contact surface ∈ [TBD] Explanation: The Armor Module detects an impact as follows: when its detection surface detects a contact force greater than α N at any position, an “impact event” begins. The event ends when the contact force at all positions falls below α N. The contact duration is the duration of one “impact event.” Impact area S is the sum of the areas where instantaneous contact force exceeds α N over the entire “impact event” (the areas need not be contiguous). Cumulative impulse on the contact surface is ∬F dt dS, the double integral of contact force over contact duration and impact area during the “impact event.”
Energy Mechanism Armor Module	Greater than 12m/s	-	-

- In an actual match, projectile speed decreases and the projectile may not strike perpendicular to the attack surface. Thus, the projectile's velocity component normal to the Armor Module's attack surface. Damage detection is based on the normal component of the projectile's velocity when it contacts the Armor Module's attack surface.

- Teams must not hit the Energy Mechanism with 42mm projectiles, darts or Aerial Robots.
- If one side's Heavy Robot has not fired a 42mm projectile for 4 consecutive seconds, all Armor Modules on the opposing robots, Outpost and Base block 42mm projectile damage until the Heavy Robot fires a 42mm projectile.
- To ensure detection accuracy and reduce false detections, the Referee System may need to wait for 42mm projectile launch information to match in some cases. Therefore, damage settlement may be delayed briefly after an Armor Module detects a 42mm projectile hit.

When an Armor Module on a Ground Robot, Base or Outpost is struck, the hit may be judged to be projectile damage. Except for Aerial Robot impacts, teams must not damage opponents by striking them, including by colliding with robots or throwing objects.

Without any Buffs, the base damage values are as follows:

Table 5-2 HP deduction mechanism for attack damage or collision

Damage type / Target	42mm projectile	17mm projectile	Aerial Robot Impact	Dart	Collision
Robot Armor Module	200	20	-	-	2
Base Armor Module	200	20	200	-	-
Base Dart Detection Module	-	-	-	Fixed target: 200 Random fixed target: 300 Random moving target: 800 End moving target: 1,250	-
Outpost center Armor Module	200	20	200	-	-
Outpost Dart Detection Module	-	-	-	1000	-

5.1.2 Initial Launch Speed Exceeding the Limit

If a Ground Robot exceeds the muzzle velocity limit, its launching mechanism is locked and unlocks after a specified time. The time does not accumulate.

Locking and unlocking caused by exceeding the muzzle velocity limit are independent of locking and unlocking caused by other circumstances. Let the robot's muzzle velocity limit be V0 (m/s), and let the actual muzzle velocity detected by the Referee System be V1 (m/s). The lock duration is shown below:

Table 5-3 Penalty mechanism for Initial Launch Speed exceeding the limit

17mm projectile	42mm projectile	Lock time
$0 < V_1 - V_0 < 5$	$V_0 < V_1 \leqslant 18$	15 s
$5 \leqslant V_1 - V_0 < 10$	-	20 s
$10 \leqslant V_1 - V_0$	$18 < V_1$	Remainder of the round, no unlocking

5.1.3 Barrel Heat Exceeding the Limit and Cooling

If a Ground Robot exceeds its Shooter Heat limit, its launching mechanism is locked. Locking and unlocking caused by exceeding the Shooter Heat limit are independent of locking and unlocking caused by other circumstances. The specific mechanism is as follows:

Let the robot's Shooter Heat limit be Q0 and its current Shooter Heat be Q1. Each 17mm projectile detected by the Referee System increases Q1 by 10, regardless of its muzzle velocity. Each 42mm projectile detected increases Q1 by 100, regardless of its muzzle velocity. Shooter Heat cools at a settlement frequency of 10 Hz; the cooling value per detection cycle is the per-second cooling value divided by 10.

When Q2 > Q1 > Q0, the first-person video feed on the corresponding Operator interface is disconnected and the launching mechanism is locked until Q1 = 0. The first-person feed then returns to normal and the launching mechanism unlocks.

When Q1 > Q2, the robot's launching mechanism remains locked for the rest of the current round and does not unlock.

For a 17mm launching mechanism, Q2 = Q0 + 100.

For a 42mm launching mechanism, Q2 = Q0 + 200.

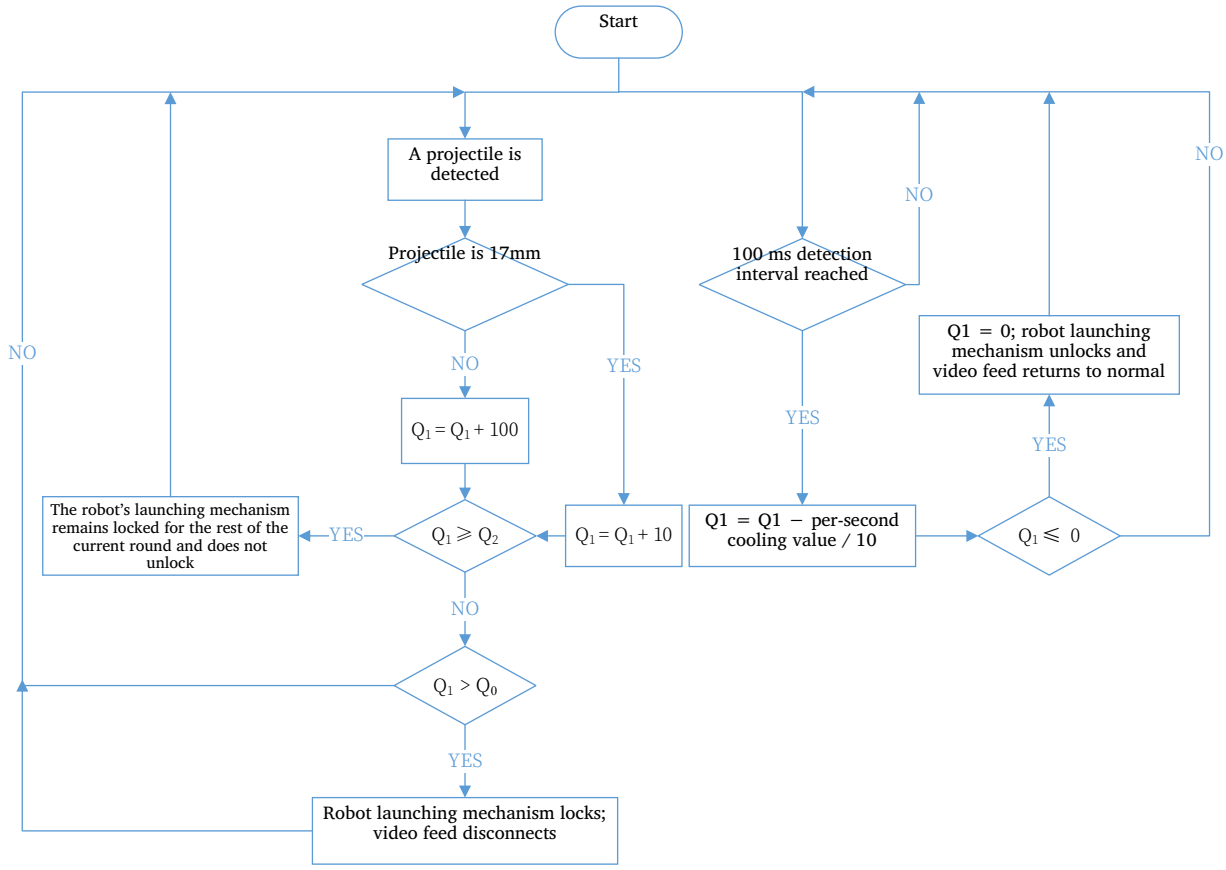


Figure 5-1 Barrel Heat exceeding the limit and cooling mechanism

When the video feed is disconnected because of excessive Shooter Heat, the minimap and Custom UI remain available.

5.1.4 Chassis Power Exceeding the Limit

The Referee System continuously monitors robot chassis power, which must remain within the power limit. If a Ground Robot exceeds the chassis power limit, its buffer energy is deducted. If buffer energy is exhausted and chassis power still exceeds the limit, the chassis is powered off for 5 seconds. The Referee System settles chassis power detection at 10 Hz.

Let Z be buffer energy, Q the buffer energy limit (Q = 60 J), Pr the instantaneous chassis output power, and Pl the power limit.

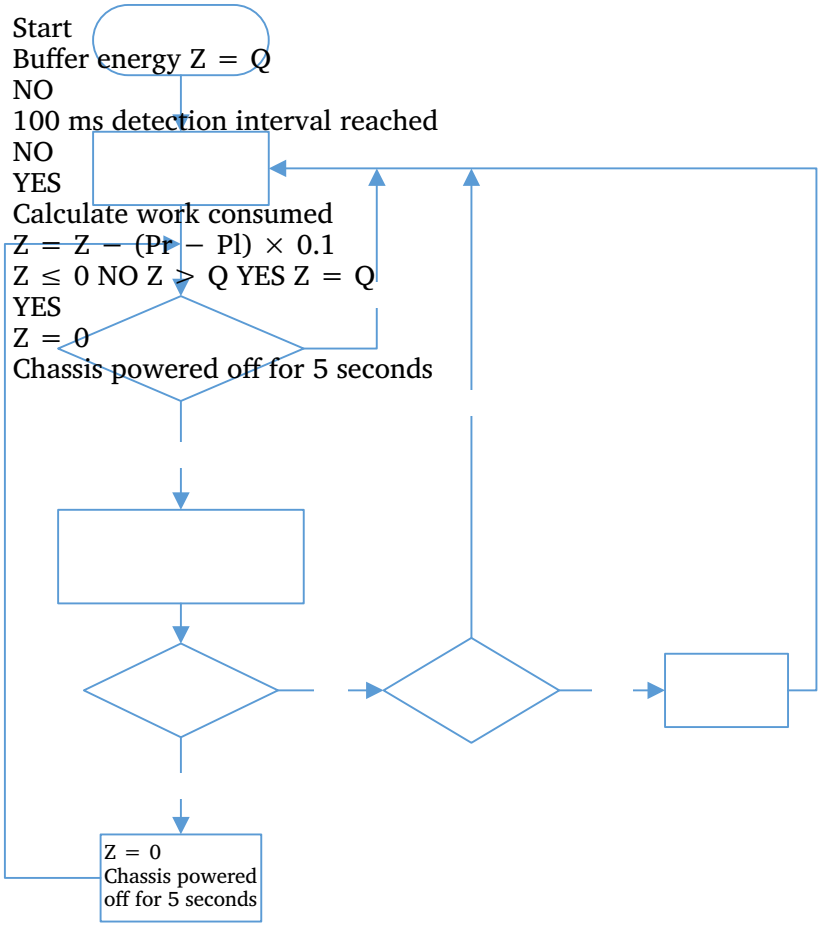


Figure 5-2 Robot Chassis Power detection and limit-exceeding penalty mechanism

5.1.5 Referee System Module Offline

Teams must install the corresponding Referee System modules on their robots in accordance with the Robot Building Specifications Manual for RoboMaster 2027 University Series, and must maintain stable connections between all Referee System modules and the server during a match.

The Referee System server checks the connection status of each module at 2 Hz.

5.1.5.1 Abnormal Offline

If a robot's Main Controller Module disconnects during a match, the robot enters the "Abnormally Disconnected" state and may reconnect. Experience and Levels continue to be settled while it is disconnected.

Table 5-4 Penalty mechanism for robot abnormal offline

Robot type	Handling while disconnected
Ground Robot	Launching mechanism (if any), gimbal and chassis are powered off; 5% of maximum HP is deducted per second until HP reaches 0. Field Interaction Module is disabled. The robot no longer detects damage from impacts or projectile hits, or HP deductions due to module disconnection. The revival countdown no longer increases. Multi-robot communication is disconnected.
Hangar	Actuators are powered off. Multi-robot communication is disconnected.
Radar System	Multi-robot communication is disconnected.
Dart System	If it disconnects while the Dart Launching Station gate is closed, the gate cannot open. If it disconnects while the gate is open, the gate closes immediately and cannot reopen.

5.1.5.2 Armor Module or Supercapacitor Management Module Offline

If a Ground Robot's Armor Module or Supercapacitor Management Module disconnects due to the robot's design or structure, the robot loses HP.

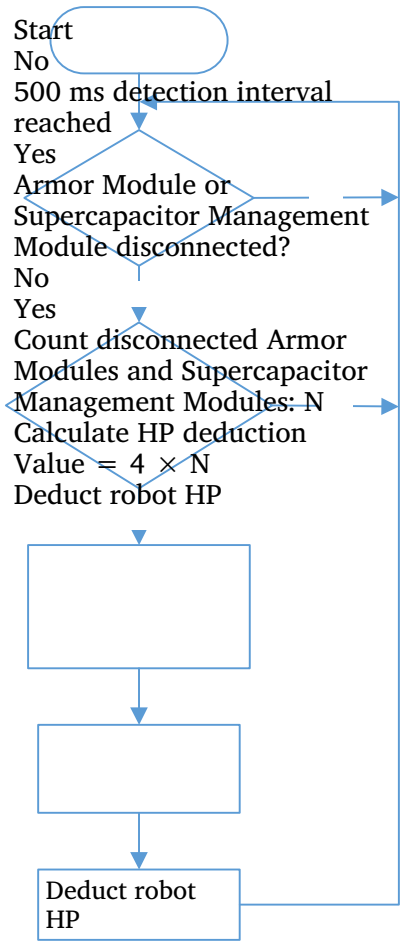


Figure 5-3 HP deduction mechanism for Armor Module and Supercapacitor Management Module offline

5.1.5.3 Speed Monitor Module Offline

If a 17mm or 42mm Speed Monitor Module installed on a Ground Robot disconnects, the corresponding 17mm or 42mm launching mechanism locks immediately. Locking and unlocking caused by a Speed Monitor Module disconnection are independent of locking and unlocking caused by other circumstances.

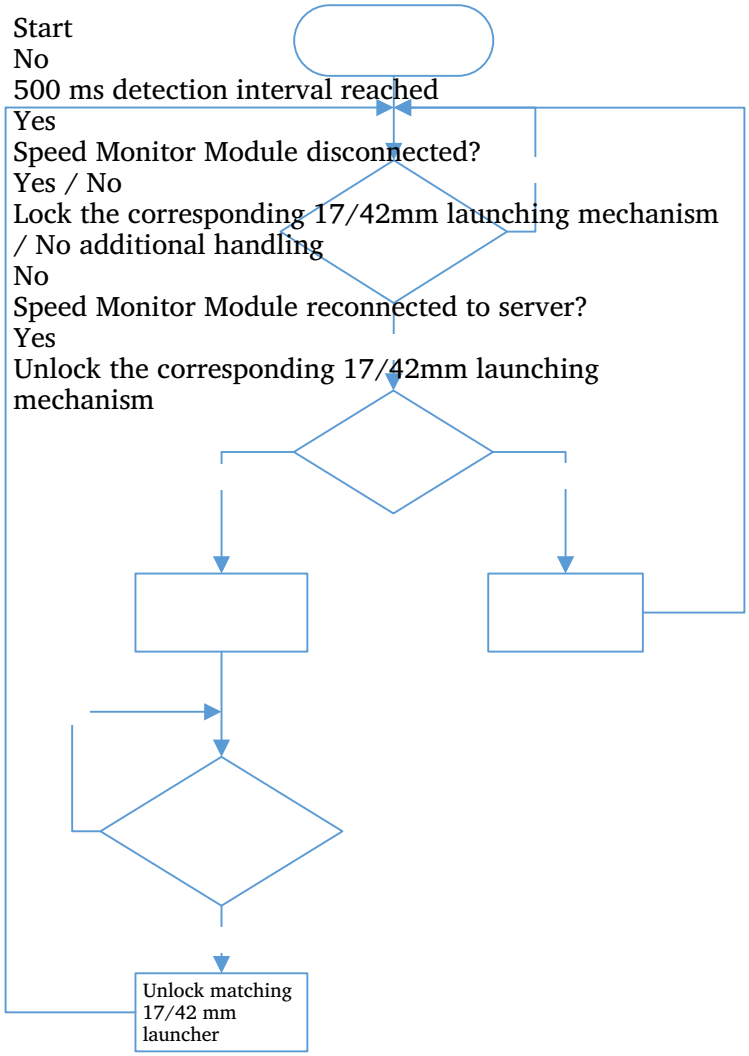


Figure 5-4 Power-off mechanism for Speed Monitor Module offline

5.1.5.4 Positioning Module Offline

If the Positioning Module installed on a Ground Robot disconnects, the opposing Radar always treats its mark of that robot as “accurate,” while the friendly Radar always treats its mark as “inaccurate.”

5.1.6 Penalty Damage

When a Ground Robot receives a penalty such as a yellow- or red-card warning, it may lose HP. See “7.1 Penalty System”.

5.2 Adaptive Armor Mechanism

During a match, a team may choose to reinforce a specific area on all of its Armor Modules. Projectile damage to a reinforced area is reduced, while damage to an unreinforced area is increased. The area that can be reinforced depends on both sides' projectile hit accuracy: the higher a side's relative accuracy, the larger its reinforced area. The specific mechanism is as follows:

At match start, all armor plates are in the "Default" state. After the match starts, each side gains one cumulative reinforcement opportunity per minute. Once a side has hit at least 10 projectiles in the current round, its Radar may send a command to the Referee System server at any time to spend one reinforcement opportunity and select or change one center point. Until the next center-point change, a circular area of size A centered on the selected point on all of that side's robot, Outpost and Base Armor Modules is considered "Reinforced"; the remaining area is "Unreinforced."

Projectile damage multipliers for unreinforced, reinforced and default areas are shown below:

Table 5-5 Damage multipliers for each region of the Armor Module

Region type	Damage multiplier
Hit on unreinforced region	*1.5
Hit on reinforced region	*0.5
Hit on default-state region	*1

This multiplier applies to the projectile's base damage, so it is multiplied with attack and defense Buffs.

The hit position (x, y, and hit-area category) of each projectile is sent to the side that was hit through the Referee System's standard link and to the attacking side through the Radar wireless link.

The area that can be reinforced is calculated as follows:

- Hit rate η: $\eta = \frac{\text{Number of projectiles that hit the opposing side}}{\text{total number of projectiles fired by the team}}$
- Spread-circle radius σ: The same Armor Module coordinate system is used for all armor plates. Each projectile hit produces a hit coordinate (x, y).
  Let x0 and y0 be the respective medians of the horizontal and vertical coordinates of all hit projectiles. Then σ is the median distance from each hit point to (x0, y0).

  Armor Module coordinate system: A Cartesian plane with the lower-left corner of the Armor Module detection surface as the origin O, the horizontal direction to the right as the positive x-axis, and the vertical direction upward as the positive y-axis.

- Relative accuracy ρ: $\rho = \frac{\Lambda_{Own}}{\Lambda_{Enemy}}$, where Λ is accuracy and $\Lambda = \frac{\eta}{\sigma}$
- Reinforceable area A: A = armor plate area $\times \frac{\rho^2}{2+\rho^2}$

5.3 HP Recovery and Revival Mechanism

Except when Abnormally Disconnected or penalized off the field, Ground Robots can recover HP and revive.

Except as described below, the specific HP recovery and revival parameters are also affected by Base HP. See “5.6.1 Base and Outpost Mechanics”.

5.3.1 HP Recovery Mechanism

HP recovery in the Supply Zone

A Ground Robot may activate its side’s Supply Zone Buff Point to gain HP recovery at 10% of maximum HP per second. After 4 minutes of the match have elapsed, a robot that is out of combat gains HP recovery at 25% of maximum HP per second when it activates its side’s Supply Zone Buff Point.

If the robot is not out of combat, the 25%-per-second HP recovery effect ends immediately.

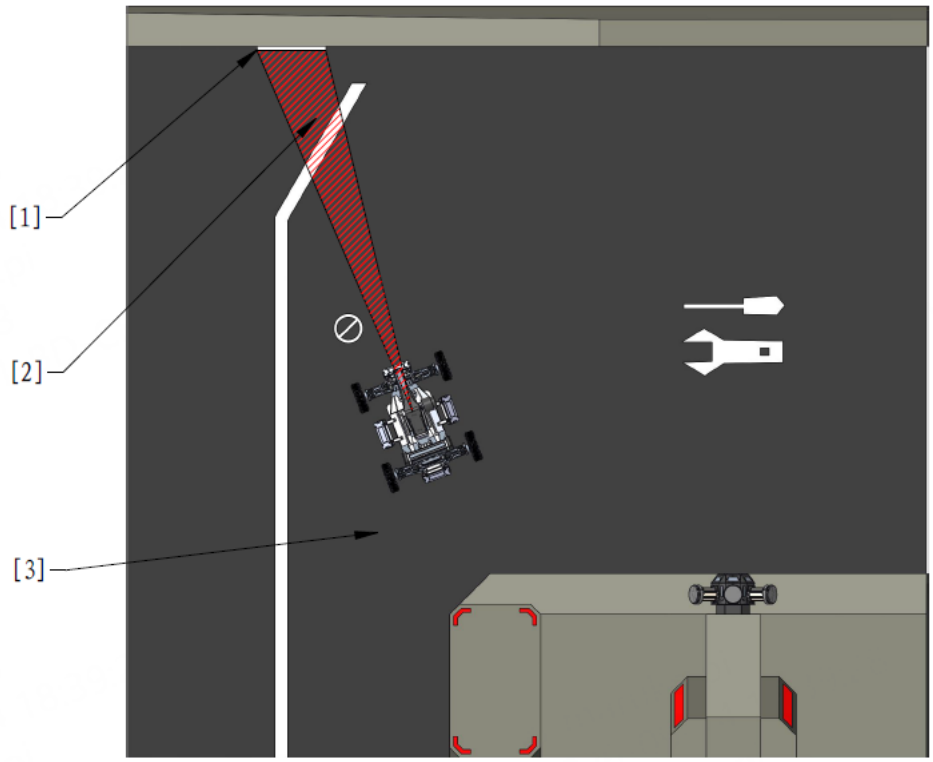
HP recovery at the Outpost

When a Ground Robot activates the Outpost Buff Point corresponding to an Outpost fully controlled by its side, it gains HP recovery at 50% of the rate it would receive in the Supply Zone at that time.

Remote HP exchange

During a round, an out-of-combat Ground Robot may spend Coins to exchange for HP remotely. Six seconds after the exchange is confirmed, the robot gains 60% of its maximum HP at that time, up to its maximum HP.

- If the robot is defeated within 6 seconds after a remote HP exchange is confirmed, the exchange is invalid and the Coins are not refunded.
- A robot is considered to be in the Supply Zone when its Field Interaction Module detects a Supply Zone Field Interaction Module card, or when the robot is in the Supply Zone as defined by the rules and its Camera Video Transmitter Module recognizes the Supply Zone visual marker. A surviving robot in its side’s Supply Zone is considered to have activated the Supply Zone Buff Point.
- A Camera Video Transmitter Module can recognize a visual marker only when the marker is in the center of its image and the module has no obvious roll angle.



[1] Visual marker | [2] Robot video range | [3] Recognizable area

Figure 5-5 Schematic of the relative position of the robot Video Transmitter Module and the visual tag

5.3.2 Revival Mechanism

When a robot is defeated, its Shooter Heat is reset to 0 and its buffer energy is reset to the buffer energy limit. A defeated robot can revive by completing the revival countdown or, while the countdown is in progress, by spending Coins for an instant revival.

- For revival by completing the countdown, the required countdown depends on the match time remaining when the robot is defeated and the number of instant revivals it has already purchased. The formula is:

$$\text{Required revival countdown} = 10 + \frac{420 - \text{match time remaining}}{} + 20 \times \text{number of instant revivals purchased}$$

Match time remaining is measured in seconds. Round the fractional part to the nearest integer.

Example: A robot is defeated with 120 seconds remaining in the match, after it has already purchased 2 instant revivals. Its revival countdown for this revival is 10 +

The revival countdown starts immediately when a robot is defeated. Starting at that moment, its countdown progress increases by 1 per second; progress increases by 4 per second while the robot is in a Supply Zone Buff Point or its side's Base HP is below 4,000.

- Revival by completing the revival countdown:
  - The robot retains the Level and Experience it had before being defeated and is temporarily Invincible for 30 seconds.
  - HP is restored to 10% of maximum HP.
  - The robot enters the “Weakened” state.

  After that, when it detects a Field Interaction Module card for an activatable Outpost, Base or Supply Zone Buff Point, the “Weakened” state ends. If the Invincibility gained from this revival has already lasted more than 10 seconds in total, Invincibility ends as well. If it has lasted less than 10 seconds, Invincibility ends after the full 10 seconds have elapsed.

- Instant revival purchased with Coins:
  - The robot retains the Level and Experience it had before being defeated and is temporarily Weakened and Invincible for 3 seconds; both states then end.
  - HP is restored to 100% of maximum HP.
  - The chassis power limit is doubled (but does not exceed 200 W) for 4 seconds.

  After that, when the robot detects a Field Interaction Module card for an activatable Outpost, Base or Supply Zone Buff Point, the “Weakened” state ends.

A robot in the “Weakened” state:

- Has its launching mechanism locked.
- Receives no effect from activating a Buff Point.

5.4 Economy System

5.4.1 Economy System Overview

During the three-minute Setup Period, all Ground Robots may preload 42mm and 17mm projectiles.

At match start, each side receives a certain amount of starting Coins. As the match progresses, both sides periodically receive more Coins, in the amounts shown below. Coins may also be earned by assembling Energy Units during the match. Coins may be spent to exchange for 17mm/42mm Projectile Allowance, remote HP recovery or instant revival.

Table 5-6 Overview of periodic Gold increments for robots

Match countdown	Red side	Blue side
06:59	800 (initial)	800 (initial)
05:59	100	100

Match countdown	Red side	Blue side
04:59	100	100
03:59	100	100
02:59	100	100
01:59	100	100
00:59	300	300

Table 5-7 Exchange rules

Exchange item	Exchange rate	Exchange limit
17mm Projectile Allowance	Remote exchange: 150 Gold / 100 projectiles; non-remote exchange: 10 Gold / 10 projectiles	1000 projectiles/ team
42mm Projectile Allowance	Remote exchange: 150 Gold / 10 projectiles; non-remote exchange: 10 Gold / 1 projectile	100 projectiles/ team
HP (remote exchange)	50 + ROUNDUP((420 - remaining match time) / 60 * 20) Gold / 1 time	Unlimited
Immediate Revival	ROUNDUP((420 - remaining match time) / 60) * 80 + robot Level * 20 Gold / 1 robot	Unlimited

- Duration is measured in seconds.
- ROUNDUP means round up to the nearest integer.
- The exchange limit for 17mm Projectile Allowance is the maximum Projectile Allowance obtainable through Coin exchanges.

5.4.2 Projectile Allowance Mechanism

Each time a robot fires a projectile, the Projectile Allowance for that projectile type decreases by one.

In the following special cases, all Armor Modules on the opposing robots, Outpost and Base block 42mm projectile damage until the Heavy Robot is Surviving and its Projectile Allowance is greater than 0:

Three seconds after the Heavy Robot changes from Surviving to Defeated or Abnormally Disconnected.

The Heavy Robot fires a 42mm projectile when its Projectile Allowance is 0 (an over-firing event).

After the Heavy Robot fires its third 42mm projectile after being defeated.

The packet loss rate between the modular 42mm launching mechanism and the Referee System Main Controller Module is at least 5%.

Damage from 17mm projectiles is not blocked.

Buff Point exchange

When a Ground Robot activates a specific Buff Point (see “5.6.4 Battlefield Buff Mechanics”), the Operator may exchange for Projectile Allowance through the Referee System Operator-Side Client. A Sentry Robot may also exchange for Projectile Allowance itself through the Referee System serial port. The types available and minimum exchange units are shown below:

Table 5-8 Minimum exchange unit of Projectile Allowance for different projectile types (for Buff Point exchange)

17mm Projectile Allowance	42mm Projectile Allowance
10 projectiles	1 projectile

Remote exchange

If a robot is out of combat, the Operator may remotely exchange for Projectile Allowance through the Referee System Operator-Side Client. A Sentry Robot may remotely exchange for Projectile Allowance itself through the Referee System serial port. The types available and minimum exchange units are shown below:

Table 5-9 Minimum exchange unit of Projectile Allowance for different projectile types (for remote exchange)

17mm Projectile Allowance	42mm Projectile Allowance
100 projectiles	10 projectiles

After a remote exchange succeeds, the Projectile Allowance takes effect 6 seconds later.

Obtaining Allowance in the Supply Zone

After the match begins, a Sentry Robot may activate its side’s Supply Zone Buff Point once per minute to obtain 100 rounds of Projectile Allowance. Allowance not obtained this way may accumulate.

Example: If a Sentry Robot has never activated its side’s Supply Zone Buff Point and activates it with 30 seconds remaining in the match, its Projectile Allowance increases by 600.

5.4.3 Tech Core

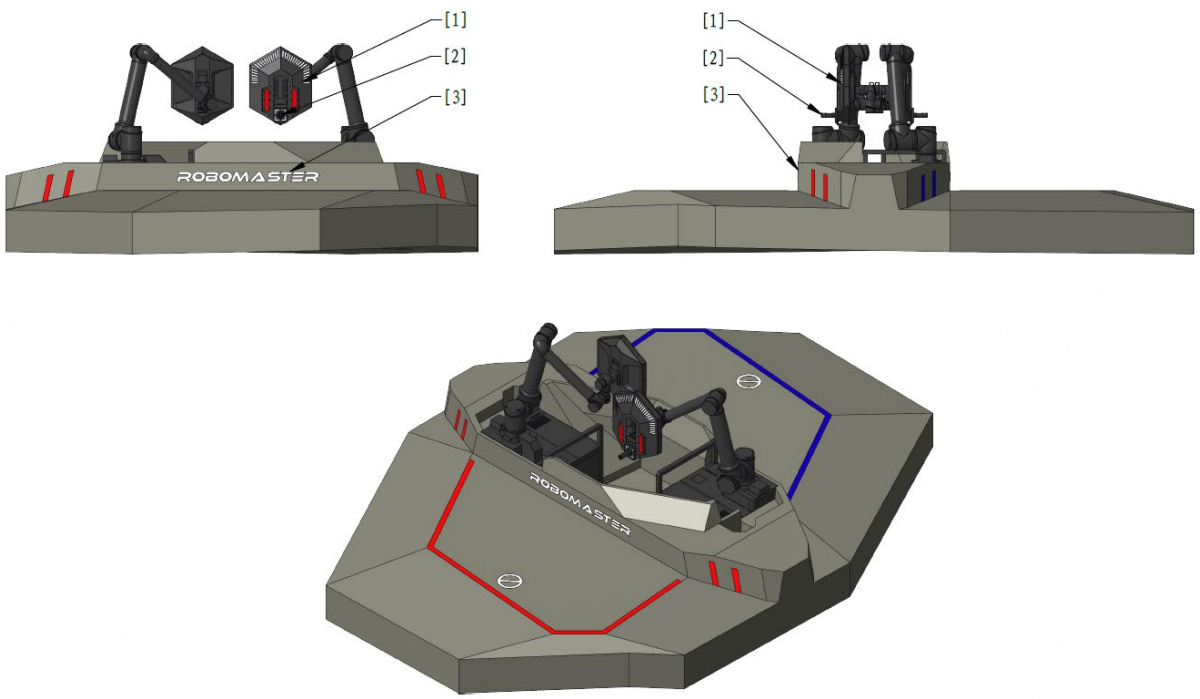
• Energy Unit Mechanism

During a match, a Heavy Robot assembles the Energy Unit it carries into the Tech Core in exchange for Coins and Team Buffs.

For each Tech Core, take the midpoint of the line segment where the pedestal’s front face intersects the Battlefield floor as origin O, and define the normal to the pedestal’s front face pointing toward the Tech Core as the negative direction of the X-axis

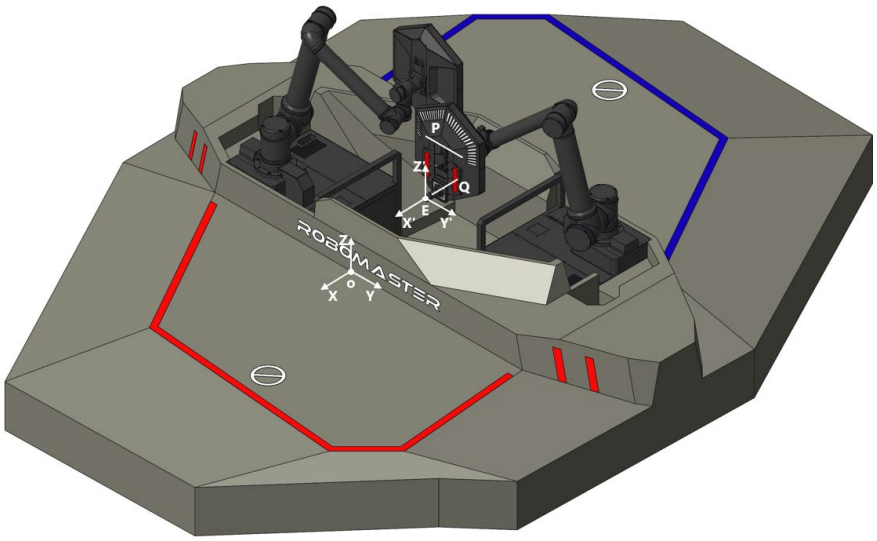
and vertical upward as the positive Z direction to establish a right-handed Cartesian coordinate system called the assembly coordinate system OXYZ. Take the geometric center of the front face of the Energy Unit assembly column as origin E to establish a right-handed Cartesian coordinate system EX′Y′Z′, whose X′, Y′ and Z′ axes are parallel to and point in the same directions as X, Y and Z. Let $\vec{e}$ be the unit normal vector of the front face of the Energy Unit assembly column. Establish spherical coordinates (r, θ, φ) based on the Tech Core coordinate system, where θ is the angle between the projection of $\vec{e}$ onto the X′Y′ plane and the positive X′ axis, $\theta \in [-90, 90]$; φ is the angle between $\vec{e}$ and the positive Z′ axis, $\varphi \in [0, 90]$. Also define α as the Tech Core's rotation angle around $\vec{e}$; counterclockwise rotation is positive, and $\alpha \in [-45, 45]$.

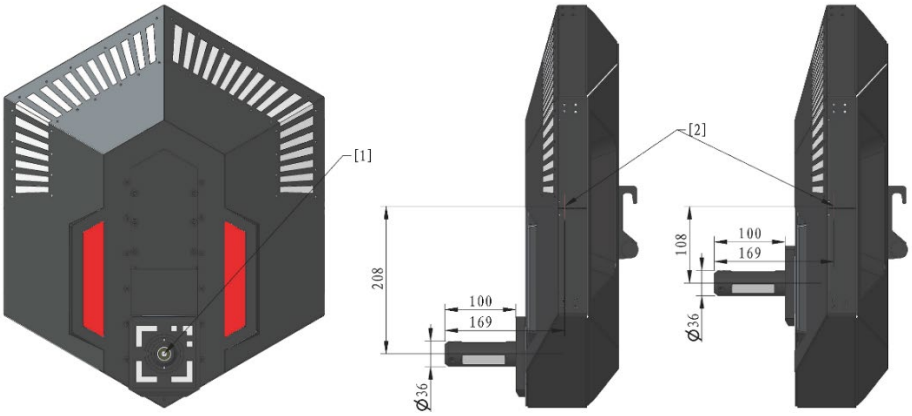
The combination (x, y, z, θ, φ, α) of the coordinates (x, y, z) of the center point E of the Energy Unit assembly column's front face in coordinate system OXYZ, the orientation coordinates (θ, φ) of unit normal vector $\vec{e}$ in the Tech Core spherical coordinate system, and the Tech Core's rotation angle α around $\vec{e}$ is called the Tech Core pose.

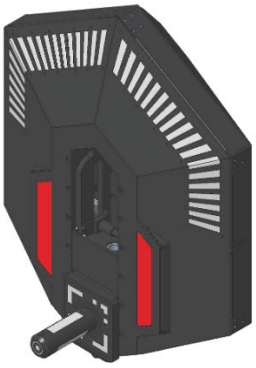


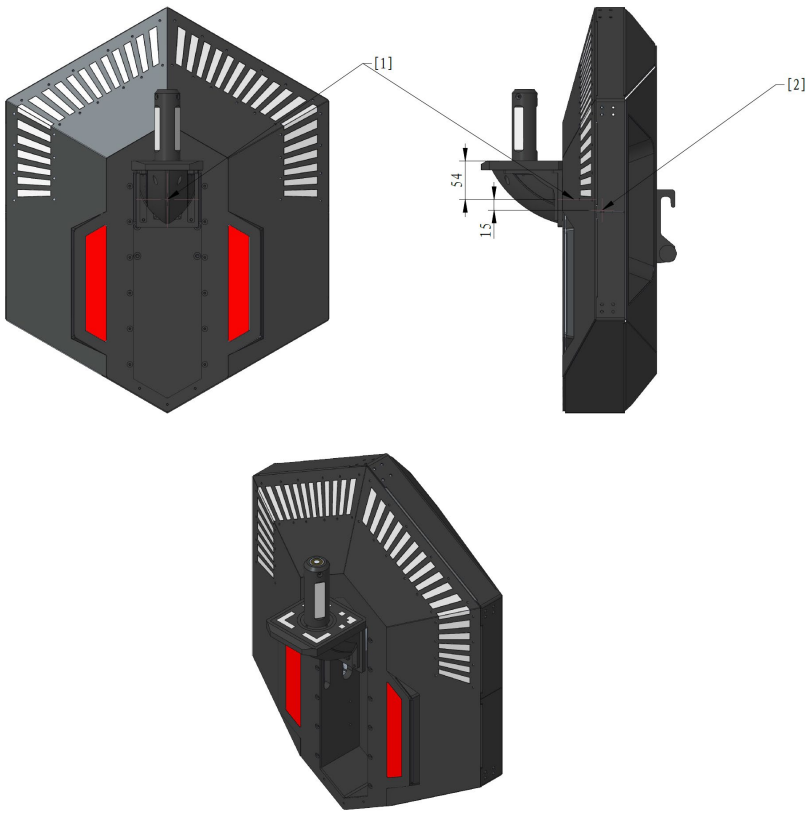
[1] Tech Core [2] Assembly column [3] Pedestal

Figure 5-6 Schematic of the Tech Core


P
Z
Q
X
Y
ROBOMASTER
X
O
Y


[1]
[2]
208
100
169
Ø36
108
100
169
Ø36





[1] Q axis [2] P axis

Figure 5-7 Schematic of Tech Core coordinates

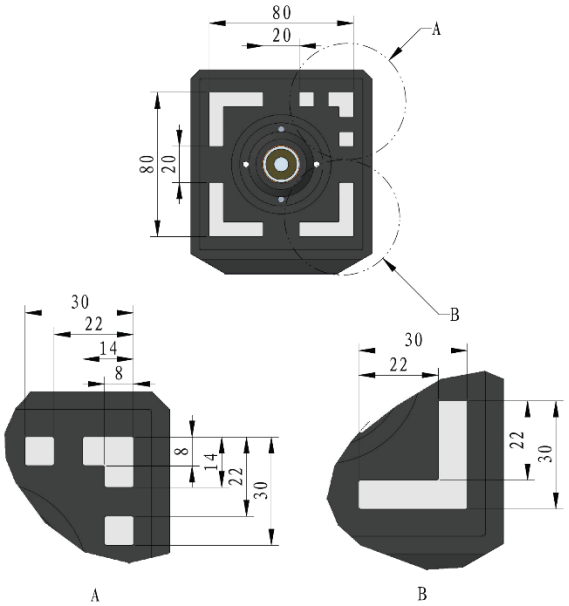


Figure 5-8 Schematic of Tech Core MARK light dimensions

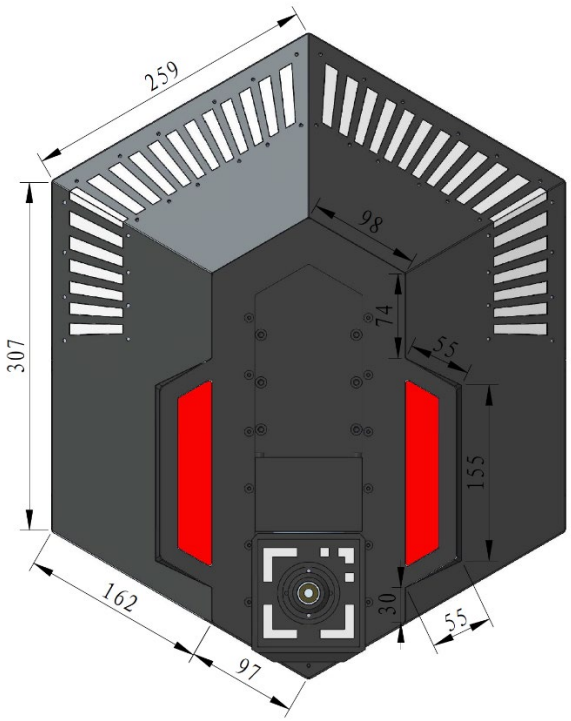


Figure 5-9 Schematic of Tech Core dimensions

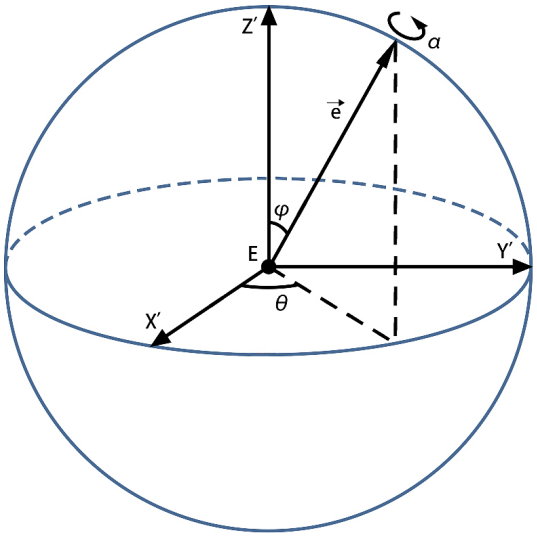


Figure 5-10 Schematic of the spherical coordinate system

The Tech Core's initial pose is $x = -200$, $y = 200$, $z = 800$, $\theta = 0$, $\varphi = 90$, $\alpha = 0$. Lengths above are in mm and angles are in degrees.

The Tech Core pose range must satisfy the following conditions:

- No part of the Tech Core may extend beyond the plane of the pedestal's front face.
- The Tech Core pose depends on the difficulty level, as follows:

Table 5-10 Relationship between Tech Core pose and difficulty level

Level	x	y	z	θ	ϕ	α	Translation requirement	Rotation requirement
Level 1	[-100,0]	[100,300]	[500,700]	0	90	0	No translation requirement	No rotation requirement
Level 2	[-100,0]	[100,300]	[500,700]	[-90,90]	[0,90]	[-45,45]	After insertion, translate	
Level 3	[-100,0]	[100,300]	[500,700]	[-90,90]	[0,90]	[-45,45]	After insertion, translate	After the translation is complete, rotate around the Q axis; see "Figure 5-7 Tech Core coordinate system diagram"; within a range of up to 90° clockwise or counterclockwise around Q, rotate to the specified angle with an accuracy of 5° .
Level 4	[-100,0]	[100,300]	[500,700]	[-90,90]	[0,90]	[-45,45]	After insertion, translate 100 mm	

During each match, the red and blue Tech Cores undergo the same pose changes at the same difficulty level. The pose ranges for Levels 3 and 4 have the same translation and rotation requirements. At Level 4, however, the opposing side's Tech Core temporarily moves to that side's position. The positional relationship between the opposing Tech Core coordinates x′y′z′ and that side's coordinates xyz is: $x' = x$, $y - y' \leq 400$, $z' = z$. The rotation requirement is random, and the assembling side must assemble both Tech Cores. After Level 4 is selected, the Heavy Robot must complete assembly within 45 seconds after the Tech Core reaches its position.

Level 4 assembly priority

- If one side is not currently attempting assembly and the other side selects Level 4, the Tech Core immediately moves to the specified position.
- If one side is attempting Level 1, 2 or 3 assembly, the other side cannot select Level 4.
- If one side is attempting Level 4 assembly, the other side cannot enter the assembly process.

The Heavy Robot may choose a difficulty level independently, but must complete assembly at the previous difficulty level before selecting the next one. Available difficulty levels are subject to certain conditions, as shown in the table:

Table 5-11 Heavy Robot assembly difficulty level unlock schedule

Match time	Selectable difficulty levels
At match start	Both sides can select Level 1 only

Match time	Selectable difficulty levels
1 minute after match start	Both sides can select Level 1 and 2
2 minutes after match start	Both sides can select Level 1, 2 and 3
3 minutes after match start	Both sides can select Level 1, 2, 3 and 4

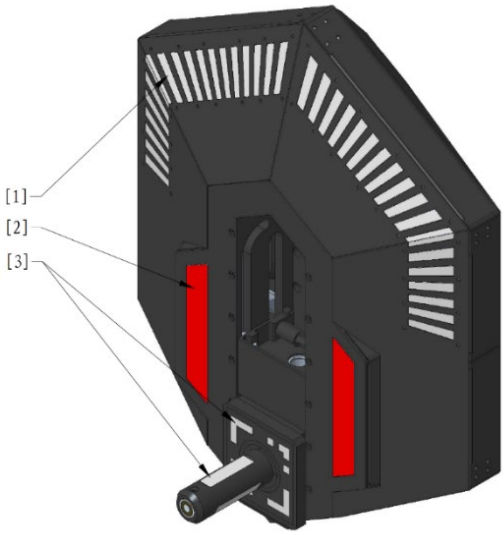
After a successful assembly, the team receives different rewards depending on whether it is the first time completing that level, as shown below:

Table 5-12 Heavy Robot assembly difficulty rewards

Difficulty completed	First-time reward	Reward for subsequent completions
Level 1	The Heavy Robot reaches Level 5. Afterward, that side gains 40 Coins every 10 seconds.	Afterward, that side gains 4 additional Coins every 10 seconds.
Level 2	The Heavy Robot reaches Level 7. Unlock the robot Level cap by completing the assembly task at this difficulty: Before completion, that side's robot Level cap is Level 5; after completion, the Level cap is raised to Level 7. Afterward, that side gains 20 additional Coins every 10 seconds.	Afterward, that side gains 8 additional Coins every 10 seconds.
Level 3	The Heavy Robot reaches Level 10. Ground Robots, the Outpost and the Base on that side gain a 25% Defense Buff lasting until the end of the match. The Heavy Robot on that side unlocks use of the modular 42mm launching mechanism. The robot Level cap for that side increases to Level 10. Afterward, that side gains 20 additional Coins every 10 seconds.	Afterward, that side gains 12 additional Coins every 10 seconds.
Level 4	Ground Robots, the Outpost and the Base on that side gain a 50% Defense Buff lasting until the end of the match. Afterward, that side gains 40 additional Coins every 10 seconds.	Can be completed only once

Robots no longer gain Experience after reaching the Level cap.

Example: If one side successfully completes five assembly tasks in a round at Levels 1, 2, 3, 2 and 1, respectively, it gains 40 + 20 + 20 + 8 + 4 = 92 Coins every 10 seconds.



[1] First light group [2] Second light group [3] Third light group

Figure 5-11 Schematic of Tech Core light effects

Table 5-13 Summary of Tech Core light effects

State	Light effect
Match in progress; assembly process not entered	First light group off Second and third light groups steadily lit in the team's color (red or blue)
Match in progress; assembly difficulty selected, mechanical arm not in position	First light group displays a flowing-light effect Second and third light groups steadily lit in the team's color (red or blue)
Match in progress; assembly difficulty selected, mechanical arm in position	First light group flashes white at 1 Hz Second and third light groups flash at 1 Hz in the team's color (red or blue)
Step 2 complete	First light group flashes white at 1 Hz Second light group flashes at 1 Hz in the team's color (red or blue)

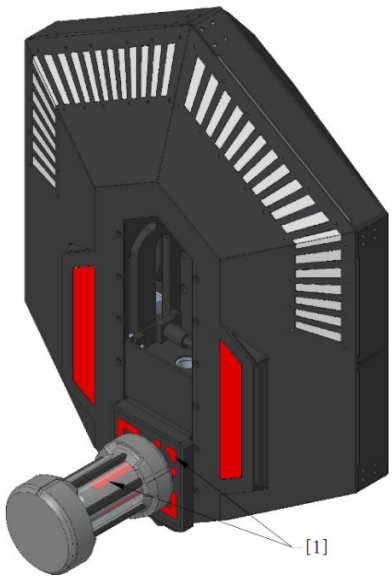
State	Light effect
	Third light group flashes at 3 Hz in the team's color (red or blue)
Step 3 complete	First light group flashes white at 1 Hz Second and third light groups flash at 3 Hz in the team's color (red or blue)
Step 4 complete	First light group flashes white at 1 Hz Second and third light groups steadily lit in the team's color (red or blue)
Step 5 in progress	First light group behaves as follows: At the specified angle, the team's color (red or blue) is lit steadily. At other angles, a white light is steadily lit at the current rotation angle of the Energy Unit. Second and third light groups steadily lit in the team's color (red or blue)
Step 5 complete	At the specified angle, a green light in the first group is steadily lit. Second and third light groups steadily lit in the team's color (red or blue)
Assembly confirmed; Energy Unit being recovered	First light group flashes white at 3 Hz Second and third light groups flash at 3 Hz in the team's color (red or blue)

To complete Energy Unit assembly, the following steps must be completed:

1. After the Heavy Robot activates the Assembly Zone Buff Point, the Operator selects an assembly difficulty on the Referee System Operator-Side Client.

2. After the Tech Core moves to the corresponding pose (assembly-ready state), the corresponding light effect appears. The Heavy Robot aligns the opening under the Energy Unit with the Energy Unit assembly column and inserts it 100 mm in the negative x direction. When it reaches the bottom, the corresponding light effect appears and Step 2 is complete, as shown below.

The Energy Unit can be inserted only after the Tech Core reaches the corresponding pose. Inserting it earlier will prevent the subsequent assembly process from proceeding normally.

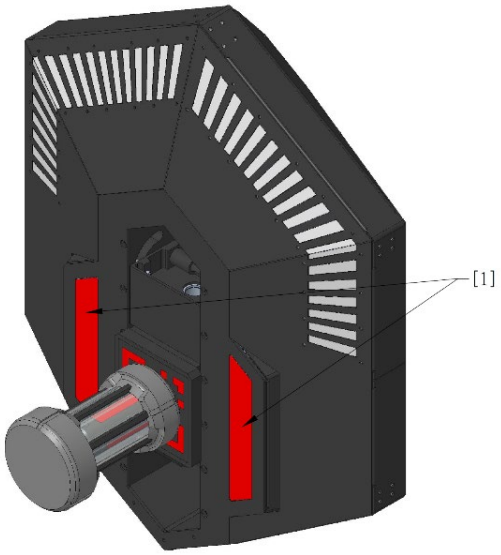


[1] Flashes at 3 Hz

Figure 5-12 Schematic of Step 2

3. The Heavy Robot moves the Energy Unit fixed to the assembly column 100 mm in the positive z direction. When it reaches the 100 mm position, the corresponding light effect appears and Step 3 is complete, as shown below.

- The Energy Unit is opaque in reality; the illustration only shows the light effect.
- During this step, besides the force needed to overcome the Energy Unit's weight, the resistance to movement in the positive z direction is at most 10 N.



[1] Flashes at 3 Hz

Figure 5-13 Schematic of Step 3

4. The Heavy Robot rotates the Energy Unit upward by 90° around the P axis. When it reaches 90°, the corresponding light effect appears and Step 4 is complete, as shown below.

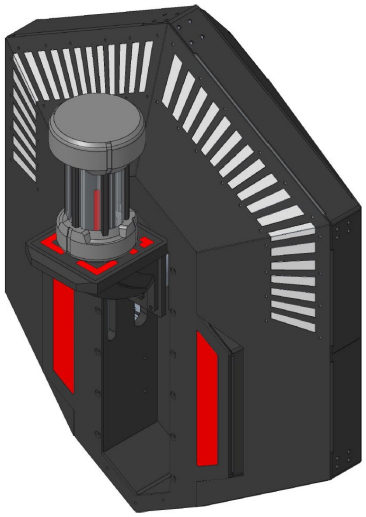


Figure 5-14 Schematic of Step 4

5. The Heavy Robot rotates the Energy Unit around the Q axis to the specified angle until the corresponding prompt appears on the Operator-Side Client (it appears after the Energy Unit remains in the correct position for at least 0.3 s and no more than 0.5 s). Step 5 is then complete, as shown below.

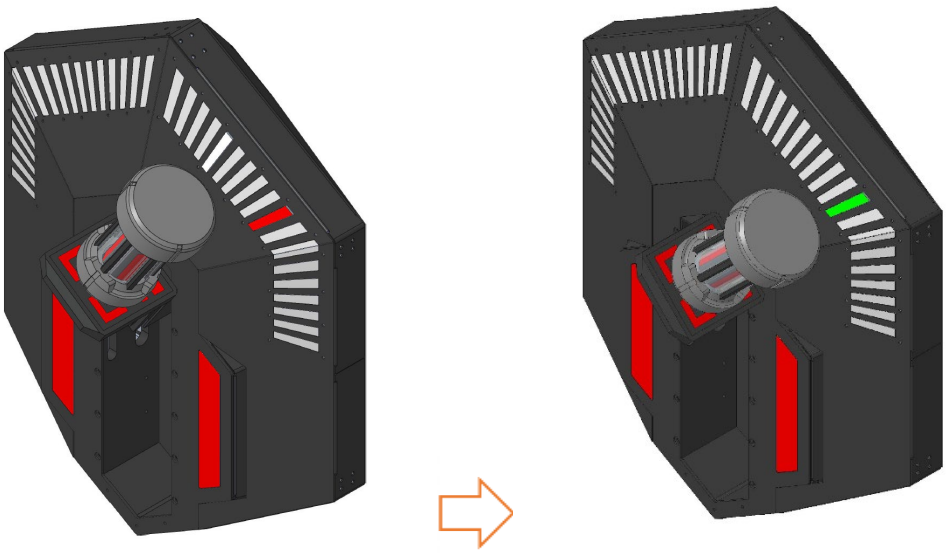


Figure 5-15 Schematic of Step 5 Tech Core light effects

- If the Operator presses “Confirm Assembly” but the Tech Core does not detect that the Energy Unit has been correctly assembled as required, it will not accept another “Confirm Assembly” command for 1 second.
- During this step, the Energy Unit assembly column naturally tends to return to 0° with a maximum torque of 0.3 N·m.

- For safety, after the Energy Unit remains in the correct position for more than 2 seconds, the rotation mechanism actively holds it in place with about 0.5 N·m of torque. The Heavy Robot mechanism can then disengage from the Energy Unit without changing its position.

6. The Operator presses the corresponding button to confirm completion of assembly. If the Tech Core detects that the Energy Unit has been fully inserted and correctly assembled according to the translation and rotation requirements, it recovers the Energy Unit and awards the corresponding rewards to that side.

For Level 1, completing Steps 1, 2 and 6 counts as successful assembly.

For Level 2, completing Steps 1, 2, 3 and 6 counts as successful assembly.

For Level 3, completing Steps 1, 2, 3, 4, 5 and 6 counts as successful assembly.

For Level 4 assembly, the Heavy Robot must complete Steps 1, 2, 3, 4, 5 and 6, and within 5 seconds after each assembly step other than

Step 1 and Step 4, it must complete the same step at the other Tech Core. After Step 6 is completed at both Tech Cores, the Tech Core recovers both Energy Units and rewards that side only if it detects that both Energy Units have been fully inserted and correctly assembled according to the translation and rotation requirements within the required time. Otherwise, assembly fails.

Operators must also observe the following during assembly:

- After an Operator selects an assembly difficulty, if an Energy Unit is already inside the Tech Core, the difficulty must not be changed. Until the Energy Unit is successfully assembled, the Tech Core pose remains the same for that difficulty.
- After the Heavy Robot Operator confirms Energy Unit assembly, they must not further operate or interfere with the Energy Unit, or assembly may fail.
- During assembly, the Heavy Robot may remove the Energy Unit. In addition to overcoming gravity, it may apply up to 10 N of outward force along the Energy Unit's rotational symmetry axis to remove it. If the Tech Core sensor detects removal, assembly immediately fails.
- If the Heavy Robot does not detect an Assembly Zone Interaction Module card for 15 consecutive seconds during assembly, that assembly attempt fails and exits automatically.
  The Tech Core then immediately recovers the Energy Unit.
- The Tech Core has no obstacle-detection or active obstacle-avoidance function while operating. If its motion path is obstructed, the system triggers an overload alarm and briefly hovers, then attempts to resume movement. If the Heavy Robot's mechanism is damaged by contact or collision with the Tech Core, the participating team bears all resulting losses.
- If the Heavy Robot is defeated after grasping an Energy Unit, that assembly attempt immediately fails. After the Operator manually cancels assembly, the Tech Core recovers the Energy Unit.
- If a side's Heavy Robot is defeated during assembly, only its 42mm launching mechanism is powered off; its gimbal and chassis remain powered on. The Operator must control the defeated robot to disengage from the Tech Core within 15 seconds. After 15 seconds, the Tech Core recovers the Energy Unit.

- During assembly, the Heavy Robot must manipulate the Energy Unit, not the Tech Core's movable structures or other body structures.
- Facing the Tech Core, the Energy Unit may rotate 90° clockwise or counterclockwise around the Q axis during assembly.

5.5 Experience and Performance System

5.5.1 Experience System

At match start, the Infantry Robot is Level 1 and can level up by gaining Experience. The Experience system does not apply to the other robots.
The Heavy Robot is Level 1. For how it levels up, see "Table 5-12 Heavy Robot Assembly Difficulty Rewards".

During a match, a robot can gain Experience in multiple ways, as shown in the table below:

Table 5-14 Experience gain and settlement mechanism

Behavior Type	Experience Gained
Firing projectiles	Infantry Robot: Gains 1 Experience point for each projectile fired.
Dealing attack damage	Dealing attack damage to a robot: The attacker gains 4 Experience points for each point of damage dealt. Dealing attack damage to an Outpost Armor Module: The attacker gains 2 Experience points for each point of damage dealt. Dealing attack damage to a Base Armor Module: The attacker gains 1 Experience point for each 2 points of damage dealt; round up if the damage is odd. If a side's robot, Base or Outpost takes attack damage but the Referee System does not detect its source, the source cannot gain Experience, or the source is a penalty or a friendly robot, then: for projectile damage, Experience is divided equally among all surviving robots on the opposing side that can normally deal that type of projectile damage and can gain Experience; for non-projectile damage, Experience is divided equally among all surviving Infantry Robots on the opposing side at that time. Round the average to one decimal place. Example: A Blue Infantry Robot takes 120 points of 17mm projectile damage, but the system does not detect the source. If the Red side has 2 surviving Infantry Robots, each gains $4 \times 120 / 2 = 240$ Experience points.
Robot defeated	In the calculations below, the Sentry Robot is considered Level 1. If there is a killer and it can gain Experience: When the defeated robot's Level is greater than or equal to the killer's Level, Experience gained by the killer = 50 × defeated robot's Level × (1 + 0.2 × the difference between the defeated robot's and killer's Levels).

Behavior Type	Experience Gained
	When the Level of the defeated robot is lower than the killer's Level, treat the Level difference as 0. Experience is calculated as follows: Experience gained by the killer = 50 × Level of the defeated robot If the Referee System does not detect the killer, or the killer cannot gain Experience: For projectile damage, the killer's Level is treated as the Level corresponding to the average Experience of all surviving robots on the opposing side that can normally deal that type of projectile damage, and Experience is divided equally among those robots. For non-projectile damage, Experience is divided equally among all surviving Infantry Robots on the opposing side at that time. Round the average to one decimal place. If a robot becomes Defeated for a reason other than an Armor Module being attacked, becomes Abnormally Disconnected, or the Referee System cannot detect the killer, the killer is considered not found. Example 1: A Level 2 Infantry Robot defeats an opposing Level 6 Infantry Robot. The killer gains 50 × 6 × (1 + (6 − 2) × 0.2) = 540 Experience. Example 2: A Level 9 Red Infantry Robot is Defeated and no killer is detected. At this time, two Blue Infantry Robots survive with 1,250 and 4,000 total Experience. Their average Experience is 2,625, corresponding to Level 5. Each Blue Infantry Robot gains 50 × 9 × (1 + (9 − 5) × 0.2) / 2 = 405 Experience.
Activate Small Power Rune	See "5.6.3 Power Rune Mechanism".
Activate Large Power Rune	See "5.6.3 Power Rune Mechanism".
Terrain Crossing	When a robot obtains the Terrain Crossing Buff (Road), Terrain Crossing Buff (Highland), Terrain Crossing Buff (Tunnel), or the High Point Buff Point for the first time, it gains 300 Experience each time.
Dart hits Outpost/Base	See the Dart hit rewards part of "5.7.3 Dart System Special Mechanisms".

Table 5-15 Infantry Robot Levels and Experience

Level	Total Experience Required to Level Up
1	0
2	550
3	1100
4	1650
5	2200
6	2750
7	3300
8	3850
9	4400
10	5000

5.5.2 Performance System

After the three-minute Setup Period begins, the Operator of an Infantry Robot may choose the robot's chassis and launching mechanism types. The Performance System does not apply to other robots.

Once the seven-minute Round begins and the robot's chassis and launching mechanism types have been selected, they cannot be changed for the rest of the round. If no chassis or launching mechanism type is selected, after the seven-minute Round begins the defaults are "HP Priority" for the chassis and "Cooling Priority" for the launching mechanism. When a robot levels up, the increase in Max HP is immediately added to its current HP.

Table 5-16 Heavy Robot Overall Attributes

Match stage	Max HP	Chassis power limit	Barrel Heat Limit	Barrel Cooling Rate	Initial launch speed limit (m/s)
Default at start	400	50	-	-	-

Match stage	Max HP	Chassis power limit	Barrel Heat Limit	Barrel Cooling Rate	Initial launch speed limit (m/s)
After completing Assembly Level 1	450	50	-	-	-
After completing Assembly Level 2	500	50	-	-	-
After completing Assembly Level 3 or after successfully installing the modular 42mm launching mechanism	600	50	300	100	16.5

Table 5-17 Infantry Robot Chassis Attributes

Chassis Type	Level	Max HP	Chassis Power Limit (W)
Power Priority	1	150	60
	2	175	65
	3	200	70
	4	225	75
	5	250	80
	6	275	85
	7	300	90
	8	325	95
	9	350	100
	10	400	100
HP Priority	1	200	45
	2	225	50

Chassis Type	Level	Max HP	Chassis Power Limit (W)
	3	250	55
	4	275	60
	5	300	65
	6	325	70
	7	350	75
	8	375	80
	9	400	90
	10	400	100

Table 5-18 Infantry Robot 17mm Launching Mechanism Attributes

Launching Mechanism Type	Level	Barrel Heat Limit	Barrel Cooling Rate (per second)
Burst Priority	1	170	5
	2	180	7
	3	190	9
	4	200	11
	5	210	12
	6	220	13
	7	230	14
	8	240	16
	9	250	18
	10	260	20

Launching Mechanism Type	Level	Barrel Heat Limit	Barrel Cooling Rate (per second)
Cooling Priority	1	40	12
	2	48	14
	3	56	16
	4	64	18
	5	72	20
	6	80	22
	7	88	24
	8	96	26
	9	104	28
	10	120	30

5.6 Field-Related Mechanisms

5.6.1 Base and Outpost Mechanisms

The Base has 10,000 HP and the Outpost has 1,000 HP.

When the Red side fully controls the Red Outpost Position (and likewise for Blue), the Red Base gains a 100% Defense Buff; otherwise, the Base loses this 100% Defense Buff.

When an Outpost Position changes from full control by one side to full control by the other side because the Outpost is destroyed, the Outpost is "rebuilt" as an Outpost owned by the other side, returns to the surviving state and has 500 HP.

For definitions and details of full control and temporary control, see "5.6.2 Battlefield Control Mechanism".

After a side fully controls an Outpost Position, it may activate the Outpost Buff Point corresponding to that position. See "5.6.4.6 Outpost Buff Point Mechanism".

As a side's Base HP decreases, its logistics capabilities are reduced to a certain extent, as follows:

Define Δ = HP lost by the side's own Base, and the logistics coefficient $\lambda = 1 - \Delta /$ ~~20,000~~. The original parameter for each item in the table below is denoted by x:

Table 5-19 Logistics Capability Parameter Calculation Logic

Logistics Capability Item	Relationship Between the Item's Parameter and the Logistics Coefficient
Progress bar required for Revival	$\overline{\lambda}$
Supply Zone HP Recovery rate	$x * \lambda$
HP Recovery ratio of remote exchange for HP	$x * \lambda$
Robot interaction data bandwidth	
Video transmission resolution	x * λ^2 (x is 1920 × 1080, the original video transmission resolution. After Base HP decreases, the video transmission picture reduces its clarity by area-average downsampling followed by bilinear upscaling. The clarity of the original Operator-side UI is not affected.)

- For the definition of robot interaction data in multi-robot communication, see the Communication Protocol.
- Video transmission resolution includes the video feeds from the official Operator-Side Client and the Custom Terminal.

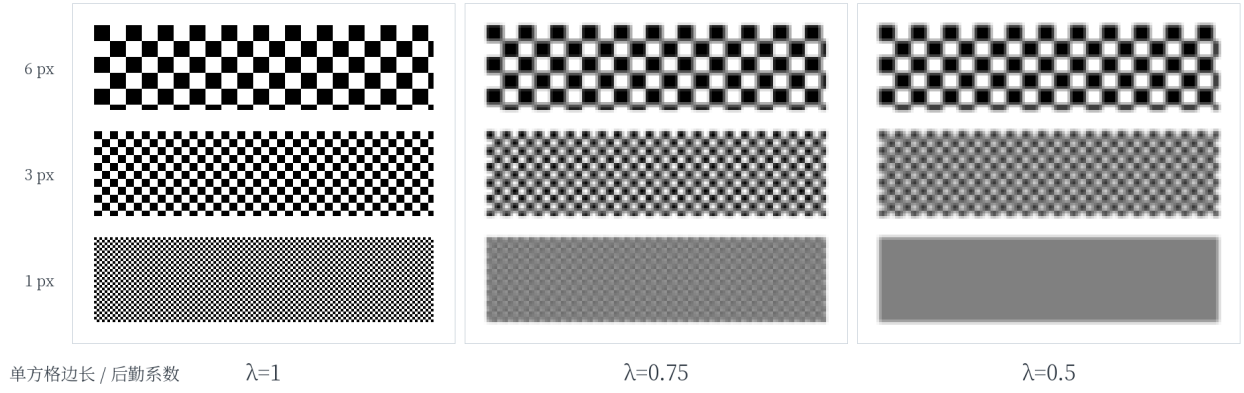


Figure 5-16 Illustration of video transmission clarity at $\lambda = 1, 0.75, 0.5$

The Outpost's center Armor Module can rotate. For its initial position, see "Figure 4-34 Outpost Illustration". After the Round begins, the center Armor Module starts rotating and reaches 0.8π rad/s within 5 seconds, then rotates at a constant speed in a random direction. In each round, the Red and Blue Outposts rotate in the same direction, which remains fixed.

A side's Outpost armor stops rotating when any of the following conditions is met:

- That side's Outpost is destroyed for the first time.
- The opposing Base Armor is deployed.

- Three minutes after the match begins.

Outpost armor that stops rotating does not rotate again during that round.

- If both Outposts stop rotating at the same time, their center Armor Modules may take slightly different times to reach their initial positions.
- The Dart Guidance Lights on the Outpost and Base stay lit continuously during the seven-minute Round.

5.6.2 Battlefield Control Mechanism

During a match, the two sides must contest Positions to demonstrate and improve their control of the Battlefield and thereby increase their control degree.

Each side has two Positions: the Outpost Position and the Base Position. A Position can be temporarily controlled or fully controlled. Their relationship is shown below:

Table 5-20 Relationship Between Positions and Control

Position	Composition	Supports Temporary Control	Supports Full Control
Outpost Position	Outpost Buff Point + Central Highland Buff Point	Yes	Yes
Base Position	Base Buff Point + Fortress Buff Point	Yes	Only the initial owner

At match start, each side fully controls its own Base Position and Outpost Position.

After the match starts, Position control may change in the following ways:

Table 5-21 Position Control State Transition Mechanisms

Transition Result	Transition Method	Additional Mechanisms
Convert a Position fully controlled by the opponent into a Position temporarily controlled by the own side	The own side continuously activates any Buff Point composing that Position for 5s	If continuous activation stops before reaching 5s, the occupation progress regresses evenly (at the same rate as it increased) instead of resetting to zero. When both sides activate Buff Points in the same Position, the occupation progress is paused until only one side's robots still have activated Buff Points in that Position.
Convert a Position temporarily controlled by the opposing side into a Position fully controlled by the own side	The opposing robots do not activate any Buff Point comprising the Position for 5 consecutive seconds	
Convert an Outpost Position fully controlled by the own side into an Outpost Position temporarily controlled by the own side	Destroy the Outpost on the Position	
Convert an Outpost Position temporarily controlled by the own side into an Outpost Position fully controlled by the own side	Destroy the Outpost on the Position	

When a side controls all 2 Outpost Positions, its control degree increases by 1 per second.

When a side controls all 2 Base Positions, its control degree increases by 3 per second.

5.6.3 Power Rune Mechanism

Robots can activate the Power Rune by hitting it with projectiles. Both sides may hit the Power Rune at the same time. Once activated, the Red side's Power Rune provides Buffs only to Red, and likewise for Blue.

The Power Rune rotates continuously during a match, but its rotation strategy changes according to its state:

The Red and Blue Power Runes rotate coaxially: when the Red Power Rune rotates clockwise, the Blue Power Rune rotates counterclockwise (direction is determined as viewed facing that side's Power Rune). Before each round, the rotation direction is random; it remains the same throughout that round.

- The Small Power Rune's rotation speed is fixed at $-\pi$ rad/s.
- The rotation speed of a Large Power Rune that is not activating is fixed at $\frac{1}{-}\pi$ rad/s.
- The rotation speed of a Large Power Rune in the activating state varies periodically according to a trigonometric function. The target speed function is: spd = a × sin(ω × t) + b, where spd is in rad/s, t is in s, a ranges from 0.780 to 1.045, and ω ranges from

  1.884 to 2.000; b always satisfies b = 2.090 − a. Whenever a Large Power Rune enters the activatable state, all parameters reset: t resets to 0, and a and ω reset to arbitrary values within their ranges. The time error between the Power Rune's actual rotation speed and the target speed function is within 500 ms.

After a side activates its Power Rune, all surviving robots on that side receive certain Buffs until the effect duration ends, and that side's Power Rune enters the activated state. When the Buff effect ends, the Power Rune returns to the inactive state. Both sides may have Power Rune Buffs at the same time.

The Power Rune has two stages based on time: the Small Power Rune and the Large Power Rune. From the start of the match until 3 minutes have elapsed, it is the Small Power Rune; after 3 minutes, it is the Large Power Rune.

- Small Power Rune: The states of the two sides' Small Power Runes are independent. At match start and 1 minute 30 seconds after match start, each side receives one chance to activate the Small Power Rune; chances can accumulate. An Infantry Operator through the Referee System Operator-Side Client, or a Sentry Robot through the Referee System serial port, may trigger an inactive Small Power Rune to enter the activating state. If it is not activated within 20 seconds, it returns to the inactive state. When a side's robot successfully activates the Small Power Rune, all of that side's robots, Outpost and Base gain a 25% Defense Buff for 45 seconds. During this Buff, all Infantry Robots gain bonus Experience equal to 100% of the Experience they would otherwise gain. A side can gain at most 1,200 bonus Experience this way during one Small Power Rune Buff.

- Large Power Rune: The states of the two sides' Large Power Runes are independent. At 3 minutes, 4 minutes 15 seconds and 5 minutes 30 seconds after match start, each side receives one chance to activate the Large Power Rune; chances can accumulate. An Infantry Operator through the Referee System Operator-Side Client, or a Sentry Robot through the Referee System serial port, may trigger an inactive Large Power Rune to enter the activating state. If it is not activated within 20 seconds, it returns to the inactive state. Each Armor Module of the Large Power Rune is divided into rings 1–10. When one side's Large

Power Rune is activated, the system gives all robots on that side Attack, Defense and Shooter Heat cooling Buffs of varying duration and effect according to the average ring number hit and the number of activated light arms. It gives the Outpost and Base on that side the corresponding Defense Buff. See “Table 5-22 Average Ring Number and Corresponding Buffs”. At the same time, 750 Experience is divided equally among all surviving Infantry Robots on the activating side.

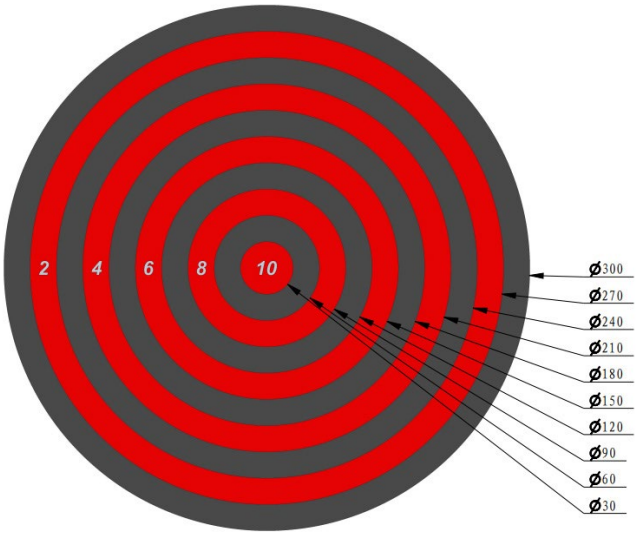


Figure 5-17 Illustration of Power Rune hit zones

The Power Rune Armor Module detects projectile hit position with a radial accuracy of 1 mm.

Table 5-22 Average Ring Number and Corresponding Buffs

Average Ring Number Range	Attack Buff	Defense Buff	Barrel Cooling Buff
[1,3]	150%	25%	None
(3,7]	150%	25%	2×
(7,8]	200%	25%	2×
(8,9]	200%	25%	3×
(9,10]	300%	50%	5×

Table 5-23 Number of Activated Light Arms and Corresponding Buff Duration

Number of Activated Light Arms	Buff Duration (s)
5	30

Number of Activated Light Arms	Buff Duration (s)
6	35
7	40
8	45
9	50
10	60

Each time a side successfully activates its Large Power Rune, that side's effective hit area shrinks. Only Armor Modules with some ring numbers can continue detecting projectile hits, as shown below:

Table 5-24 Relationship Between Large Power Rune Activation Count and Effective Detected Rings

Cumulative Activation Count of the Side's Large Power Rune	Rings That Can Still Be Detected
1 time	Rings 4, 5, 6, 7, 8, 9, 10
2 times	Rings 7, 8, 9, 10

5.6.3.1 Power Rune States

The Power Rune has four states: inactive, activating, activated and activation failed.

1. Inactive state

When the Power Rune enters the inactive state, it appears as follows:

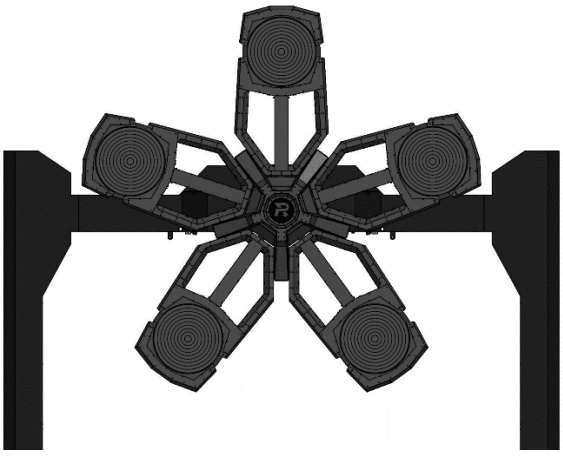


Figure 5-18 Illustration of the Power Rune inactive state

2. Activating state

Small Power Rune

When the Small Power Rune enters the activating state, it randomly lights 1 of its 5 Armor Modules. The lit Armor Module displays a special light effect, and the center axis of its corresponding light arm displays an arrow-shaped flowing light effect. If a projectile hits any lit Armor Module within 2.5 seconds, that light arm lights up completely and displays the corresponding effect. At the same time, the Power Rune randomly lights one of the other 4 Armor Modules, and so on.

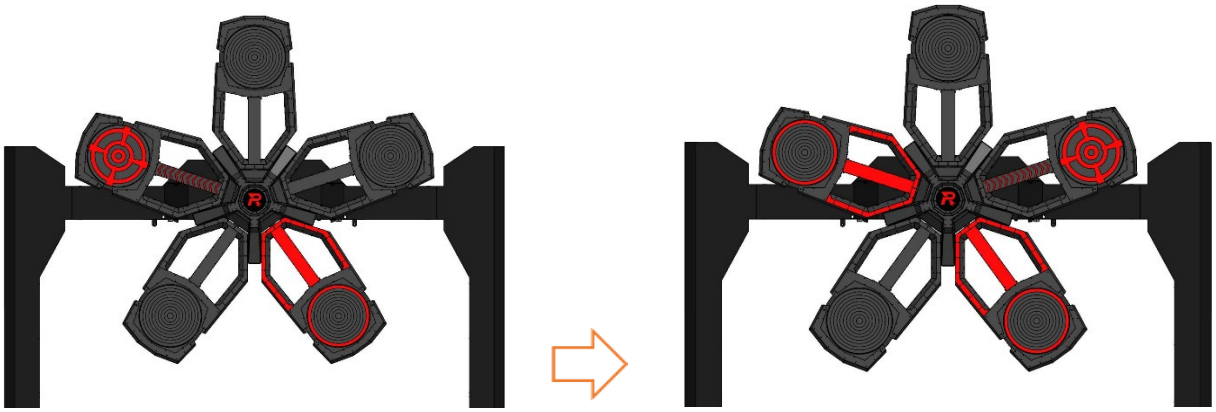


Figure 5-19 Illustration of the Small Power Rune activating state

Large Power Rune

When the Large Power Rune enters the activating state, it randomly lights 2 of its 5 Armor Modules. The lit Armor Modules display a special light effect, and the center axis of each corresponding light arm displays an arrow-shaped flowing light effect. If a projectile hits the detectable zone of any lit Armor Module within 2.5 seconds, the corresponding light arm changes its light effect. Within the following 1 second, the robot may hit the other lit Armor Module. Whether or not it succeeds, the Power Rune randomly lights 2 of its 5 Armor Modules again, and so on. The light effect in the middle of the Large Power Rune's light arms indicates activation progress. For example, when the first group is activated and the second group is being activated, the middle light arm lights up about 1/5; after that, if the second group is successfully activated (hitting the first lit Armor Module of the second group counts as successful activation, whether or not the second Armor Module in that group is hit; the light effect changes when activation of the third group begins), the middle light arm lights up about

The specific light effects are shown in the figure below:

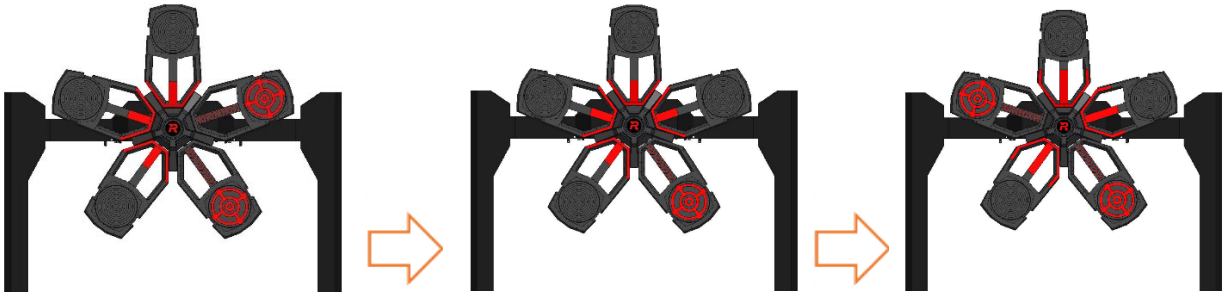


Figure 5-20 Illustration of the Large Power Rune activating state

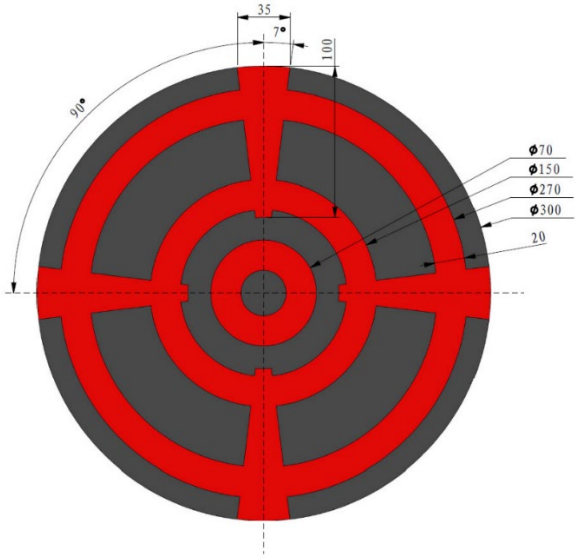


Figure 5-21 Illustration of the Power Rune activatable indicator

To ensure timing alignment, during activation the light arms and Armor Module lights may go dark for up to 200 ms before the Power Rune randomly lights the next group of Armor Modules.

3. Activated state

If all 5 groups of light arms are activated, the Power Rune is in the activated state, as shown below:

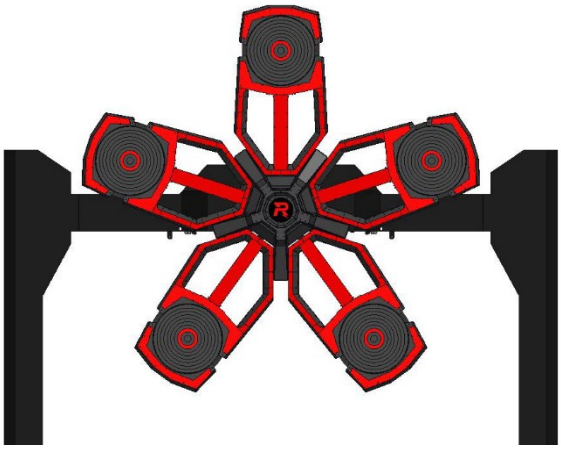


Figure 5-22 Illustration of the Power Rune activated state

4. Activation failed

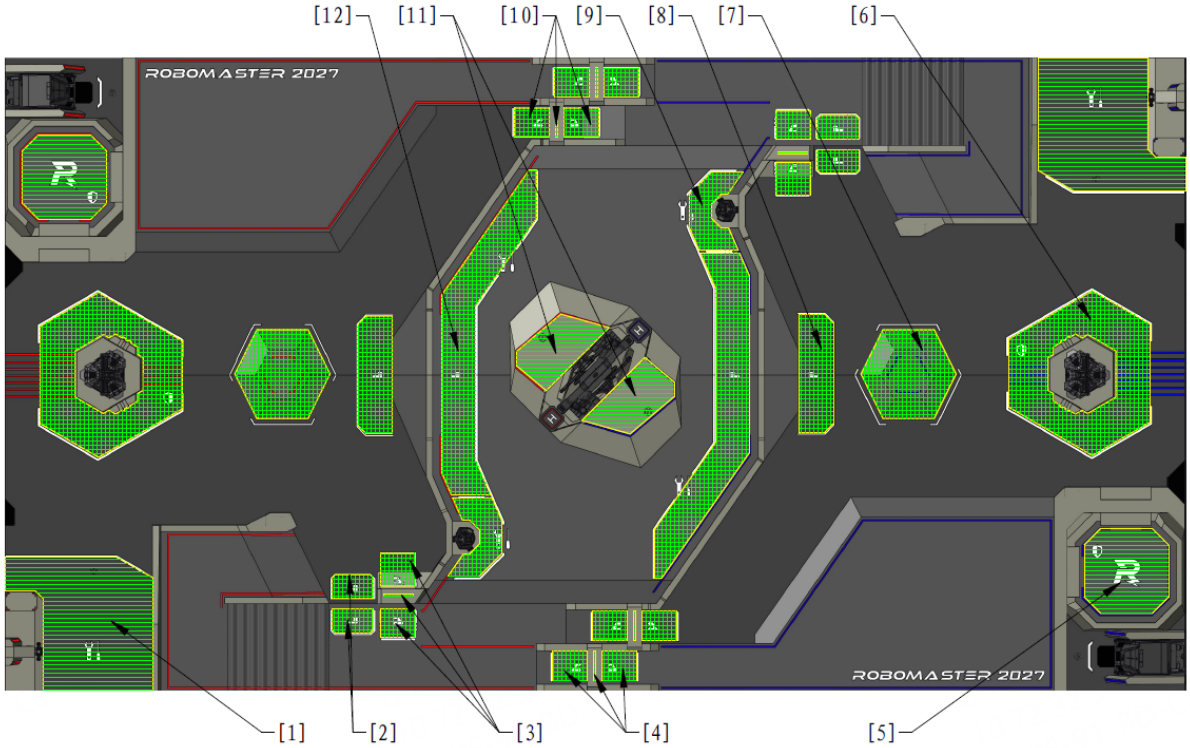
If any of the following occurs during activation, the attempt fails and the Power Rune resets to the activating state:

- Failing to hit a lit Armor Module within 2.5 seconds.
- Hitting an Armor Module that was not randomly lit.

5.6.4 Field Buff Mechanisms

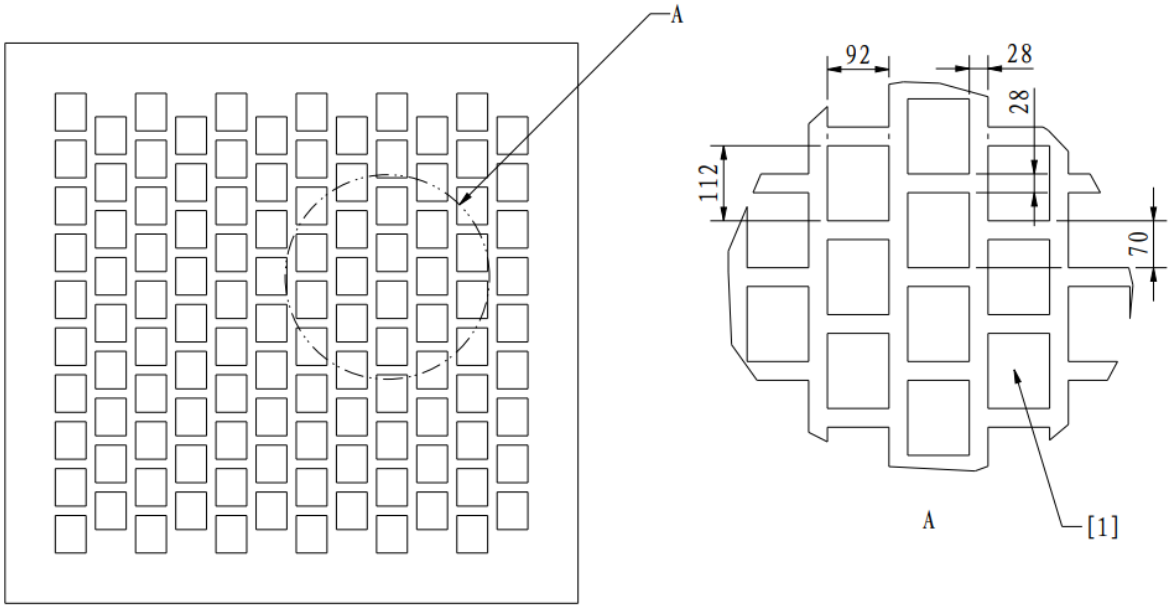
5.6.4.1 Field Buff Point Overview

The distribution of all Buff Points on the field is shown below. Field Interaction Module cards are laid at all Battlefield Buff Points. These cards may have dead zones, so teams must adapt as necessary. Detection by a Field Interaction Module may be delayed; in all cases, deactivation of a robot's activated state is delayed by 2 seconds.



[1] Supply Zone Buff Point	[2] Terrain Crossing Buff Point (Road)	[3] Terrain Crossing Buff Point (Tunnel)	[4] Terrain Crossing Buff Point (Tunnel)
[5] High Point Buff Point	[6] Base Buff Point	[7] Fortress Buff Point	[8] Terrain Crossing Buff Point (Highland)
[9] Outpost Buff Point	[10] Terrain Crossing Buff Point (Tunnel)	[11] Assembly Zone Buff Point	[12] Central Highland Buff Point and Terrain Crossing Buff Point (Highland)

Figure 5-23 Illustration of Battlefield Buff Point distribution



[1] Field Interaction Module card mounting position

Figure 5-24 Illustration of Field Interaction Module card installation

Buffs can be divided into different types according to their effects; see the table below:

Table 5-25 Buff Effect Types

Type	Description
Attack Buff	Increases the damage dealt by projectile attacks or Aerial Robot impacts; the damage becomes: original damage * attack Buff
Defense Buff	Changes the damage taken from projectile attacks, collisions and Aerial Robot impacts, so that the attack damage taken by robots, the Base and the Outpost becomes: original damage * (1 - defense Buff). A negative defense Buff is called "vulnerability". Note: Defense Buffs do not apply to HP deductions caused by rule-violation penalties, Referee System module offline, Darts, etc.
Barrel Cooling Buff	Increases the Barrel Cooling Rate.
Buffer Energy Buff	Gains extra Chassis Power Buffer Energy.
HP Recovery Buff	The robot recovers a certain amount of HP per second until reaching Max HP.
Chassis Power Buff	Increases the robot's Chassis Power limit.

For base damage, see “Table 5-2 HP Deduction Mechanism for Attack Damage or Collisions” and “5.2 Adaptive Armor Mechanism”.

By source, Buffs can be divided into field persistent Buffs, field temporary Buffs and non-field temporary Buffs. Different Buffs

behave differently when a robot's state changes; see the table below:

Table 5-26 Buff Source Types and Behavior

Type	Buff Characteristics	Included Buff Sources	Behavior	Notes
Field persistent Buff	A Buff type that requires the robot to continuously activate the Buff Point and has no concept of duration	- Supply Zone Buff • High Point Buff • Base Buff • Fortress Buff • Outpost Buff • Assembly Zone Buff • Central Highland Buff	Removed immediately once the robot no longer activates the corresponding Buff Point	-
Field temporary Buff	A Buff type that does not require the robot to continuously activate a Buff Point and can be triggered once certain conditions are met; it has a concept of duration	Terrain Crossing Buff	Removed when the robot is Defeated or the duration ends	-
Non-field temporary Buff	A Buff type not obtained through Field Interaction Module cards	- Power Rune Buff • Buffs from special robot mechanisms (Sentry posture Buff, Heavy Robot defense Buff, Radar vulnerability) • Defense Buff from Assembly Level	Not removed when the robot is Defeated	As long as the Buff source exists, all robots/field props to which the Buff applies have the Buff. For example, an Outpost rebuilt due to a control change immediately gains the owner's current Power Rune Buff

Example 1: If a Red robot has a 200% Attack Buff, a Blue robot has a 25% Defense Buff, and the Blue robot's marking progress by the opposing Radar is greater than 100, then when Red fires one 42mm projectile that hits the Blue robot, the damage taken by the Blue robot is:

Example 2: If a Red robot has a 200% Attack Buff, and the Blue Heavy Robot's marking progress by the opposing Radar is greater than 100, then when Red fires one 42mm projectile that hits the Blue Heavy Robot, the damage it takes is: 200 × 200% × (1 + 15%) = 460.

When a robot obtains more than one Buff of the same type, the strongest effect applies. This includes Attack, Defense, Vulnerability, HP Recovery and Shooter Heat cooling Buffs.

Example 1: A Red Infantry Robot gains a 25% Defense Buff by activating the Large Power Rune. If it then activates its own Base Buff Point, it has a 50% Defense Buff. If it subsequently gains 15% Vulnerability, it has a 35% Defense Buff.

Example 2: A Blue Infantry Robot gains a 25% Defense Buff by activating the Large Power Rune. If it then activates the opposing Fortress Buff Point, it has 75% Vulnerability. If it subsequently gains another 15% Vulnerability, it still has 75% Vulnerability.

If the final damage or HP Recovery value after applying Buffs is not an integer, round it to the nearest integer.

5.6.4.2 Base Buff Point Mechanism

Ground Robots on either side may activate a Base Buff Point. Multiple robots on the same side may activate a Base Buff Point simultaneously.

During the seven-minute Round, a robot that activates its own Base Buff Point gains a 50% Defense Buff, may exchange for Projectile Allowance and has the "Weakened" state removed. For "Weakened," see "5.3.2 Revival Mechanism".

5.6.4.3 High Point Buff Point Mechanism

All robots on a side may activate that side's High Point Buff Point. Multiple robots on the same side may activate it simultaneously.

When a side's robot activates its High Point Buff Point, that robot gains a 50% Defense Buff and may exchange for Projectile Allowance non-remotely.

5.6.4.4 Central Highland Buff Point Mechanism

Heavy Robots, Infantry Robots and Sentry Robots may activate the Central Highland Buff Point. Activation of separate, unconnected Central Highland Buff Points is independent, and their Field Interaction Module card coverage matches the higher parts of the two Terrain Crossing Buff Points (Highland). Multiple robots on the same side may activate the Central Highland Buff Point simultaneously. If one side's robot activates it, the opposing side's robots cannot activate it at the same time.

A robot that activates the Central Highland Buff Point gains a 25% Defense Buff.

5.6.4.5 Terrain Crossing Buff Point Mechanism

There are two Terrain Crossing Buff Points (Highland) on the Central Highland, two Terrain Crossing Buff Points (Road) in each side's Road Zone, and nine Terrain Crossing Buff Points (Tunnel).

All Ground Robots may activate Terrain Crossing Buff Points. Multiple robots on the same side, as well as robots from both sides, may activate them simultaneously.

Within X seconds, the same robot must detect the Field Interaction Module card at every Terrain Crossing Buff Point of the same type on one side, and it must not detect any other Field Interaction Module card along the way.

For Terrain Crossing Buffs (Road/Highland), the robot must detect the lower Buff Point before the higher one. Terrain Crossing Buff Points (Tunnel) have two independent Buff forms, depending on the type and order of Field Interaction Module cards detected when passing through the Tunnel. X is 3 for Terrain Crossing Buff Points (Road) and (Tunnel), and 5 for Terrain Crossing Buff Points (Highland). The Buffs are as follows:

Terrain Crossing Buff (Highland)

- 25% Defense Buff lasting 30 seconds.

Terrain Crossing Buff (Road)

- 25% Defense Buff lasting 5 seconds.
- A robot cannot obtain the Terrain Crossing Buff (Road) again within 15 seconds after obtaining it.

The two types of Terrain Crossing Buff (Tunnel) are:

Table 5-27 Tunnel Buff Trigger Logic and Effects

Buff Type	Field Interaction Module Card Detection Type and Order	Buff Effect
Terrain Crossing Buff (Tunnel Crossover)	Only the Field Interaction Module cards at one end and then the other end of the tunnel are detected, in order	25% defense Buff, lasting 30 seconds
Terrain Crossing Buff (Tunnel Passthrough)	The Field Interaction Module cards at one end, the inside, and then the other end of the tunnel are detected, in order	1.5x Barrel Cooling Buff, lasting 120 seconds

If a robot holding a Terrain Crossing Buff (Tunnel Crossover, Tunnel Passthrough, Highland or Road) obtains a different Terrain Crossing Buff, it gains a 50% Defense Buff. The remaining duration of this Defense Buff is updated to the greater of the remaining duration of the held Buff and the initial duration of the newly obtained Buff.

Terrain Crossing Buffs are cleared when a robot is Defeated.

5.6.4.6 Outpost Buff Point Mechanism

All Ground Robots may activate the Outpost Buff Point, and multiple robots from either side may activate it at the same time.

At an Outpost Position fully controlled by its side, a friendly robot that activates the corresponding Outpost Buff Point may exchange for Projectile Allowance, recover HP, and remove the “Weakened” state and its corresponding Invincible state. See “5.3 HP Recovery and Revival Mechanisms”.

5.6.4.7 Assembly Zone Buff Point Mechanism

Only the friendly Heavy Robot may activate the Assembly Zone Buff Point.

A Heavy Robot that activates the Assembly Zone Buff Point and is in the Assembly process is Invincible and has its launching mechanism powered off. Invincibility lasts for a maximum cumulative 45 seconds (reactivating after leaving the Buff zone does not reset the timer). Each time the highest successfully completed Assembly Level increases, the cumulative timer resets to 0 seconds.

5.6.4.8 Supply Zone Buff Point Mechanism

All Ground Robots may activate their side's Supply Zone Buff Point. Multiple robots on the same side may activate it simultaneously.

A robot that activates its side's Supply Zone Buff Point may exchange for Projectile Allowance non-remotely and also gain faster Revival progress or an HP Recovery Buff. For the specific forms and values, see "5.3 HP Recovery and Revival Mechanisms".

After the Supply Zone Buff Point is activated, the Heavy Robot is Invincible.

5.6.4.9 Fortress Buff Point Mechanism

Heavy Robots, Infantry Robots and Sentry Robots may activate the Fortress Buff Point. Both sides may activate it simultaneously. At any one time, only one friendly robot may activate it, and only the first robot to activate it gains the Buff; multiple opposing robots may activate it simultaneously.

The Fortress Buff Point takes effect when the friendly side does not control its Outpost Position.

Friendly activation effect

Define Δ as the difference between the friendly Base's Max HP and its lowest HP during the current round.

A friendly robot that activates the Fortress Buff Point gains a 50% Defense Buff and a Shooter Heat cooling Buff of value w. The relationship between w and Δ is: $w = \Delta / 80$ (rounded down), with a maximum w of 75. This Shooter Heat cooling Buff takes the highest effect among cooling Buffs from different sources. effect applies.

Example:

A robot with a Shooter Heat cooling rate of 40 per second simultaneously gains a Fortress Buff worth 75 cooling per second and a 5× Shooter Heat cooling Buff. Since $40 \times 5 > 40 + 75$, its actual Shooter Heat cooling rate is 200 per second.

After a robot activates the Fortress, the Fortress Buff zone provides it with "Reserve Projectile Allowance," independent of its own Projectile Allowance, up to a limit of N. When firing, a robot that activated the Fortress Buff zone consumes Reserve Projectile Allowance first and then its own Projectile Allowance. Each 17mm projectile fired by an Infantry or Sentry Robot consumes 1 unit of Reserve Projectile Allowance; each 42mm projectile fired by a Heavy Robot consumes 10 units.

$N=100+\Delta / 15$ (rounded down), with N capped at 500.

Opposing-side activation effect

After 3 minutes of the match have elapsed and the opposing Outpost Position is controlled by the friendly side, Ground Robots may activate the opposing Fortress Buff Point.

A robot activating the opposing Fortress Buff Point gains 100% Vulnerability. When one side's robot has activated it for a total of 10 seconds, the opposing Base Armor is deployed. The activation timer may be retained for 3 seconds after the robot dies or loses its activated state; the timer pauses while retained. After the opposing Base Armor is deployed, the robot activating the opposing Fortress Buff Point no longer gains Vulnerability.

5.7 Robot Special Mechanisms

5.7.1 Heavy Robot Special Mechanisms

During the first 3 minutes of the match, and before the modular 42mm launching mechanism is successfully installed, the Heavy Robot has a 50% Defense Buff.

Installing the modular 42mm launching mechanism

After meeting either of the following conditions, the Heavy Robot unlocks permission to use the launching mechanism and may go to the Workbench in the Supply Zone to install the modular 42mm launching mechanism. After it successfully fires one 42mm projectile detected by the Referee System, it is considered to have “successfully installed the modular 42mm launching mechanism.”

The Heavy Robot may perform the following operations, but only at the Workbench:

Split into sub-robots or multiple subsystems connected by flexible cables, or merge back together from those states.

Install or remove the modular 42mm launching mechanism.

- This 42mm projectile does not count toward Shooter Heat, muzzle velocity or damage.
- Before permission to use the modular 42mm launching mechanism is unlocked, the Heavy Robot’s launching mechanism is not powered on.

Ways to unlock permission to use the modular 42mm launching mechanism:

- Complete Assembly Level 3.
- After (300 − ~~M~~/50 − 2N/3) seconds of the match have elapsed. M is the maximum value during the current round of the difference between the opposing Base’s HP and the friendly Base’s HP; N is the maximum value during the current round of the difference between the opposing side’s control degree and the friendly side’s control degree.

  $$\frac{M}{}$$

  the opposing side’s control degree and the friendly side’s control degree.

The Heavy Robot’s attributes differ by match stage. See “Table 5-16 Heavy Robot Overall Attributes”.

5.7.2 Sentry Robot Special Mechanisms

After the three-minute Setup Period begins and before the 15-second Referee System Self-Check Period, the Commander may select automatic or semi-automatic control for the corresponding Sentry Robot. If no mode is selected within the specified time, the default after the seven-minute Round begins is automatic control. A Sentry Robot may independently exchange for Projectile Allowance, remotely exchange for Projectile Allowance or HP, confirm revival, exchange for instant revival, and switch posture by sending messages to the Referee System server.

In addition, the Commander may intervene in the Sentry Robot’s actions through commands on the Referee System Operator-Side Client, including:

- Forward minimap marks
- Purchase Projectile Allowance in the Supply Zone or at a Base or Outpost Buff Point
- Confirm revival
- Remotely exchange for Projectile Allowance

- Remotely exchange for HP
- Remotely exchange for instant revival
- Switch posture

For a semi-automatic Sentry, these actions do not cost Coins. For an automatic Sentry, each action by the Commander costs 50 Coins.

The Sentry Robot has a “posture” system that strengthens some abilities while weakening others. It starts in the Movement posture by default.

Posture switching has a 5-second cooldown. In each round, if a Sentry Robot spends more than 3 minutes cumulatively in one posture, that posture’s effect is reduced. A Sentry may enter an enhanced posture to strengthen its abilities further, but each enhanced posture may be entered for at most 15 seconds cumulatively. Timers for enhanced postures are independent; once the limit is exceeded, that enhanced posture cannot be entered again. The details are shown below:

Table 5-28 Sentry Robot Posture Effects

Posture	Enhanced posture effect	Default effect	Reduced effect
Attack posture	Current Shooter Heat is always 0; firing projectiles does not increase Shooter Heat. Does not increase Shooter Heat; the chassis power limit is reduced to 1/2 of the original; gains 25% Vulnerability	Gains a 3× Shooter Heat cooling Buff; the chassis power limit is reduced to 1/2 of the original and gains Gains 25% Vulnerability	Gains a 2× Shooter Heat cooling Buff; The chassis power limit is reduced to 1/2 of the original to 1/2 of the original; gains 25% Vulnerability
Defense posture	Gains a 99% Defense Buff; chassis power limit is reduced to 1/2 of the original; Shooter Heat cooling rate is reduced to 1/3 of the original	Gains a 50% Defense Buff; chassis power limit is reduced to 1/2 of the original; Shooter Heat cooling rate is reduced to 1/3 of the original	Gains a 25% Defense Buff; chassis limit is reduced to 1/2 of the original; Shooter Heat cooling rate is reduced to 1/3 of the original
Movement posture	Chassis power limit is increased to 200 W; gains 25% Vulnerability; Shooter Heat cooling rate is reduced to 1/3 of the original	Gains a 1.5× Chassis Power Buff; gains gains 25% Vulnerability; Shooter Heat cooling rate is reduced to 1/3 of the original	Gains a 1.2× Chassis Power Buff; gains 25% Vulnerability; Shooter Heat cooling rate is reduced to 1/3 of the original

- All the above calculations use integer values.
- When an effect reduces Shooter Heat cooling or chassis power, it is multiplied by the value after the Sentry Robot’s Shooter Heat cooling Buffs and Chassis Power Buffs have been calculated.

Example 1: An automatic Sentry in the default Defense posture has a base Shooter Heat cooling rate of 30 per second. It then activates the Large Power Rune

and gains a 3× Shooter Heat cooling Buff, then gains a Shooter Heat cooling Buff of 75 per second by activating the Fortress Buff Point.

Buff. First take the larger of these two Buffs (that is, +75 per second), giving the automatic Sentry a Shooter Heat cooling rate of 30 + 75 = 105,

Then multiply by the cooling-rate attenuation coefficient for Defense posture (1/3), giving the automatic Sentry Robot an actual Shooter Heat cooling rate of

$105 \times 1/3 = 35.$

Example 2: In the current round, the Sentry Robot has cumulatively been in Movement Posture for 2 minutes 50 seconds and has not entered another posture. Thereafter, if it switches to

Enhanced Movement Posture, then after 15 seconds it automatically switches back to Movement Posture, and the reduced Movement Posture effect applies.

5.7.3 Dart System Special Mechanisms

In each round, the Dart System can load 4 darts. The Dart Launching Station gate has 2 opportunities to open: 1 at 30 seconds after the match starts and 1 at 4 minutes after the match starts. The Commander may choose suitable times to use them, and unused opportunities accumulate.

Dart Launching Station gate opening and closing

During the match, the Commander can use the keyboard and mouse on the Referee System Client to open the launching station gate. While the gate is opening or closing, the Commander cannot control dart launching. The Referee System Client displays the opening and closing status of the gate.

Note: The gate takes about 7 seconds to open fully.

After the launching station gate is fully open, the Commander can launch darts by controlling the Dart System for 30 seconds. At the same time, the dart detection module of the opponent's Outpost or Base refreshes its detection window, which lasts 40 seconds. A launched dart must hit the dart detection module within the detection window; otherwise, the attack is invalid.

After the launching station closes its gate for the first time, the Dart Launching Station enters a 15-second cooldown. Only after the cooldown ends can the gate be opened a second time. During the cooldown, the Dart Launching Station gate cannot be opened.

Dart target selection

Before controlling the Dart Launching Station gate to open, the Commander can use the Referee System Client to change the target for this round of dart attacks to "Outpost", "Base fixed target", "Base random fixed target", "Base random moving target", or "Base terminal moving target". If the target is selected as "Outpost", then during the same gate opening, after one dart hits, the target can be switched only once from "Outpost" to one of the several Base targets.

- If "Fixed Target" is selected, the dart detection module remains at its initial position.
- If "Random Fixed Target" is selected, when the gate opens, the Base dart detection module starts moving and reaches another random position relative to its initial position before the gate fully opens. It returns to its initial position after the detection window ends. If a dart hits during the detection window, it moves to a new position.
- If "Random Moving Target" is selected, when the gate opens, the Base dart detection module starts moving. After detecting a dart launch, it moves to a new position different from its position at launch, completing the move and coming to rest within 600ms. During the detection window, after a hit or 10 seconds after detecting the first launch, it starts moving again until it detects the next launch.

- If "terminal moving target" is selected, then when the Dart Launching Station gate opens, the Base's dart detection module starts moving. 1.2 seconds after detecting that any dart has been launched, it starts moving to a new position different from the position at the time of launch, completing the move and coming to rest within 600ms. During the detection window, after a dart hits, or 10 seconds after the first dart launch is detected, the dart detection module starts moving again, until the next dart launch is detected.

The dart detection module and the dart guide light keep the same relative position while moving, and always move in the same plane relative to the initial position. The movement range is [-280mm,280mm] relative to the initial position, in the direction parallel to the wide side of the field.

When the target is selected as "random fixed target", "random moving target", or "terminal moving target", a dart hit is valid only after the dart detection module has finished moving and come to rest. Information that the dart detection module has finished moving is displayed on the Referee System Client.

Dart hit rewards

The following are the rewards obtained after a dart hits its target, in addition to the damage it causes to the target itself:

- When a dart hits the opponent's Outpost or the Base "fixed target" or "random fixed target" for the first, second, third, and fourth time, all of the opponent's operator interfaces, robot positions on the mini-map, and custom UI are blocked for 10 seconds (first), 5 seconds (second), 3 seconds (third), and 2 seconds (fourth), respectively. When a dart hits the opponent's Base, one's own surviving Infantry Robots equally split 200 (fixed target) or 600 (random fixed target) EXP.

- When a dart hits the opponent's Base "random fixed target", the terrain traversal and Power Rune buffs currently held by the opponent temporarily become ineffective: the timers continue to run, but the effects are disabled. The duration of ineffectiveness is the same as the corresponding blocking duration above.

- When a dart hits the opponent's Base "random moving target" or "terminal moving target", all of the opponent's operator interfaces, robot positions on the mini-map, and custom UI are blocked for 10 seconds; one's own surviving Infantry Robots equally split 2500 EXP; and the terrain traversal and Power Rune buffs currently held by the opponent become ineffective.

- If the Base's dart detection module is hit after selecting "random moving target" or "terminal moving target", each of the opponent's surviving Ground Robots immediately loses HP equal to 20% (random moving target) or 50% (terminal moving target) of its current maximum HP. This is counted toward the opponent team's total damage, but this HP deduction itself, and any robot deaths it causes, do not generate EXP. If hits occur consecutively, the blocking durations are added together.
- If all 4 darts hit the Base after selecting "fixed target" or "random fixed target", or any dart hits the Base after selecting "random moving target" or "terminal moving target", the opponent's Base armor immediately deploys.
- If a dart hits the Base or Outpost, its corresponding Buff Point temporarily becomes ineffective for 30 seconds; consecutive hits refresh the duration. After each dart hit, the detection window of the corresponding target closes for 2 seconds.

- While operator interfaces, robot positions on the mini-map, and custom UI are blocked, operators can choose to control robots through semi-automatic control. Specifically, the Referee System Client displays a large map interface similar to the one available to the Commander, and operators can click on the large map to send commands to control robots. The interval between two commands must be no less than 3 seconds.

- The custom terminal can still receive data such as video transmission normally.

5.7.4 Radar System Special Mechanisms

The functions of the Radar System include radar recognition and radar decoding of information waves, as follows:

Radar recognition

The Radar can recognize the positions of both sides' Ground Robots and send the robots' coordinates to the Referee System server. Each Ground Robot has a "marked progress" P, with a value range from 0 (initial and minimum value) to 150 (maximum value). The Referee System updates P in real time according to the accuracy of the Radar's recognition: consecutive accurate marking makes P rise at an accelerating rate; conversely, consecutive erroneous marking or data interruption makes P decay at an accelerating rate. When P reaches certain thresholds, the corresponding target receives certain effects. In addition, by continuously marking to maintain the opponent's vulnerable state, the Radar can also unlock the "double vulnerability" effect.

Affected by the wireless environment at the match venue, or when the Positioning Module is covered over a large area, the coordinates detected by a robot's Positioning Module may, with small probability, exhibit intermittent or persistent drift.

Parameter settings:

- d: recognition error, that is, the straight-line distance between the "coordinates sent by the Radar" and the "robot's actual planar coordinates", divided into three levels:
  - Accurate: $d < 0.8m$
  - Semi-accurate: $0.8m \leq d < 1.6m$
  - Erroneous: $d \geq 1.6m$
- x: the influence value of the current marking (that is, the progress change amount), with an initial value of 0.
- Judgment frequency: determined by the Radar's sending frequency.

Calculation logic of progress P:

Each time the Referee System receives Radar data, or the data is interrupted, x is updated first, then P is updated.

Table 5-29 Calculation Logic of "Marked Progress" P

Marking state	Update logic of x	Update logic of P
Accurate	If the previous marking was accurate or semi-accurate, $x = x + 1.0$; otherwise $x = 1.0$	$P = P + x$. If the result is $P < 0$, P is set to 0; if the result is $P > 150$, P is set to 150.
Semi-accurate	If the previous marking was accurate or semi-accurate, $x = x + 0.5$; otherwise $x = 0.5$	
Erroneous	If the previous marking was erroneous/interrupted, $x = x - 0.8$; otherwise $x = -0.8$	
Data interruption	Each time the Referee System accumulates 0.5 seconds without receiving data, it is treated as triggering one "erroneous" judgment	

Example: If one side's Radar continuously sends the coordinates of the opponent's Heavy Robot to the Referee System server at a frequency of 5Hz with 100% accuracy, then within 0 to 1 seconds, at 0.2-second intervals, the opponent Heavy Robot's marked progress is 1, 3, 6, 10, 15 respectively, and at second 3.0 the robot's marked progress reaches 120. After that, if the Radar sends the robot's coordinates to the Referee System server at a frequency of 5Hz with 0% accuracy, then within 3.0 to 4.0 seconds, at 0.2-second intervals, the robot's marked progress is 119.2,

117.6, 115.2, 112, 108. After that, if the Radar does not send the Heavy Robot's coordinates for 1.0 consecutive seconds, then within 4.0 to 5.0 seconds the robot's marked progress will be 103.2, 97.6 (judged as two additional consecutive errors on top of the previous errors). After that, if the Radar sends the Heavy Robot's correct coordinates twice consecutively, the robot's marked progress will be 98.6, 100.6.

Thresholds of P and corresponding effects:

Table 5-30 Thresholds of P and Effects Obtained by the Corresponding Target

Target type	Value range of P	Mini-map display effect	Buff (Ground Robots only)
Opponent robot	[0,100)	The position corresponding to the robot coordinates sent by one's own Radar	No additional effect
	[100,120)	- The actual position detected by the robot's Positioning Module - Special marker	15% vulnerability
	[120,150]	- The actual position detected by the robot's Positioning Module - Special marker	20% vulnerability
Own robot	[0,50)	The position corresponding to the robot coordinates sent by one's own Radar	No additional effect
	[50,150]	- The actual position detected by the robot's Positioning Module - Special marker	No additional effect

After a robot is destroyed and Revived, its P value is retained, but its current x value is reset to 0.

Double vulnerability mechanism: Each time the Radar accumulates 1 minute of making opponent robots vulnerable (when multiple robots are vulnerable at the same time, the time is not added up), it gains 1 opportunity to trigger "double vulnerability". The Radar can consume the opportunity by actively sending a command through the Referee System, which doubles all current vulnerability values originating from the Radar for 30 seconds. In each round, the Radar can trigger "double vulnerability" at most 2 times.

Radar decoding of electromagnetic waves

During the match, an official transmitting source will be placed on each side's Radar Base. This source transmits electromagnetic waves, including an information wave carrying the opposing team's information and an interference wave of a similar frequency band. The red side's broadcast source/interference source is placed on the red side's Radar Base and contains information about the blue team, and vice versa. Participating teams can build and install an antenna and a demodulation circuit on the Radar to decode and recognize the information wave and the interference wave, and to provide their own side with information and advantages.

The interference wave has three difficulty levels. Decoding difficulty rises with the difficulty level. Specific parameters will be released later.

The information wave will contain the following information about the opponent:

- Positions of each robot
- HP of each robot
- Remaining projectile allowance of each robot
- Buffs currently held by each robot
- Team's remaining Economy
- Team's total Economy
- Activation status of each of the opponent's Buff Points
- Armor hit information

For the specific communication protocol structure, see: "RoboMaster 2027 University Series Communication Protocol".

The interference wave contains no specific information, but contains a 6-character (alphanumeric combination) key customized by the opponent. The key is reset to a random value at the start of the round and whenever the difficulty of the corresponding side's interference wave changes.

At the start of a round, both sides' interference waves default to difficulty level 1.

The Radar can decode this key and send it to the event engine server. If decoding succeeds, the opponent's interference wave decoding difficulty subsequently rises by 1 level, that is, the difficulty of the interference source placed on one's own Radar Base rises by 1 level. When the opponent's interference wave difficulty is higher than one's own, the opponent's Radar will not produce vulnerability or accumulate "double vulnerability" progress against one's own robots, even if it marks them accurately.

5.8 Competition Format and Victory Conditions

The official matches of RMUC 2027 are divided into two stages: the group stage and the knockout stage. The group stage format is BO2 or BO3, and the knockout stage format is BO3 or BO5.

The outcome of a single round is determined item by item in the priority order of the following numbered list, until a final result is obtained:

1. When one side's Base is destroyed, the side with the higher remaining Base HP wins.
2. When the round time runs out, if neither side's Base is destroyed, the side with the higher cumulative control degree wins.
3. When the round time runs out, if neither side's Base is destroyed and the cumulative control degrees are equal, the side with the higher team attack damage wins.
4. When the round time runs out, if neither side's Base is destroyed, the cumulative control degrees are equal, and the team attack damage is equal, the side with the higher total team remaining HP wins.

5. If the winner cannot be determined under the above conditions, the round is considered a draw. In BO3 and BO5 matches, a draw immediately results in an additional round, until a winner is determined.

6 Match Procedure

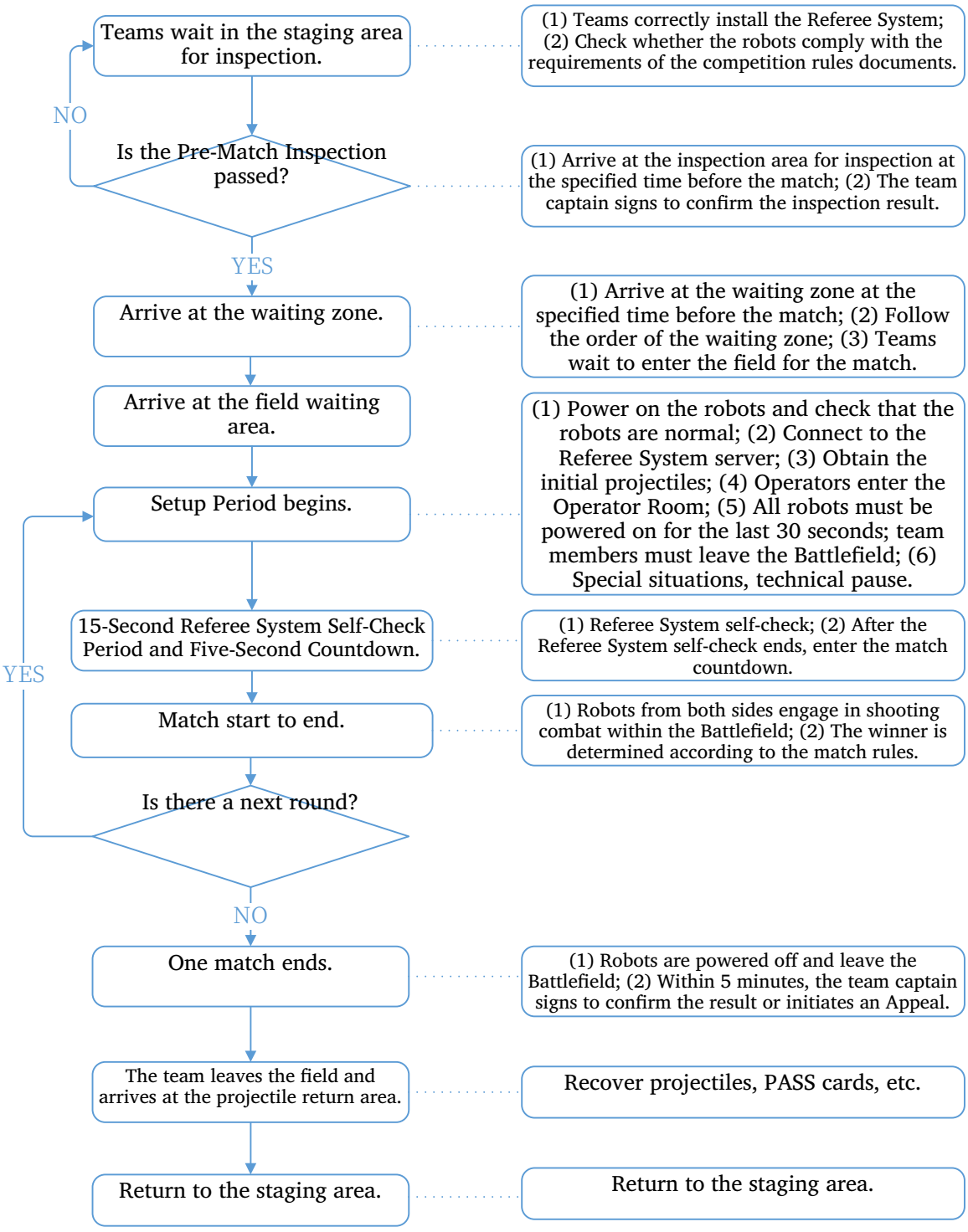


Figure 6-1 Single Match Flowchart

6.1 Pre-Match Inspection

- The inspection results of the pre-inspection and adaptive training are for reference only and are not the basis for passing the Pre-Match Inspection of official matches.
- The inspection result of an official match is valid only for that match.
- Passing the inspection only proves that the robot met the inspection pass criteria at the time of inspection. Teams must themselves ensure that their robots always meet the requirements of the Robot Building Specifications Manual. During inspection, all functions of the robot must be proactively demonstrated to the inspectors.

To ensure that the robots built by teams comply with the requirements of the "RoboMaster 2027 University Series Robot Building Specifications Manual",
teams are advised to arrive at the inspection area 60 minutes before the start of each match. If a team has not started inspection 55 minutes before the start of each match, the team's eligibility to field spare robots is cancelled. If a team has not started inspection 50 minutes before the start of each match, the eligibility of 1 Ground Robot to take the field is cancelled. Robots that have not arrived at the inspection area 45 minutes before the start of the match are stopped from entering the inspection area. The inspection procedure is as follows:

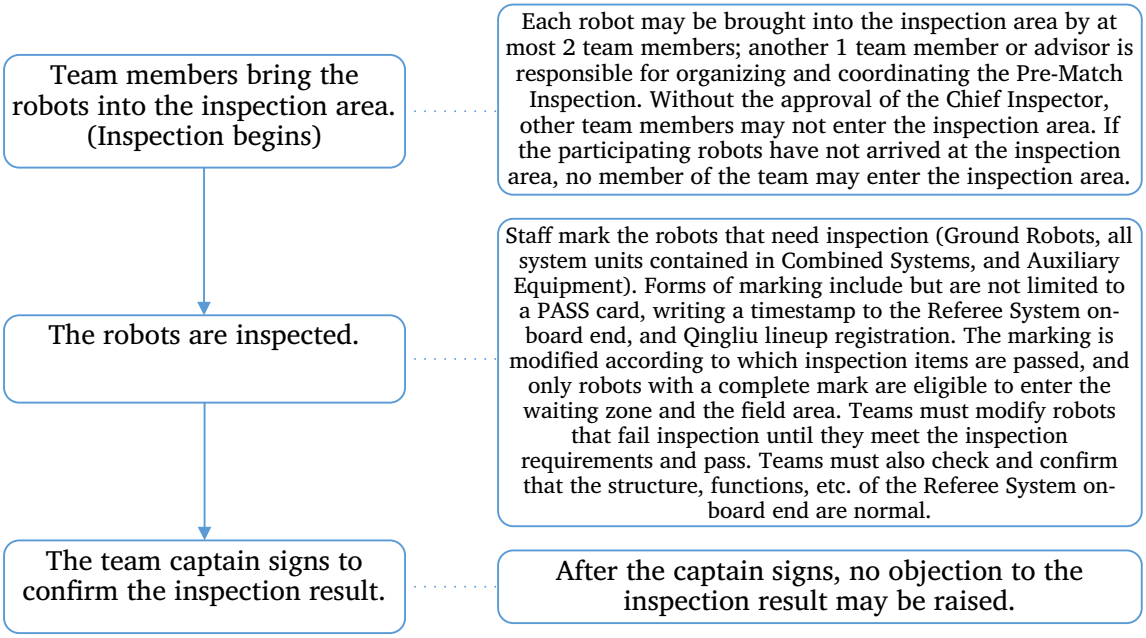


Figure 6-2 Pre-Match Inspection Flowchart

Spare robot regulations are as follows:

- At pre-inspection, each team must bring all participating equipment (including spares) for registration and pre-inspection. In subsequent matches within the same event stage, both the team's robots and spare robots must be on the registration list, and no replacement may be made within the same event stage. Registered robots are marked on the main body structure with a unique special identifier (marking methods include but are not limited to tamper-evident labels and laser marking; the specific form is subject to the on-site arrangement), and the identifier is checked in subsequent match inspections.
  If maintenance of a robot during the competition affects its unique special identifier, the relevant change information must be submitted through the designated channel.

  After the Revival Tournament ends, the robot list may be re-registered. The specific schedule is subject to the latest notice from the RMOC.

- For each match, each team may bring a specified number of main robots and spare robots.
- Each team may borrow the Referee System of at most 2 spare robots.

6.2 Waiting Zone

Teams must arrive at the waiting zone at least 10 minutes before the start of each match. The staff in the waiting zone will check the participating robots' PASS cards and the on-site personnel information.

The limits on on-site personnel for each team are as follows:

- 1 advisor.
- 2 team mentors.
- 20 official team members.

One of the on-site personnel must wear the captain-designated mark and perform the captain's duties. After a team enters the waiting zone, if a robot needs repair, it needs the Referee's permission before the robot can leave the waiting zone for repair. After repair is completed, the robot must return to the inspection area for inspection and pass the Pre-Match Inspection again before returning to the waiting zone. If the resulting delay causes the team to fail to arrive at the waiting zone on time, the robot cannot take part in the match, and the team bears the consequences.

Captain-designated mark: any on-site person wearing the captain-designated mark performs the captain's duties during the match. The captain must oversee the team's match procedure, confirm results, file Appeals, etc.

After the team is sent from the waiting zone to the field, it enters the field's waiting area and places its robots. After the Referee's approval, the team brings its robots to the Battlefield entrance and stands by. According to the match procedure, the Referee opens the door and guides the team members in. The countdown of the three-minute Setup Period starts at the same time the door opens.

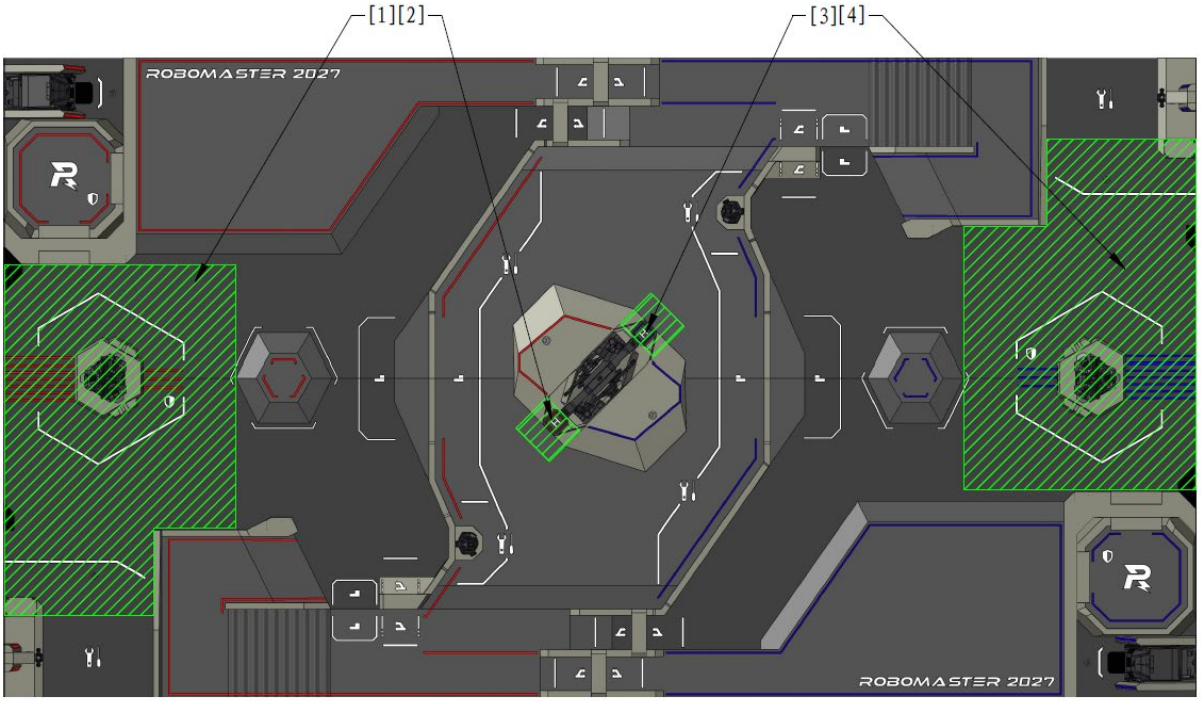
6.3 Three-Minute Setup Period

- After the second and fourth rounds of a BO5 match end, both teams have 10 minutes to debug their robots.

  Immediately after the 10 minutes end, the three-minute Setup Period of the next round begins.

- When a team arrives at the field, it may collect at the designated location all the 42mm light-emitting projectiles needed for that match. For the limit on the quantity of projectiles collected, see "4.5.2 Projectiles"; for example, in a BO3 match, 300 42mm light-emitting projectiles can be collected at one time.
- After the previous match ends and the Radar has been withdrawn from the Radar Base, the team may deploy the Radar on the Radar Base in advance.

During the three-minute Setup Period, on-site personnel must place the robots in their respective starting areas, check that the robots' Referee Systems operate normally, pre-load projectiles into the robots, debug the modular 42mm launching mechanism, load darts into the Dart System, deploy the Radar on the Radar Base, and install the aerial system on the Hangar platform. On-site personnel may repair robots or replace their equivalent parts, but must meet the requirements of the specification documents. On-site personnel must debug the robots near their own robots' starting areas. Each team is allowed only 2 on-site personnel to leave the Base and the vicinity of its starting area to assist with debugging, and they may not cross the Central Highland. The starting areas are shown in the figure below:



[1] Red Base and the vicinity of its starting area

[2] Red Hangar platform and the vicinity of its starting area

[3] Blue Hangar platform and the vicinity of its starting area

[4] Blue Base and the vicinity of its starting area

Figure 6-3 Setup Period Debugging Area Diagram

- Equivalent parts: standard modules or parts of the same material, form, and function, such as motors of the same model or self-made friction wheel modules. The main structures carrying the robot body cannot be replaced as equivalent parts, such as a robot's chassis or an Aerial Robot's airframe.

- If an "equivalent part" contains a Referee System module, the team must itself ensure that the Referee System on-board end module can be used normally (including version number, sensor calibration, installation standards, etc.). When the Referee System on-board end module in an "equivalent part" is abnormal, the team bears the consequences, and no official technical pause will be triggered.

When 1 minute 30 seconds remain in the three-minute Setup Period, operators are advised to enter the Operator Room to debug the keyboard and mouse. Check and confirm that robot control and the official equipment operate normally. If the official equipment cannot operate normally, on-site personnel must raise it before 15 seconds remain in the three-minute Setup Period, otherwise the Referee will not grant a technical pause.

Otherwise the Referee will not grant a technical pause. During the three-minute Setup Period, in the Operator Room, besides the operators and Commander corresponding to the robots taking the field, at most 1 additional person is allowed to enter for pre-match tactical deployment. Before the three-minute Setup Period ends, anyone other than the operators and Commander corresponding to the robots taking the field must leave the Operator Room.

When 30 seconds remain in the three-minute Setup Period, all robots in the Battlefield must be powered on, and personnel in the Battlefield leave in an orderly manner. Robots that are still not powered on when 15 seconds remain in the three-minute Setup Period must be moved out of the Battlefield. After the three-minute Setup Period ends, on-site personnel must place the Sentry Robot's and the Radar's Remote Controllers in the designated area at the Battlefield entrance or the Operator Room entrance.

6.3.1 Official Technical Pause

During the three-minute Setup Period, when the Referee System or official equipment malfunctions, or a robot needs temporary inspection (the malfunction situations are shown in the table below), the Head Referee may initiate an official technical pause and pause the countdown. The time at which the pause is initiated is decided by the Head Referee according to the situation.

During an official technical pause, on-site personnel must cooperate with the Referee's requirements to eliminate faults related to the Referee System or official equipment, and may not repair other faults.

If the Referee System malfunctions and needs replacement, the time limits for replacing the Referee System on-board end in the "RoboMaster 2027 University Series Robot Building Specifications Manual" must be followed.

After the faults related to the Referee System or official equipment are eliminated and the Head Referee resumes the countdown, on-site personnel must follow the standards of the three-minute Setup Period and leave the Battlefield at the specified time points.

Table 6-1 Malfunction Situations

Item	Description
1	Official equipment in the Operator Room malfunctions, or key match props in the Battlefield suffer structural damage or functional abnormality.
2	During the three-minute Setup Period of the first round, a Referee System on-board end module malfunctions, for example: an Armor Module is damaged, or a Speed Monitor Module is offline.
3	During the three-minute Setup Period, the Referee System main controller cannot connect to the server, or the robot cannot transmit images back to the Operator Room normally.
4	Other situations that the Head Referee judges to require an official technical pause.

Because it cannot be determined whether a malfunction is due to a fault in the Referee System module itself, a defect in the participating robot's circuit or structural design, or a result of robot combat in earlier matches, except for the situation in item 3 above, ordinary battle damage does not trigger an official technical pause, and the Referee will provide a spare Referee System module. The team may apply for a "team technical pause" to repair the robot.

If, after investigation by the Referee, the malfunction situations described in items 2 and 3 above are caused by the team, the Referee will explain the situation and end the official technical pause.

6.3.2 Team Technical Pause

If a robot's mechanical structure, software system, or the Custom Terminal and other equipment brought into the Operator Room malfunctions, the team may apply for a "team technical pause" only before the 15-second countdown of the three-minute Setup Period, with on-site personnel applying to the Referee in the Battlefield or Operator Room and stating the reason for the application.

Once a team technical pause application has been initiated and conveyed to the Head Referee, this technical pause cannot be withdrawn or modified.

After the Head Referee confirms a team technical pause, regardless of which side initiated it, the Referee will notify both teams at the same time. During the team technical pause, both teams may continue to debug and repair robots as normal.

After the Head Referee confirms that both teams are ready, the technical pause may be ended early. Even if a team did not enter the Battlefield or the technical pause was ended early, the opportunity consumed is still the one corresponding to the time declared when the team applied.

To ensure that the subsequent schedule proceeds on time, in the same three-minute Setup Period the two sides together may initiate only 1 team technical pause, on a first-come, first-served basis. The post-match result confirmation form records whether a technical pause opportunity was consumed during the match.

The team technical pause arrangements for different schedule stages within the same event stage are shown below:

Table 6-2 Team Technical Pause Arrangements

Event stage	Schedule stage	Arrangement
Regional Competition	Group stage	2 two-minute technical pauses.
Regional Competition	Knockout stage	1 three-minute technical pause. If the technical pause opportunities in the group stage are not used up, they can carry over to the knockout stage.

6.4 15-Second Referee System Self-Check Period

After the three-minute Setup Period ends, the match enters the 15-Second Referee System Self-Check Period. During the self-check, the Referee System server automatically checks the connection status of the Referee System Clients, the status of the robots' Referee System modules, and the status of the field props, and resets all robots' HP, ensuring that all robots are at full HP when the match starts.

6.5 Five-Second Referee System Self-Check Period

After the 15-Second Referee System Self-Check Period ends, the match enters the Five-Second Countdown. During this time, the Referee System Client does not respond to robot control commands (including the Custom Terminal). After the countdown ends, the Referee System Client resumes responding to robot control commands, and the match starts immediately.

6.6 Seven-Minute Round

In the seven-minute Round, the robots of the two teams engage in shooting combat in the core competition venue (the Battlefield).

6.7 Match End

A round ends when the round time runs out or one team triggers the Victory Conditions early. For the Victory Conditions, see "5.8 Competition Format and Victory Conditions". A match ends when the winner of the match is determined or all rounds are finished.

6.8 Result Confirmation

In a match, the Referee records on the result confirmation form the main penalty situations of each round, key settlement information at the end of the match, win/loss results, and the use of the team technical pause opportunity.

The team captain must sign to confirm the match result within 5 minutes after a match ends. If the captain neither signs to confirm the result nor files an Appeal within 5 minutes, the match result is deemed confirmed.

6.9 Leaving the Field

After a match ends, the team must power off its robots and move them out of the Battlefield, and go to the projectile return area to return projectiles. In the projectile return area, the team must follow the staff's guidance, proactively return the relevant materials, empty the projectiles loaded in the robots, and proactively return all projectiles used in the match.

7 Violations and Penalties

To ensure the fairness of the competition and strict discipline, teams, participants, and robots must strictly follow the match rules. In case of violations, the Referee will impose corresponding penalties. Some violation penalties issued before the official start of the match will be executed after the official start. Major penalties during the match and all Appeals will be publicly announced.

The penalties corresponding to all violation provisions in this chapter are determined by the Head Referee according to the actual situation of the match. If a situation arises during the match that affects the fairness of the match but is not covered by the detailed penalty rules and serious violations, it is determined by the Head Referee according to the actual situation.

If one side's behavior directly causes the other side to commit a violation, the other side is not considered to have violated the rules, but must terminate the violating behavior promptly.

7.1 Penalty System

7.1.1 Penalty Methods

During the event, the Referee System or Referees will impose penalties on participants and robots that do not comply with the event rules. The penalty methods are as follows:

Table 7-1 Penalty Methods

Penalty method	Description
Referee System automatic penalty	Everything in "5.1 HP Deduction and Exceeding-the-Limit Penalty Mechanism" other than attack damage or collision and Penalty Damage belongs to Referee System automatic penalties.
Referee System manual penalty	A penalty issued by a Referee operating the server against a robot's violation.
Referee manual penalty	Applies to scenarios where the Referee System cannot be used to impose a penalty, such as verbal warnings and disqualification from the match.

7.1.2 Penalty Types

The types of manual penalties during the match are shown in the table below:

Table 7-2 Penalty Types

Penalty type	Description
Verbal warning	Verbal caution
Yellow Card Warning	- One-side Yellow Card:

Penalty type	Description
	- If the violating robot is a Sentry Robot, its chassis is powered off for 2 seconds, and other robots' operation interfaces are blocked for 2 seconds. - If the violating robot is not a Sentry Robot, its operation interface is blocked for 5 seconds, and other robots' operation interfaces are blocked for 2 seconds. - The Referee System automatically deducts 15% of the violating robot's current Max HP, and the other surviving robots are deducted 5% of their current Max HP. If a robot receives another Yellow Card Warning within 30 seconds of each Yellow Card Warning, the percentage of current Max HP deducted is 2 times the previous one, and the other surviving robots are deducted 5% of their current Max HP. - In each round, when all robots have cumulatively received 3 Yellow Card Warnings, a Red Card Warning is received at the same time. - Both-side Yellow Card: All operators' operation interfaces are blocked for 2 seconds, and all robots are deducted 5% of Max HP, which does not count toward a single robot's cumulative Yellow Card count. - If Yellow Card Warnings are received consecutively, the interface blocking times are added together, and the 30-second timer is reset. - If a robot's remaining HP is less than or equal to the HP to be deducted by the penalty, the robot's HP is reduced to 1. - The durations of chassis power-off, interface blocking, etc. caused by violation penalties are independent of match mechanisms and do not stack with them. - When the interface is blocked due to a Yellow Card Warning, the video transmission feed, mini-map, and custom UI are all unavailable.
Red Card Warning (Ejection)	- Ejected robot: • If a robot is ejected before the 15-Second Referee System Self-Check Period, the violating robot is not allowed to take the field and must be moved out of the Battlefield, and no other robot may substitute for it in all rounds of that match. • If a robot is ejected during the 15-Second Referee System Self-Check Period, the Red Card Warning is issued after the match starts. • If the aerial system is ejected during the match, the Hangar is powered off. If the Aerial Robot is in flight, the safety officer must immediately take control of the Remote Controller and stop the propellers, and the Aerial Robot may not fly again. If the Dart System is ejected during the match, the dart launch button is immediately disabled and the Dart Launching Station gate cannot be opened;

Penalty type	Description
	- If the gate is already open, it is closed immediately. - If the Radar is ejected during the match, the Radar's multi-machine communication is disconnected (it can only receive one-way and can no longer send). - If a Ground Robot is ejected during the match, the robot's HP immediately becomes 0, its video transmission feed is disconnected, its multi-machine communication is disconnected, and it cannot be Revived. - Ejected Auxiliary Equipment: the corresponding equipment is not allowed to take the field. - Ejected on-site personnel: the Referee requires the ejected on-site personnel to leave the competition area immediately, and no other on-site personnel may substitute for them in all rounds of that match. If an operator is ejected, the robot they operate is ejected for that round and is not allowed to take the field in all rounds of that match, and no robot may substitute for it; if the Commander is ejected, both the aerial system and the Dart System are ejected and are not allowed to take the field in all rounds of that match, and no robot may substitute for them.
Loss by forfeit	- If a Loss by forfeit penalty is issued for a round (hereinafter "round forfeit"), the details are as follows: • If the forfeit penalty is issued before the seven-minute Round (including the three-minute Setup Period and the 15-Second Referee System Self-Check Period), the violating side's Base HP is 0 and control degree is 0, and its other robots have full HP. The other side's Base and robots have full HP, and its control degree is 1680. Both teams' total team damage is considered 0. The round time is considered 420s. • If the forfeit penalty is issued during the seven-minute Round, the round ends immediately. The violating side's Base HP is 0, its other robots' HP is as of the end of the match, and its control degree is the value at the end of the match. The other side's Base and robots' HP is as of the end of the match, and its control degree is the value at the end of the match. The round time is the actual match time elapsed when the forfeit was issued. • If the forfeit penalty is issued after the seven-minute Round ends, the violating side's Base HP is 0, its other robots' HP is as of the end of the match, and its control degree is the value at the end of the match. The other side's Base and robots' HP is as of the end of the match, and its control degree is the value at the end of the match. The round time is the actual match time elapsed when the forfeit was issued. • If a forfeit penalty is issued for a match (hereinafter "match forfeit"), that is, a forfeit penalty is issued for all rounds of that match, the HP settlement of each round follows the description above.
Disqualification from awards	- Participants are disqualified from awards. • Teams are disqualified from awards.

Penalty Type	Description
Disqualification	- Participants are disqualified from the competition and from awards. • Teams are disqualified from the competition and from awards. The team's results for this season are retained as a reference for the advancement of other teams.

7.2 Penalty Details

This section describes the detailed violation penalties. Items numbered R# are rules that explicitly state what participating teams, participants and robots must follow.

7.2.1 Personnel

7.2.1.1 General Requirements

R1 Participating teams must meet the requirements of the RoboMaster 2027 Super RoboMaster Championship Participant Manual.

Violation penalty: up to disqualification of the violating party.

R2 Participants and their actions must not interfere with the normal operation of official equipment or competition procedures, or the normal work of RMOC staff.

Violation penalty: up to disqualification of the violating party.

R3 Participating teams must not set up their own wireless devices or use walkie-talkies in competition-related areas (including but not limited to the preparation area, the Pre-Match Inspection area, the waiting area and the competition area).

Violation penalty: up to disqualification of the violating party.

R4 Participating teams must not damage official equipment (including but not limited to equipment located in the competition area, waiting area, preparation area and Pre-Match Inspection area).

Violation penalty: up to disqualification of the violating party, and the violating party must pay compensation at the equipment's price.

R5 Except for field personnel who enter the waiting area and competition area for the needs of the match, other participants must not enter the waiting area and competition area without special reason.

Violation penalty: up to disqualification of the violating personnel.

R6 Field personnel who have entered the waiting area or competition area for the needs of the match must not leave the waiting area or competition area, or change field personnel, without the Referee's consent.

Violation penalty: the violating personnel may not enter the waiting area and competition area for this match, up to disqualification of the violating personnel.

R7 Except for projectiles preloaded in the Pre-Match Inspection area, participating teams must not bring their own projectiles for use in the match into the waiting area or competition area.

Violation penalty: the projectiles are confiscated, up to disqualification of the violating personnel.

R8 After a match ends, field personnel must immediately power off the robots and move them out of the competition area, and empty the projectiles inside the robots in the projectile return area.

Violation penalty: the violating robot is held in the projectile return area until its projectiles are emptied.

R9 After a match ends, field personnel must return all projectiles used in the match in the projectile return area.

Violation penalty: the projectiles are confiscated and the violating personnel lose the right to enter the competition area in subsequent matches in the current regional competition. Up to disqualification of the violating personnel.

R10 Except in emergencies, participating teams must arrive at the Pre-Match Inspection area for Pre-Match Inspection at least 55 minutes before the start of each match, and must arrive at the waiting area and be ready at least 10 minutes before the start of each match.

Violation penalty: up to Loss by forfeit for the match.

R11 Without the Referee's permission, after a robot passes inspection and before the match ends, the robot must not be powered on for debugging or repaired outside the Battlefield.

Violation penalty: up to Loss by forfeit for the match.

R12 The identity and number of personnel each team sends into designated areas such as the preparation area, Pre-Match Inspection area, waiting area and competition area must meet the requirements.

Violation penalty: up to disqualification of the violating party.

R13 Field personnel must wear the corresponding badges, and the badges must not be covered. One person must wear the badge designated for the team captain.

Violation penalty: up to disqualification of the violating personnel.

R14 Without the Referee's permission, field personnel who have entered the competition area must not communicate with the outside in any way.

Violation penalty: up to disqualification of the violating personnel.

R15 Except for the Radar, field personnel must not use official equipment power supplies to power their own equipment in the competition area, but may bring their own power supplies.

Violation penalty: up to disqualification of the violating personnel.

R16 Except in special circumstances, field personnel are prohibited from entering the competition area wearing slippers.

Violation penalty: up to a Red Card Warning for the violating personnel.

R17 During an Appeal, the appealing party must provide valid evidence or the relevant provisions of the normative documents relied upon.

Violation penalty: up to disqualification of the violating personnel.

R18 The team captain has management responsibilities and must ensure the following during preparation and the competition:

- Participating robots must pass the review of complete finished robots and the open-source robot usage specifications.
- Team advisors, reserve-squad members and other persons not eligible for awards must not commit violations during the competition.
- Other violations must be stopped promptly.

Violation penalty: the captain's management responsibility is determined according to the violation, up to disqualification of the violating personnel.

7.2.1.2 Battlefield Requirements

R19 Field personnel must wear safety goggles when entering the Battlefield.

> Violation penalty: verbal warning, and the violating personnel are ejected from the Battlefield. If the warning is ineffective, a Red Card Warning is issued to the violating personnel.

R20 During an official technical pause, field personnel must not repair faults other than those of Referee System related modules.

> Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating personnel.

R21 After the three-minute Setup Period ends, field personnel must return to the designated area outside the Battlefield. During the match, field personnel must not leave this area without the Referee's permission.

> Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating personnel.

R22 After 30 seconds remain in the three-minute Setup Period, team members are prohibited from continuing to debug robots inside the competition area.

> Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating robot after the match starts.

R23 After the three-minute Setup Period ends, the Remote Controllers of the Sentry Robot and the Radar must be placed in the designated area at the Battlefield entrance or Operator Room entrance, and the Remote Controllers of the other robots must be placed inside the Operator Room or at the designated position at the Battlefield entrance.

> Violation penalty: if before the seven-minute Round, a verbal warning; if the warning is ineffective, a Red Card Warning is issued to the violating robot. If during the seven-minute Round, a Red Card Warning is issued to the violating robot.

R24 After the Five-Second Countdown begins, field personnel must not operate the Remote Controllers corresponding to the robots on the field that are located outside the Operator Room.

> Violation penalty: a Red Card Warning is issued to the violating robot, up to Loss by forfeit for the round.

R25 Field personnel must ensure that their own robots operate safely and do not cause harm to any personnel or equipment in the competition area.

> Violation penalty: the violating party bears the corresponding responsibility.

7.2.1.3 Operator Room Requirements

R26 During the three-minute Setup Period, in addition to the Operators and Commander corresponding to the robots on the field, at most 1 additional person is allowed in the Operator Room, and persons other than the Operators and Commander corresponding to the robots on the field must leave the Operator Room before the three-minute Setup Period ends.

> Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating personnel.

R27 Without the Referee's permission, during the 15-Second Referee System Self-Check Period and the seven-minute Round, Operators must be in the corresponding Operator Room, operate the control devices of the corresponding robot, wear the corresponding headset, and must not change position after the three-minute Setup Period ends.

> Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating personnel.

R28 An Operator has not arrived at the Operator Room before the Five-Second Countdown, or leaves the Operator Room during the seven-minute Round.

> Violation penalty: a Red Card Warning is issued to the violating personnel.

R29 During the seven-minute Round, the Commander may be configured with the Hangar Remote Controller, the Dart System Remote Controller and a Custom Terminal at the same time. Apart from this, each Operator must configure control devices and a Custom Terminal according to the control method requirements of the corresponding robot.

Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating personnel.

R30 During the match, the safety officer must watch the flight of the Aerial Robot and may operate the Remote Controller used for emergency propeller stop only when an urgent safety problem occurs (for example, the robot is about to fly out of the Battlefield).

Violation penalty: if the safety officer operates the emergency propeller stop Remote Controller, a Red Card Warning is issued to the Aerial System.

R31 Using your own headsets or computers (other than the Custom Terminal) in the Operator Room is prohibited.

Violation penalty: verbal warning. If the warning is ineffective, Loss by forfeit for the round.

R32 Each operating device in the Operator Room provides at most 1 RJ45 network port, and this port may only be connected to the Custom Terminal.

Violation penalty: verbal warning, and use of the violating device is prohibited. If the device is affected, up to disqualification of the violating party.

R33 During the match, communication with the outside of the Operator Room is not allowed.

Violation penalty: verbal warning. If the warning is ineffective, Loss by forfeit for the round.

7.2.2 Robots

7.2.2.1 General Requirements

R34 Robots and related equipment used in the match must pass Pre-Match Inspection.

Violation penalty: Loss by forfeit for the round.

R35 In the first round of a match, the robots must meet the minimum lineup for taking the field.

Violation penalty: Loss by forfeit for the match.

R36 Robots must meet the requirements of the RoboMaster 2027 University Series Robot Building Specifications Manual.

Violation penalty: up to disqualification of the violating party.

- The RMOC will conduct spot checks on robots from time to time.
- If a robot is reported as not complying with the Robot Building Specifications, the reporter must provide corresponding evidence.
- During the match, if the Positioning Module or Speed Monitor Module is not installed as required, the corresponding module is regarded as offline for that round.

R37 In case of a dispute, participating teams are responsible for showing the RMOC the robot's mechanical and circuit design drawings and related code files, and for answering related technical inquiries.

Violation penalty: up to disqualification of the violating party.

R38 While waiting in the waiting area, participants must not take robots out of the waiting area without permission.

Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating personnel and robot, up to disqualification of the violating personnel.

R39 Robots must not have or develop safety hazards including but not limited to short circuits, crashes, smoke, open flames, parts falling to the ground and gas cylinder explosions;

if a safety hazard exists or occurs, participants must cooperate with the Referee in carrying out the corresponding operations.

Violation penalty: before the Battlefield is cleared, field personnel must resolve the safety problem as the Referee requires, otherwise the violating robot may not take the field. If during the match, a verbal warning; if the warning is ineffective, a Red Card Warning is issued to the violating personnel or the violating robot. If the safety hazard is serious, the Head Referee handles it according to "8 Abnormal Situations".

R40 No robot may continuously launch projectiles out of the Battlefield.

Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating robot.

R41 The Dart System must not launch darts out of the Battlefield.

Violation penalty: a Red Card Warning is issued to the violating robot.

R42 From the three-minute Setup Period until the Battlefield is cleared, robots inside the Battlefield must not leave their corresponding debugging area. After the Battlefield is cleared, through the 15-Second Referee System Self-Check Period and the Five-Second Countdown, robots inside the Battlefield must not leave their corresponding starting area.

Violation penalty: up to a Red Card Warning for the violating robot.

R43 During the three-minute Setup Period, if projectiles need to be launched, they must be launched into the projectile clearing bag.

Violation penalty: up to a Red Card Warning for the violating robot.

R44 During the three-minute Setup Period, modules or parts replaced on a robot must meet the "equivalent part" requirements.

Violation penalty: up to a Red Card Warning for the violating robot.

7.2.2.2 Ground Robots

R45 During the match, no surviving robot may use its own structure to block any of its own Armor Modules, and no robot may block more than 1 Armor Module of other surviving robots on its own side. When a Heavy Robot grasps or carries Movable Props, the carried Movable Props and the related parts of its own structure are allowed to block 1 of its own Armor Modules, and the blocked Armor Module may be a different one each time, but blocking multiple Armor Modules at the same time is not allowed.

- Infantry Robots and Sentry Robots must not block their own Armor Modules while carrying Movable Props.
- If a blocked Armor Module goes offline, it is not considered a violation.

Violation penalty: if 1 Armor Module is blocked, a warning is issued to the violating party according to subjective intent: if the blocking is active, a Yellow Card Warning and a verbal warning are issued when the violation occurs, and if the warning is ineffective, a Red Card Warning is issued; if the blocking is passive, 1 Yellow Card Warning is issued when the violation occurs.

If the number of blocked Armor Modules is greater than 1, a Yellow Card Warning and a verbal warning are issued when the violation occurs, and if the warning is ineffective, a Red Card Warning is issued.

R46 The subsystems or individual units into which a Heavy Robot is separated must not leave the Workbench.

Violation penalty: up to a Red Card Warning for the violating robot.

7.2.2.3 Combined Systems

R47 During the three-minute Setup Period, field personnel may debug the Aerial System near the Hangar platform and its starting area, but must not start the propellers.

Violation penalty: up to a Red Card Warning for the violating personnel and the violating robot.

R48 During the match, the Aerial Robot must not fly out of the Battlefield.

Violation penalty: a Red Card Warning is issued to the violating robot. This Aerial System may not take the field in the other rounds of the same match.

R49 During the match, if the Aerial System develops a fault, or suffers battle damage because of an unreasonable design of the power system and power supply system, the system may continue to take the field only after passing the Referee's inspection and after the Head Referee confirms there are no safety hazards.

Violation penalty: a Red Card Warning is issued to the violating robot. The violating robot may not take the field in the other rounds of the same match.

For safety reasons, if the Aerial Robot flies unstably during the match, the Head Referee will eject the Aerial System. The safety officer should land the Aerial Robot and stop the propellers as soon as possible, and must not operate that Aerial Robot again.

R50 Except during the seven-minute Round, the Dart System must not be in the ready-to-launch state.

Ready-to-launch state: the energy storage element used to give the dart its initial kinetic energy is tensioned, inflated or rotating. Energy storage elements include but are not limited to rubber bands, cylinders and friction wheels.

Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating robot.

7.2.3 Interaction

7.2.3.1 Robot Interaction

R51 A robot of one side must not use any of its own structure to collide with a robot of the other side. If a Defeated robot blocks a key movement path, it may be pushed away slowly.

Violation penalty: a warning is issued to the violating robot according to subjective intent and the severity of the collision.

Table 7-3 Collision Violation Penalty Standards

Violation Level	Description
Yellow Card Warning	- Actively causing a head-on, fast collision • Actively pushing so that the other side's robot moves noticeably • Actively pushing the other side's robot so that its normal movement is obstructed
Red Card Warning	- Actively causing a head-on, fast, repeated or violent collision • Actively pushing so that the other side's robot moves a considerable distance • Actively pushing the other side's robot so that its normal movement is severely obstructed

If the collision has serious consequences for the other side's robot or affects the fairness of the match, in the next round of the same competition stage the violating robot (by robot number only) may not take the field, and no other robot may substitute for it.

R52 A robot of one side must not, through active interference, blocking, collision or similar behavior, cause any of its own structure to become fixedly attached to a robot of the other side (except robots that cannot be revived).

Violation penalty: timing starts from the moment the attachment can be determined, and warnings are issued to the violating robot according to the violation duration: the first Yellow Card Warning is issued when the violation exceeds 10 seconds, and then 1 Yellow Card Warning every 20 seconds until the violating robot is ejected. Whether or not the violating robot is alive, when the violation time exceeds 90 seconds, up to Loss by forfeit for the round is issued according to the subjective intent of the violating party's behavior.

R53 A robot of one side must not actively attack the other side's Dart launcher or Radar.

Violation penalty: up to disqualification of the violating party.

R54 A robot of one side, and its actions, must not block a robot of the other side from entering its Supply Zone.

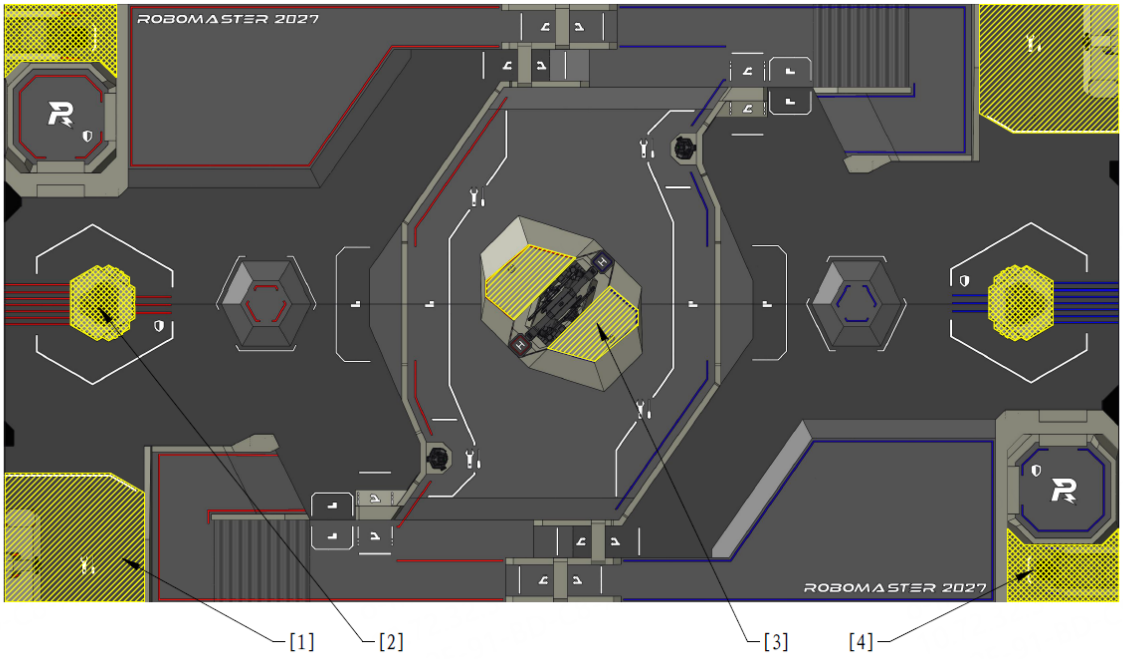
Violation penalty: a Yellow Card Warning is issued to the violating robot. If the warning is ineffective, a Red Card Warning is issued to the violating robot.

R55 Robots must not actively carry the other side's darts or the other side's Aerial Robot.

Violation penalty: verbal warning. If the warning is ineffective, a Red Card Warning is issued to the violating robot.

7.2.3.2 Robot Interaction with Field Props

Multiple Restricted Zones are set up in the Battlefield. A Restricted Zone covers the ground and the vertical airspace above it. If a robot body or anything it carries (such as Movable Props) touches or enters this space, it is judged to have entered the Restricted Zone. The extent of the Restricted Zones is shown in the figure below.



[1] Supply Restricted Zone
[2] Base Restricted Zone
[3] Assembly Restricted Zone
[4] Dart Launching Station Restricted Zone

Figure 7-1 Battlefield Restricted Zones

Robot interaction with Restricted Zones must follow the rules below, otherwise it is regarded as a violating entry into the Restricted Zone.

One-side Restricted Zone: only robots of the owning side may enter; all robots of the other side are prohibited from entering.

Shared Restricted Zone: all robots of both sides are prohibited from entering.

Table 7-4 Battlefield Restricted Zone Access Rules

Restricted Zone	Restricted Zone Attribute
Supply Restricted Zone	One-side Restricted Zone
Assembly Restricted Zone	One-side Restricted Zone
Base Restricted Zone	Shared Restricted Zone
Dart Launching Station Restricted Zone	Shared Restricted Zone

R56 Ground Robots must not enter the Base Restricted Zone in violation of the rules.

Violation penalty: warnings are issued to the violating robot according to the duration of stay and the impact of the violating behavior. The first Yellow Card Warning is issued when the violation exceeds 3 seconds, and then 1 Yellow Card Warning every 10 seconds until the violating robot is ejected.

R57 Ground Robots must not enter the Supply Restricted Zone or Assembly Restricted Zone in violation of the rules.

If a robot is Defeated or ejected inside any Restricted Zone, the Referee may, depending on the situation on site, set the robot to a temporarily activated state and guide its Operator to operate the robot out of the Restricted Zone.

Violation penalty: warnings are issued to the violating robot according to the duration of stay and the impact of the violating behavior. The first Yellow Card Warning is issued when the violation duration exceeds 3 seconds, and then 1 Yellow Card Warning every 5 seconds until the violating robot is ejected. If a robot that is Defeated or abnormally offline stays in a Restricted Zone for more than 20 seconds, up to Loss by forfeit for the round.

R58 Robots must not place Movable Props into their own Supply Restricted Zone, the Shared Base Restricted Zones, or the other side's Dart Launching Station.

Violation penalty: a Yellow Card Warning is issued to the violating robot. If the Movable Props subsequently have a decisive effect on projectile resupply, dart launching or dart hits, or affect the normal operation of any core prop, a Red Card Warning is issued to the violating robot.

R59 Wireless debuggers and Movable Props used in the match must not be brought into the competition area. During the match, robots may only use the official dedicated projectiles and Movable Props provided in the competition area by the RMOC.

Violation penalty: items brought into the competition area are confiscated; if the match is affected, up to disqualification of the violating party.

R60 Robots must not obtain, or carry or use in any way, an Energy Unit that has already been successfully assembled.

Violation penalty: Loss by forfeit for the round.

R61 During the match, robots must not damage the Battlefield, field props or projectiles, nor affect the normal function of field props.

Violation penalty: up to disqualification of the violating party.

R62 42mm projectiles, darts and Aerial Robots must not be used to hit the Power Rune and the Hangar.

Violation penalty: up to a Red Card Warning for the violating robot.

R63 During assembly, robots must not operate the movable structures of the Tech Core (other than the Energy Unit) or other body structures.

Violation penalty: Loss by forfeit for the round.

R64 Robots and the Aerial System must not enter the Dart Launching Station Restricted Zone in violation of the rules.

Violation penalty: a Red Card Warning is issued to the violating robot.

7.3 Serious Violations

If any of the behaviors listed below occurs during the competition, it is judged a serious violation. For serious violations, the RMOC will disqualify the violating party at most from the competition.

If the behavior violates local laws and regulations, the RMOC will cooperate with the relevant authorities in pursuing the legal liability of the offender.

Table 7-5 Types of Serious Violations

No.	Type
1	Maliciously damaging official equipment such as the venue and props, or other teams' robots, equipment, etc.
2	Cheating, impersonation or other dishonest participation.
3	Modifying or damaging the Referee System, or using technical means to interfere with any detection function of the Referee System.
4	Failing to comply with the normative documents and being judged a serious violation by the Chief Referee.
5	Disobeying penalties, not cooperating with inspections, deliberate delay, disrupting order, withdrawing or boycotting without reason, and other behavior that hinders the match.
6	Passive play, match manipulation and similar behavior.
7	Giving property to others, or unlawfully soliciting or accepting property from others, in order to obtain improper competition results or improper benefits.
8	Uncivil or immoral words and actions such as slandering, abusing, making improper gestures, malicious heckling and maliciously throwing objects.
9	Publishing, spreading or disseminating to the media false or irresponsible statements.
10	Deliberately attacking or colliding with others, or acting in a way that endangers one's own or others' safety.
11	Carrying dangerous or prohibited items.
12	Other behavior that violates the spirit of the competition and is judged a serious violation.
13	Other words and actions that run counter to the core socialist values, violate sportsmanship, violate public order and good morals, violate competition conduct and discipline, cause adverse social impact, or violate laws and regulations.

8 Abnormal Situations

There may be some delay in the Referee's manual penalties and handling of abnormal situations. If this has a major impact on the match result, the Chief Referee will determine the final handling according to the actual situation.

If any of the following abnormal situations occurs, it will be handled in the corresponding way, and neither team may object. The handling is as follows:

- When a serious safety hazard or abnormal condition occurs, for example battery combustion or explosion, venue power outage, high-pressure gas cylinder explosion or conflict among personnel on the field, after the Head Referee finds and confirms it, the Head Referee will notify both Operators and eject all robots through the Referee System, the result of that round is void, and the round is restarted after the hazard or abnormal condition is cleared. The RMOC will handle safety problems first and protect the robots as far as possible, but cannot guarantee that robots will not be affected at all; any loss or impact caused by safety handling is borne by the participating team.
- During the match, if non-critical props in the Battlefield are damaged, for example damaged floor mat, damaged field Light Indicators or damaged Base lighting effects, which do not affect the fairness of the match, the match proceeds normally.
- During the match, if abnormal robot Armor Module lighting effects or Light Indicator effects, damaged Armor Modules or similar situations occur, the match proceeds normally.
- During the match, if critical official equipment or props malfunction or are structurally damaged in a way that affects the fairness of the match, for example the on-site network is abnormal and causes robots to go offline, the Buff is not triggered after hitting the Power Rune, or the field prop mechanisms cannot operate normally, the Referee will handle such faults manually through the Referee System. If the fault cannot be handled manually, the Referee will notify both Operators and eject all robots, the round ends immediately and its result is void. After the problem is resolved, the round is restarted.
- During the match, if critical official equipment or props malfunction or are structurally damaged in a way that affects the fairness of the match, and the Head Referee does not confirm and end the round in time, so that a round that should have ended continues and produces a win/loss result, then within 5 minutes after the match ends, once verified by the Chief Referee, the result of that round is void and the round must be replayed.
- If a serious violation that clearly triggers a Loss by forfeit penalty occurs but the Head Referee does not confirm and enforce it in time, then within 5 minutes after the match ends, once verified by the Chief Referee, the result of that round is void and an additional Loss by forfeit penalty is imposed on the violating party.
- During the match, if a situation that may affect the fairness of the match occurs, within 5 minutes after the match ends the Chief Referee will inform both team captains and suspend the result confirmation process, and within the following 60 minutes will verify the situation and inform both team captains of the final handling. Neither team may object to the handling result.

9 Appeals

Each participating team has 1 Appeal opportunity each in the Regional Competition, the Revival Tournament and the Final Tournament; they cannot be combined. If the Appeal succeeds, the Appeal opportunity is retained; otherwise 1 Appeal opportunity is consumed. When no Appeal opportunity remains, the arbitration committee will no longer accept any Appeal from that team. After an Appeal is accepted, the arbitration committee arbitrates on the Appeal materials and related evidence, and the Chief Referee informs both parties of the Appeal result on behalf of the arbitration committee; the arbitration committee has the final right of interpretation of the Appeal result.

The following cannot be used as grounds for an Appeal:

- Verbal warnings, Yellow Card Warnings and Red Card Warnings in violation penalties.
- The type and procedure of initiating a technical pause.
- "Normal battle damage" occurring on the Referee System onboard unit.

An Appeal cannot be initiated after signing the result confirmation form or more than 5 minutes after a match ends.

9.1 Appeal Procedure

A participating team that needs to make an Appeal should follow this procedure:

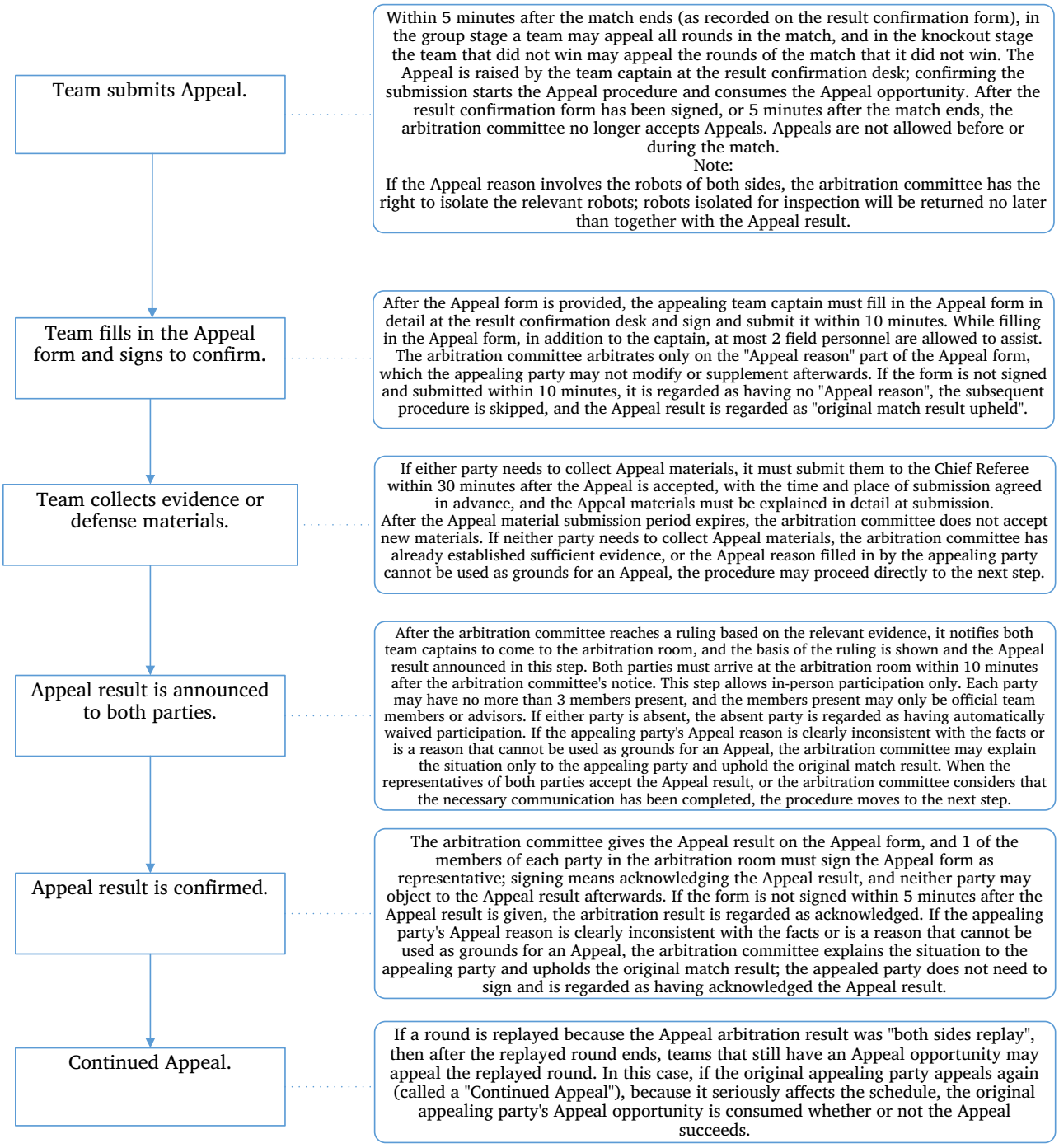


Figure 9-1 Appeal Flowchart

9.2 Appeal Materials

Each file in the Appeal materials submitted by a participating team must not exceed 500MB, and there must be no more than 10 files.

9.3 Appeal Results

Appeal results include: original match result upheld, appealed party Loss by forfeit, and both sides replay. After the arbitration committee confirms the Appeal result, neither party may object to it.

- Appeal succeeds: appealed party Loss by forfeit, or both sides replay.
- Appeal fails: original match result upheld.

If the result announced in the arbitration communication step is both sides replay, but neither side accepts the replay, the Appeal is regarded as failed and the original match result is upheld.

- Where it does not affect the overall schedule, replays are in principle scheduled after all matches of the day end, subject to actual circumstances.
- The replay procedure is the same as the regular schedule procedure, and both sides must play at the time specified by the RMOC and comply with the relevant regulations.

If the Appealed round affects the win/loss outcome of the match, the match outcome must be re-determined.

Example: A BO3 match ends 1:2, with Red winning the first round and Blue winning the second and third rounds. After the match, Red appeals the second round and the Appeal result is both sides replay. If Red wins the replayed round, the final match result is 2:0; if Blue wins the replayed round, the final match result is 1:2.

Appendix 1 Pre-Match Procedure Description

This season there will be no separate Sentry mapping session.

Pre-Inspection

Pre-Inspection is an inspection session set before the formal Pre-Match Inspection, intended to help participating teams find and resolve problems with their robots in advance. During Pre-Inspection and adaptive training, the RMOC will do its best to help teams troubleshoot and handle robot faults, but the Pre-Inspection result is not equivalent to the formal match Pre-Match Inspection result; if a robot is found at the formal Pre-Match Inspection not to meet the passing criteria, it still may not take the field.

Unique identifier registration can only be done during Pre-Inspection; after Pre-Inspection ends, only replacement re-registration of robots whose unique identifiers have already been registered is allowed.

If a team needs to repair or replace a part that carries a unique identifier, it must apply to the RMOC to report it and re-register and re-mark it. Each team may initiate at most 6 such reports in the same competition region; replacing a part that carries a unique identifier without reporting it causes the robot to lose its eligibility to take the field.

- Non-emergency reporting method: if the team's next match is more than 2 hours away, or it is outside the RMOC's working hours, the team may replace the relevant parts itself and, in [Robot Unique Identifier Re-registration], submit detailed photos of the robot before and after the replacement or a full video of the replacement process, book an inspection time through the same link, and arrive at the second Pre-Match Inspection area at the agreed time, where dedicated RMOC staff will review it. Robots that pass the review will receive their unique identifiers again.

- Emergency reporting method: if the team's next match is less than 2 hours away and it is within the RMOC's working hours, the team should contact the Referee at the second Pre-Match Inspection area to report an emergency repair. After confirmation, RMOC staff will accompany the team to the team preparation area and supervise the whole repair process. In this case, the robot may pass Pre-Match Inspection and take part in this match without all unique identifiers being completed, but after the team's next match ends, the robot must be moved to the second Pre-Match Inspection area for a second review and have all unique identifiers completed.

Field Prop Training

Prop training provides field props and wooden structures similar to those used in the formal match for participating teams to test with. Passing Pre-Match Inspection is not required before training; teams must make sure their robots' Referee System works properly, and if the on-site Referee System fails, the related time counts toward that team's training time.

Time and booking:

Each team must wait in the waiting area 20 minutes in advance; for teams that do not arrive on time, training time is still counted as normal and is not extended. Each team has two training sessions in total, each 30 minutes (including entering and leaving the field). The first session is arranged by the RMOC and the second is booked by the team itself; enter and leave strictly according to the time slots, and do not occupy other teams' training time.

Personnel and equipment:

Each team may send at most 12 people into the training area, and entrants must wear safety goggles. The RMOC provides only 2 Operator computers for the video transmission channel; if more robot player-side devices need to be connected, teams may bring their own computers, but these must not be connected to the competition network.

Props, projectiles and safety:

The RMOC provides only Movable Props for testing and does not provide 17mm or 42mm projectiles; teams must bring their own if testing requires them. When testing with 42mm projectiles, take care to avoid actively hitting the Power Rune. Because of venue limitations, the training area may be affected by ambient light, and teams must adapt on their own. If a team does not obey the on-site Referee or commits other violations, its prop training eligibility will be revoked and it will be asked to leave immediately; in serious cases its adaptive training eligibility is also revoked.

Adaptation Training

Adaptive training is a debugging and acclimatization opportunity for participating teams before the formal matches. Its main purposes include:

- Become familiar with the competition environment and procedures.
- Test and adjust robot status and strategy.

Rules for adaptive training:

- The training schedule is tight and mainly for debugging; absent a major official equipment failure, training is not paused by default.
- If the Referee System (onboard unit) is abnormal, the RMOC may grant an official technical pause of at most 2 minutes to handle the fault; the pause duration counts toward the total adaptive training time.
- No team technical pause opportunity is provided during adaptive training.

Rules for the free debugging stage:

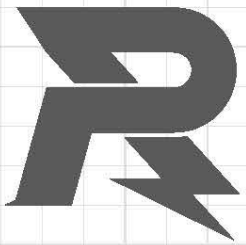
- When testing projectile launching, a team may apply to the Referee to borrow a projectile clearing bag and may launch only into the clearing bag.
- During the free debugging stage the match stage status is "Not Started"; teams must adapt to and become familiar with the switching and requirements of the match stages themselves.

Eligibility and stage division:

- Robots that did not pass Pre-Match Inspection on time or that have safety hazards may not take part in adaptive training.
- Robots that failed Pre-Match Inspection because they seriously do not meet the inspection standards are allowed to take part only in the free debugging stage of adaptive training.
- The number of robots taking part in adaptive training must not exceed the maximum robot lineup limit specified for each event.

Appendix 2 Future Rules Planning

To be updated



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